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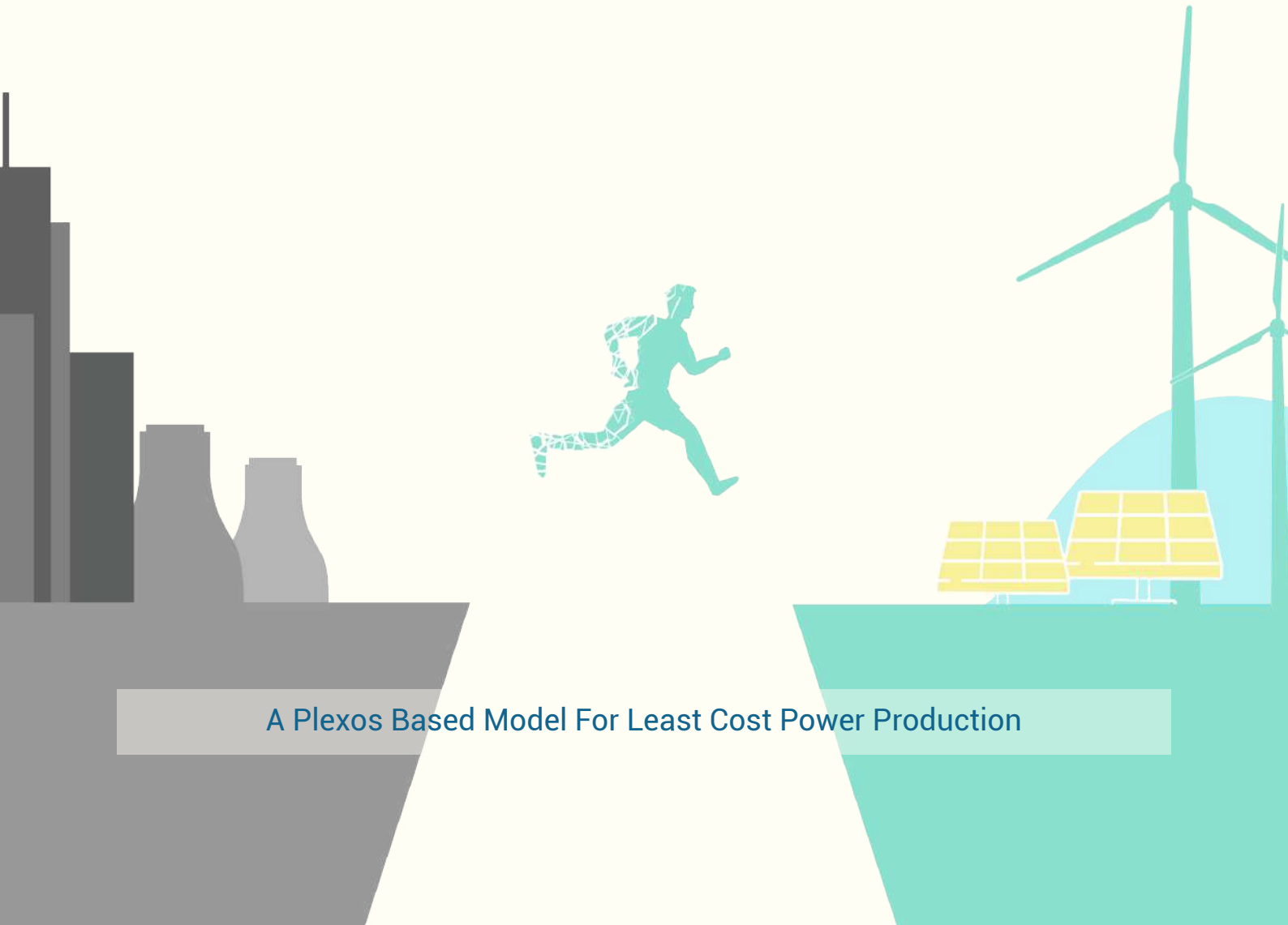
# POWERING
## PAKISTAN'S FUTURE

Pathways to Optimize Affordable and
Sustainable Electricity Generation,
Beyond IGCEP 2024-34



A Plexos Based Model For Least Cost Power Production

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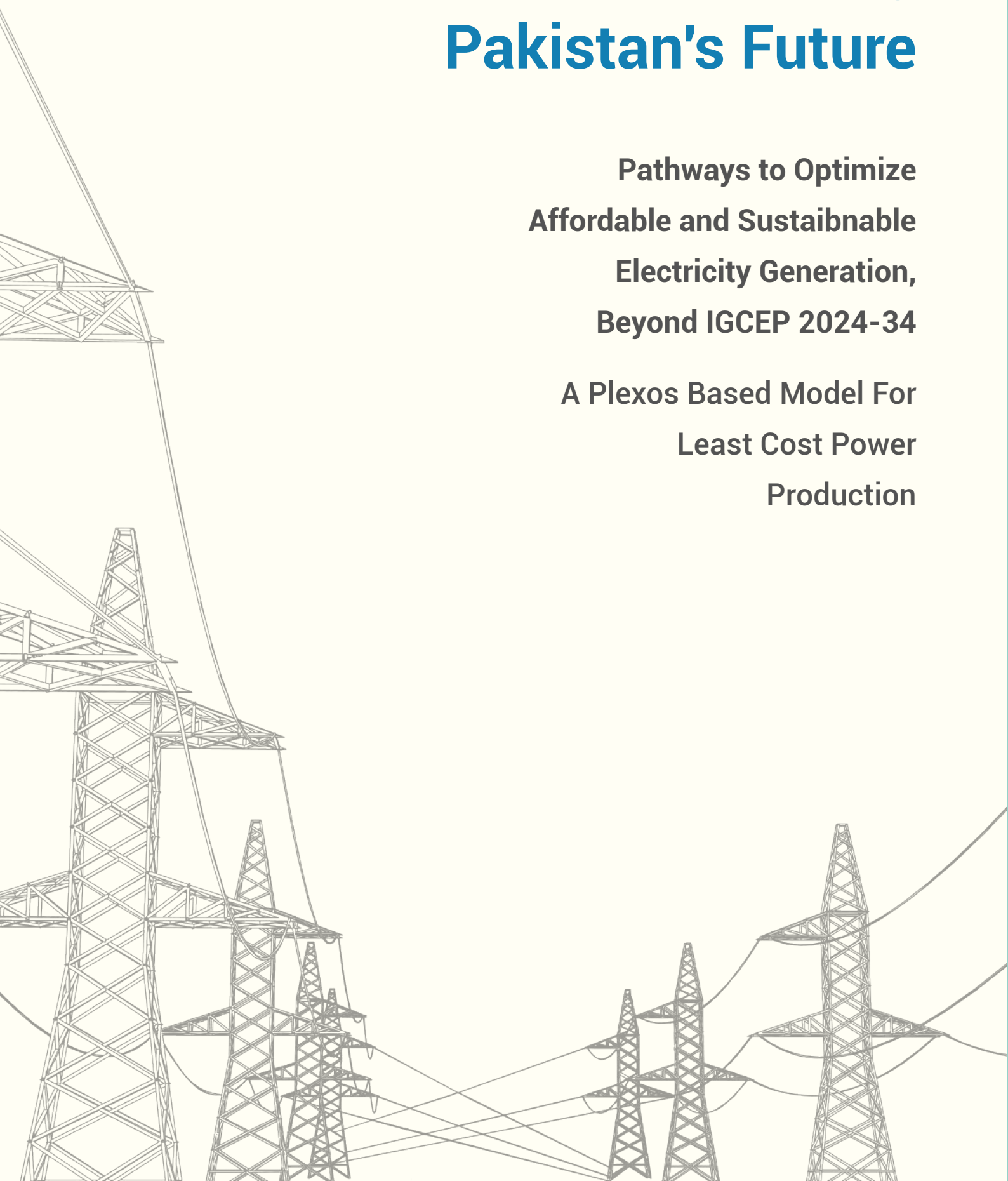

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**Renewables First (RF)** is a think tank for energy and the environment. RF's work addresses critical energy and natural resource issues with the aim of making energy and climate transitions just and inclusive through impactful research, advocacy, and strategic partnerships.

**Policy Research Institute for Equitable Development (PRIED)** is a think-tank committed to sustainable and equitable development and a low carbon future. We produce high quality research to partake in and promote a global discourse on transition to renewable sources of energy; institutionalize interaction between all energy sector stakeholders in Pakistan; provide regulatory input, policy critique and research support to the parliament, government departments, aid organizations and international financial institutions; and organize events for networking and information-sharing.

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## Disclaimer

All the information and analysis provided in this document are accurate and to the best of our knowledge and understanding, in case you identify any error, please email: info@renewablesfirst.org


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# Foreword by CEO (PRIED & RF)

Salman Faruqui, a seasoned civil servant with over seven decades of experience, recently authored his memoir, Dear Mr. Jinnah -- 70 years in the life of a Pakistani civil servant. The book chronicles his journey through Pakistan's bureaucracy, offering insightful reflections on some of the most significant events in the country's history. During a book launch in Karachi, when questioned about the hefty capacity payments being made to Independent Power Producers (IPPs), Faruqui highlighted two important details: in 1993, the government predicted that Pakistan's electricity demand would reach 55,000 megawatts by 2018, and it sought to cap the share of electricity production in this demand by IPPs at 2,500 megawatts to avoid excessive capacity payments.

As of 2024, while the total capacity of IPPs has soared to over 12,000 megawatts, contributing to massive capacity payments running into hundreds of billions of rupees annually, the country's peak demand remains well below 30,000 megawatts, far short of the 55,000-megawatt forecast for 2018. Faruqui's memoir suggests that the government has been building power plants for years to cater for a demand that was simply not there, leaving Pakistan with more electricity than it can consume and more capacity payments than it can afford.

It is notable here, however, that these erroneous "assumptions" and "plans" on part of the government were never formally conducted, never published, and never subjected to independent reviews or consultations. It was only in 2021 that the first long-term electricity generation plan, the IGCEP, was approved by the regulator. There is a consistent pattern here: governmental institutions, bureaucrats, and politicians have repeatedly made ill-informed, last-minute decisions within the hidden corridors of power, resulting in the disaster we see in the energy sector today.

Where do things stand today? The approval of IGCEP in 2021 and the Transmission System Expansion Plan (TSEP) in 2022-mark major milestones in Pakistan's history. For the first time, long-term electricity generation and transmission plans were developed through structured processes, subject to public scrutiny – at least in principle - and made publicly accessible. These plans have garnered significant attention from various stakeholders and brought greater clarity to the consumers and developers alike.

However, some issues remain which indicate that the government's mindset has not changed. While the earlier editions of the IGCEP indicated significant promise with regards to renewable energy sources and least cost planning, the latest IGCEP has reduced the optimizable candidate projects to merely 1% of the total planned. This means that the rest 99% are not subject to least cost optimization. This means that only 1% of Pakistan's electricity in the next decade will be generated from projects optimized under the least-cost principle, while the vast majority will be produced through pre-committed projects—regardless of their cost. This is in sharp contrast to the previous edition of IGCEP, where the share of candidate projects open to optimization was significantly higher.

This situation is both problematic and absurd. Despite the use of advanced modeling software like PLEXOS, policymakers have effectively rendered the entire process moot by committing to projects whose necessity is questionable and which the software is not allowed to comment on. Large shares of hydroelectric and nuclear in the form of Diamer Bhasha and C-5 have been declared as committed even when, as our study shows, they are neither least-cost nor needed for their capacity in the electric grid.

While this is the situation on the supply side, there is little effort on the demand side to reign in new consumers. Captive power plants, which can raise utilization rates of idle plants, remain outside the national grid due to industrial mistrust in the government's ability to ensure stable and reliable electricity. At the same time, there is no initiative by the government to consider new technologies such as battery storage or electric vehicles, both of which could increase electricity consumption through the grid and lower average electricity costs.

The government may have been wrong in its assumptions about the projected power demand in the past, but demand can always be increased through proactive measures and initiatives. The government must also realize that reduced demand of electricity in the grid is not independent of its own behavior. Flawed planning, coupled with erratic policies, has alienated consumers and investors alike and the recent rise in roof-top net metering is an evident example of the public's growing distrust in the government's ability to provide reliable energy.

In the current study, our team has made an attempt to discuss the above scenarios and more with the intention of improving transparency and quality in power sector planning. By shedding light on the severe shortcomings in the current plan and proposing alternative scenarios, the authors hope to provide policymakers with the insights they need to correct course. Without careful reassessment, Pakistan risks burdening itself with unnecessary electricity projects—projects that could impose unsustainable costs for generations to come, a mistake that we have continued to make in the past and are in the process of making yet again.

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# List of Abreviations

## Key Documents

| | |
|---|---|
| IGCEP | Indicative Generation Capacity Expansion Plan |
| TSEP | Transmission System Expansion Plan |
| ISP | Integrated System Plan |
| ARE Policy 2019 | Alternative & Renewable Energy Policy |

## Institutions & Entities

| | |
|---|---|
| MoE | Ministry of Energy |
| NTDC | National Transmission and Desptach Company Limited |
| CPPA-G | Central Power Purchasing Agency |
| DISCO | Distribution Company (Total 10 in Pakistan) |
| KE | K-Electric |
| AEDB | Alternative Energy Development Board |
| IPP | Independent Power Producer |
| GENCO | Government Owned Generation Company |
| CPP | Captive Power Plants |
| WAPDA | Water and Power Development Authority |
| NEPRA | National Electric Power Regulatory Authority |
| GENCOs | State-owned Generation Companies |
| NEECA | National Energy Efficiency & Conservation Authority |
| IPPs | Independent Power Producers |
| CPPs | Captive Power Producers |

## Others

| | |
|---|---|
| ISP | Integrated System Plan |
| VRE | Variable Renewable Energy |
| kW, MW, GW | Units of Power |
| kWh, MWh, GWh | Units of Energy |

# Executive Summary

Pakistan's power generation capacity expansion planning process continues to face credibility challenges despite being regularly conducted within an established regulatory framework. The newly launched Indicative Generation Capacity Expansion Plan (IGCEP) 2024-2034 (IGCEP 2024), succeeding the previous IGCEP 2022-2032 (IGCEP 2022), launched after a one-year delay, reveals significant changes in load patterns, generation patterns, and capacity expansion plan. The IGCEP is a comprehensive planning document prepared annually by the National Transmission and Despatch Company (NTDC) as per the Grid Code 2022 and approved by the National Electric Power Regulatory Authority (NEPRA).

NTDC has defined its base case for the IGCEP 2024 predicated on an average GDP growth rate of 3.5%, a figure considered low compared to the previous year's plan. In 2023-24, Pakistan's installed capacity stood at 43.7 GW, projected to scale up to 56 GW by 2033-34 according to the latest plan. While this increase corresponds to rising electricity demand, the projected peak demand is significantly lower than in the 2022 version. For instance, the financial year 2030-2031 is now expected to see a peak demand of 34 GW — about seven GW less than previously envisaged.

Additionally, total energy consumption is reduced by 18%, and peak demand projections have been notably lowered, with the high growth scenario for 2034 now equivalent to the low growth scenario from the previous plan. Notably, although overall electricity demand and consumption have dropped since last year, the generation expansion planning still needs to uphold the principles of reliable, affordable, and sustainable energy supply across the country. Alarmingly, over 99% of the capacity additions over the planning period of 2024 to 2034 are committed projects, leaving no room for least cost optimization by the power sector generation planning software PLEXOS. This report, a sequel to our previous study 'Powering Pakistan's Future' launched in 2023, endeavors to verify the outputs of the latest plan, along with bringing forward the outcomes of a true least cost generation expansion modeling exercise. We do so by first replicating the 2024 base case and extending it to various realistic and futuristic scenarios and sensitivities. Using this approach, we believe that the people of Pakistan can be better informed about the status quo of energy sector planning, its deficiencies, and the opportunity costs of paths undertaken by the government.

This assessment is also critical as the existing thermal 'take or pay' power plants are rendering the power sector financially unviable and becoming a financial burden because of their low utilization rates but persisting capacity payments. Therefore, it is indispensable to critically analyze any future capacity additions in terms of impact on the overall system cost, be they strategic or committed. The IGCEP plays a crucial role in this regard, as it outlines the specific power plants that need to be constructed following the least cost criteria, along with their respective timelines, to ensure a seamless transition to a more efficient and economically viable energy mix.

To conduct this rigorous exercise, we have leveraged our previous dataset and modified it to incorporate the latest available information. In addition, we have engaged with NTDC in the acquisition of the latest load profile data, whereas no other agency we reached out to provide the data we needed. Data acquisition and availability challenges have remained, but we have managed to bring forth this report in a condensed time of four to five months, using the best-suited assumptions wherever required.

The formation of a long-term capacity expansion plan is a praiseworthy effort as it sets a baseline direction for the power sector. However, there are certain transparency and methodology issues, which include:

- Incomplete disclosure of committed and strategic projects' cost data
- Missing technical data such as ramp-up, ramp-down rates, and scheduled and forced outages of thermal plants.
- Missing information and rationale behind macroeconomic assumptions such as forming base case on a low GDP growth rate.
- Over 19 GW capacity additions have been modeled as committed keeping the modeling tool blind to their capital costs.
- Heavily relying on hydro projects that will generate electricity from far away from the load centers. Also, the impact of dry seasons such as winter should also be considered.
- No weightage given to projected alternate and renewable energy shares in the national grid mandated by the ARE Policy 2019
- No scenario built on assessing how PLEXOS optimizes the generation expansion if committed projects are taken as candidate generation options.
- Not exploring the financial implications of delayed or early retirements of expensive thermal power plants.

The latest iteration of IGCEP specifies that about 20 GW of new capacity will be constructed and integrated into the national grid over the next decade, increasing generation capacity to 56,046 MW. This capacity expansion endeavor demands a substantial investment, estimated at USD 63.31 billion as per IGCEP 2024. The committed Diamer Basha Dam and Chashma Nuclear Unit-5 projects constitute a major portion of this required investment; but are they truly needed to meet the system demand; this is a question that we explore in this report.

This report is a continuation of an effort that started in 2021, in collaboration with Renewables First (RF) and the Policy Research Institute for Equitable Development (PRIED), to verify the generation planning process being followed in Pakistan and understand its implications for an average end consumer in terms of affordability and sustainability. We performed the same modeling exercise as NTDC did in formulating the IGCEP 2022 using the publicly available data, and assumptions, constraints, and data provided in the IGCEP 2022. Yet the results varied, and renewables were heavily optimized by our model. The 2024 iteration of the IGCEP, however, depicts an energy mix that not only contradicts our previous study results but also contradicts its previous 2022 iteration. Therefore, it is critical to evaluate IGCEP 2024 results, and what has led to such a stark shift in installed capacity by 2034 from different technologies.

The study underscores the vital need to incorporate a range of scenarios and sensitivities in power generation capacity expansion planning, integrating macroeconomic factors for sound decision-making. To further enhance this process, we aim to involve academic institutions and utilize advanced methodologies like Multi-Criteria Decision Making (MCDM) and Fuzzy Failure Mode and Effect Analysis (FFMEA). By fostering collaboration with academia, research institutions, and policy planning departments, we can ensure our strategies are both innovative and robust, ultimately strengthening Pakistan's energy resilience and sustainability for the future

Some of the major findings of our analysis are as follows:

1. **If committed projects are treated as candidate projects and evaluated on merit in the base case, nearly $5.79 billion can be saved in total system cost, and VRE share can reach 13% in the grid by 2033. In medium and high growth scenarios, the VRE share can further improve to 25% and 26% respectively.**

Committed projects are plants that have not been built or entered commercial operation to date, yet their feasibility is not evaluated by the IGCEP. In this scenario we fed both the initial capital expenditure (CAPEX) and ongoing operational expenditure (OPEX) of these projects into PLEXOS and allowed it to select, reject or optimize them. As a result, installed capacity by 2033 reaches 46.6 GW with 6.2 GW VREs in the base case, 55.4 GW with 14 GW VREs in the medium growth, 59.3 GW with 15.56 GW VREs, respectively.

2. **As net metering quantum continues to increase because of soaring electricity prices and declining solar PV prices, no capacity additions will be required in the grid except for committed projects until the year 2027.**

IGCEP 2024 projects 2.1 GW of net metering additions by 2034, however, the net metering additions have already crossed 2 GW. As electricity consumers continue to become prosumers with rooftop solar installations, large scale generators away from load centers can be avoided, bringing total installed capacity to 44.7 GW by 2033 with 4.7 GW of VRE, and enabling total system cost savings of $3.7 billion.

3. **If the load of captive power plants is shifted on to the national grid, utilization rates of existing fleet become better, and no significant capacity has to be added to cater their load in the NTDC system. Whereas, in the KE system, 2.77 GW of Solar & wind, 82 MW of hydro, and 1.22 GW of coal is optimized.**

IGCEP 2024 has no scenario which discusses the impact of shifting industry load on the national grid owing to the frequent shortage of gas. If load of industries is shifted on the national grid in phases (40% by FY 2027, 30% by FY 2029, and 30% by 2030), the installed capacity reaches 60 GW by 2033 with 10.4 GW VREs (17% VRE share). The utilization rates of the existing generation fleet also improve as a result.

4. **When EVs charging demand is considered in the system, with pessimistic estimate of 2000 MW (400 MW in KE system and 1600 MW in NTDC system), no substantial capacity is optimized in the NTDC system leveraging the surplus capacity. Whereas, in the KE system, 1 GW of wind, 82 MW of hydro, and 305 MW of coal is optimized.**

The total cost in this case is $62 billion owing to new capacity additions and utilization rates of the existing fleet also improve as a result.

5. **When transmission capacity between NTDC and KE system is increased the need for optimizing new generation options in KE system is reduced**

When tie-line transmission capacity is realistically assumed to increase to 3600 MW, the utilization of KE's fleet reduces significantly since the cheaper power from the national grid is available. Increasing the interconnection capacity also reduces the need

for additional thermal power plants in the KE system. The results further suggest that the total installed capacity by 2033 will be 56.7 GW with a total system cost of around 60 billion USD.

6. **Diamer Bhasha Dam and CHASHNPP-5 are both expensive and not needed in the system**

Treating these high-cost projects as candidate options show that they are not optimized by the model due to their high cost and lack of need in the grid, and if forcibly added, they will lead to unnecessarily incurred costs of up to $7.6 billion (Diamer) and $5.52 billion (Chashma) as well as causing a reduction in potential shares of renewable energy.

# Recommendations

**Transparent Evaluation of Committed Projects:** Rigorously assess all committed projects against least-cost criteria to ensure economic viability and alignment with policy goals, particularly in increasing renewable energy adoption.

**Focus on Renewable Integration and Grid Modernization:** Prioritize investments in grid infrastructure to accommodate renewable energy sources and BESS, enhancing flexibility, reducing losses, and preventing theft.

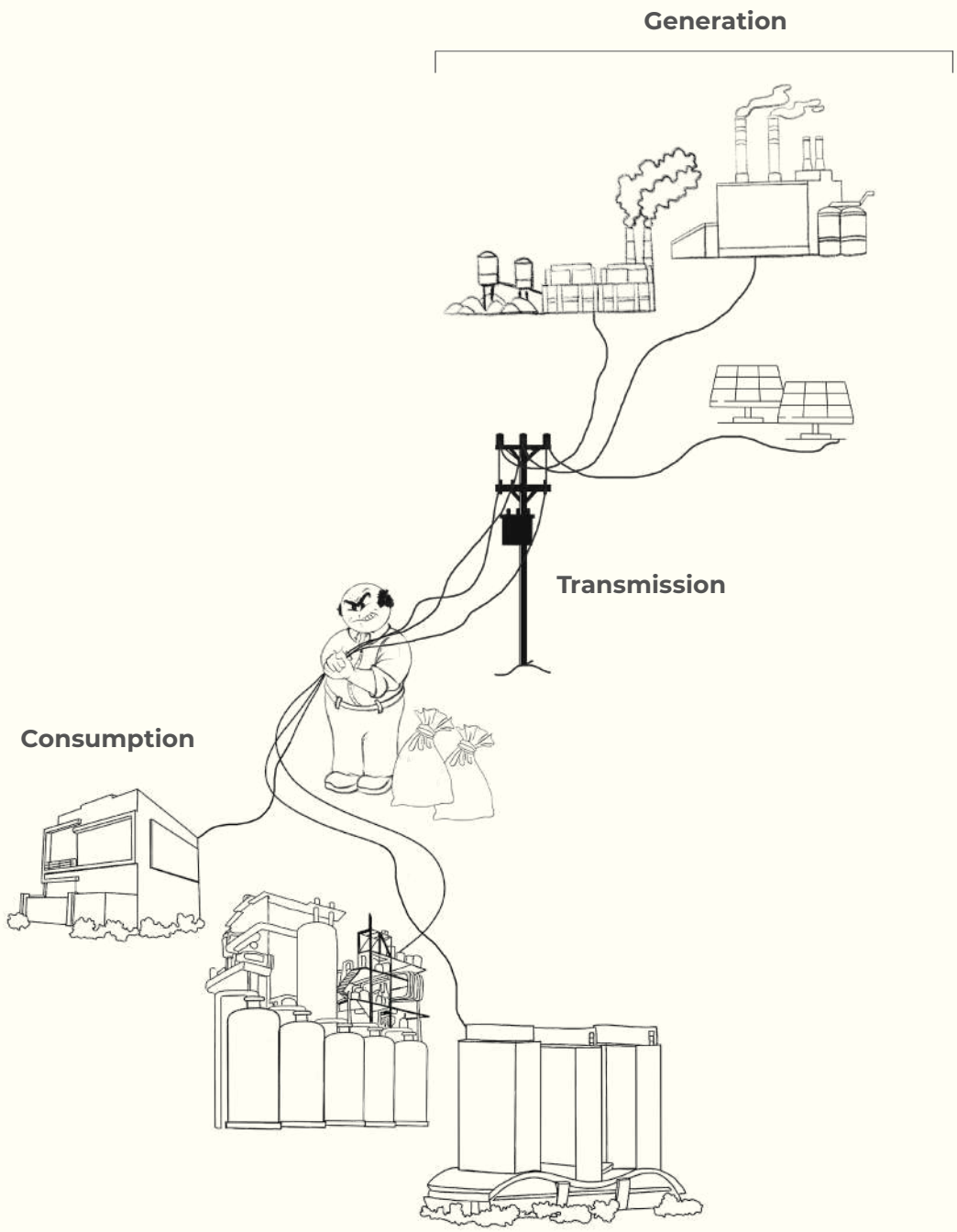
**Enhance NTDC-KE Connectivity:** Increase the tie-line capacity between NTDC and KE to supply affordable electricity, reducing KE's reliance on costly and environmentally unsustainable local coal and fossil fuels.

**Leverage Rooftop Solar Growth:** Encourage rooftop solar installations to meet energy demands without additional capacity investments, capitalizing on declining solar PV prices.

**Incorporate EV and CPP Loads Strategically:** Integrate EV charging infrastructure and shift CPP loads to the national grid to optimize existing capacity and promote renewable energy utilization.

**Avoid Unnecessary Capacity Additions:** Given surplus capacity and underutilized plants, refrain from new capacity additions unless they meet strict least-cost and sustainability criteria.

Finally, it is worth mentioning that significant resources and energy have been dedicated to conducting the above analysis. Driven by a genuine commitment to enhancing Pakistan's power sector, our team has worked year-long to address the shortcomings in the IGCEP 2024-2034 diligently. We urge the government and planning agencies to be more open to our insights and recommendations. By integrating our findings, they can better serve the needs of Pakistan's citizens and advance towards a more sustainable and efficient energy future.




# Chapter 1:

# An Overview of Pakistan's Power Sector

The power sector of Pakistan can be broadly divided into three separate categories based on the flow of electricity: generation, transmission & distribution, and consumption.

### Generation

Generation of electricity across the country is carried out by a combination of state-owned Generation Companies (GENCOs), large hydro power plants (maintained by WAPDA), nuclear plants, and Independent Power Producers (IPPs) which include thermal, hydro and renewables.

### Transmission & Distribution

Transmission of electric power across the country is the responsibility of NTDC which operates the primary transmission network of Pakistan and supplies electricity to all areas of the country except the zone of K-Electric (KE). KE, being a privately owned and vertically integrated utility, possesses its own transmission system, which is the sole network for transmitting power to Karachi and some surrounding regions. However, NTDC relates to the KE system through a 1100 MW tie-line, the capacity of which has augmented to 2050 MW, after commissioning of KANUPP – K-Electric Interconnection grid in June 2024. This way cheaper electricity can flow from NTDC's system to the KE system up to transmission capacity as and when needed.

Distribution is being carried out through eleven publicly owned Distribution Companies (DISCOs). These are responsible for electric power distribution in their respective areas. The only exception is KE, which is a vertically integrated utility and carries out all three main functions of generation, transmission, and distribution within a 6500 km2 territory including Karachi and its adjoining areas.

### Consumption

Electricity consumption (or demand) in Pakistan is distributed across four main areas: residential, commercial, agricultural, and industrial. Pakistan's daily and annual consumption patterns are highly variable, with a consistent increase in peak demand over the years and little to no increase in the base load. These features make variable renewable energy sources (VRE) such as solar, wind, and hydro attractive for Pakistan as they naturally complement each other.

### End-Users

End users are connected through electricity meters with the DISCOs infrastructure to power their homes, businesses, industries, Farms, and Government. These end users are further categorized based on their consumption patterns (e.g., peak/off-peak hours), load profiles and tariff structures (e.g., residential, commercial, industrial, agricultural), and metering arrangements (e.g., prepaid, postpaid, smart meters).

| | | |
|---|---|---|
| **42,501 MW**<br>NTDC's Installed Capacity | **3,384 MW**<br>KE's Installed Capacity | **27,333 MW**<br>Peak demand of Country<br>FY 2022 - 2023 |
| **129,030 GWhs**<br>NTDC's Total Generation<br>FY 2022 | **8,998 GWhs**<br>KE's Generation Fleet<br>FY 2022 | **16.45%**<br>Transmission and Distribution<br>Losses |

# Overseeing Bodies

**Federal**

- MOE (Ministry of Energy)
- CCOE (Cabinet Committee on Energy)

**Mediator**

- CCI (Council of Common Interests)

**Provincial**

- Provincial Energy Departments

- NEPRA (System Regulator)
- NTDC (System Planner and Operator)

## Generation

60 Fossil Fuel Plants
7 Nuclear Plants
22 Hydel Plants
14 Small Hydel (<30 MW)
12 Solar Plants
45 Wind Plants
10 Bagasse Plants

## Transmission

**Lines**

8,825 km of 500 kV (71 circuits)
11,633 km of 220 kV (166 circuits)

**Grid Stations**

19 grid stations of 500 kV
50 grid stations of 220 kV

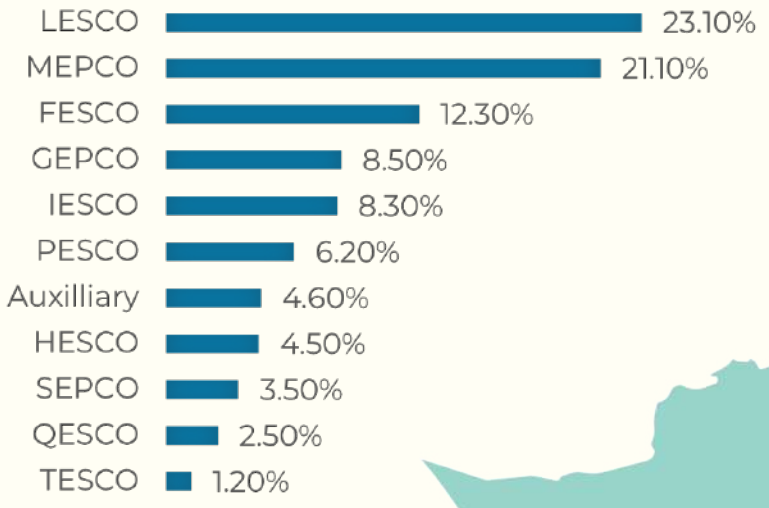
**Transformers**

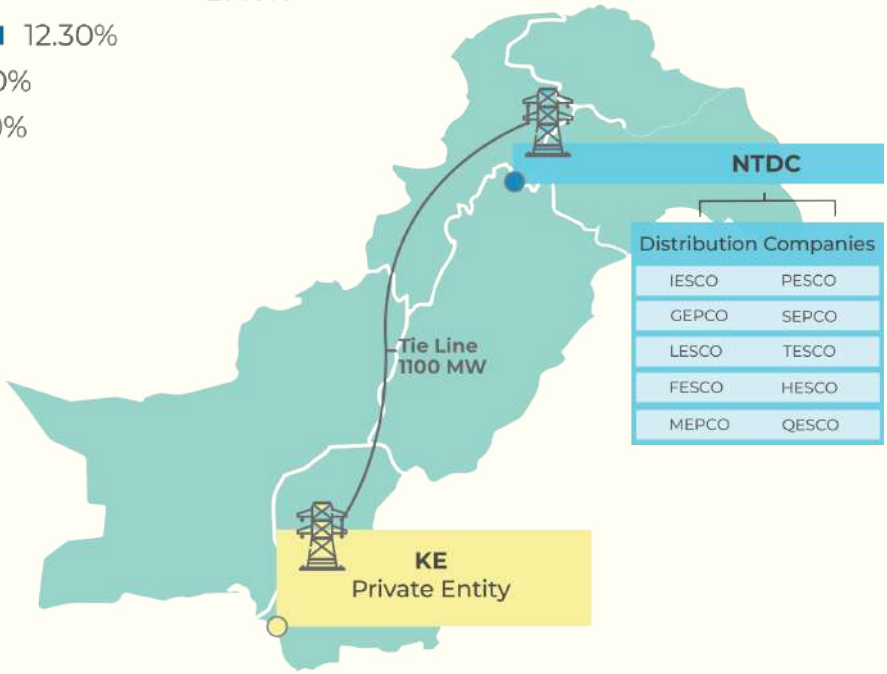
47 transformers of 500/220 kV
179 transformers of 220/132 kV

## Distribution

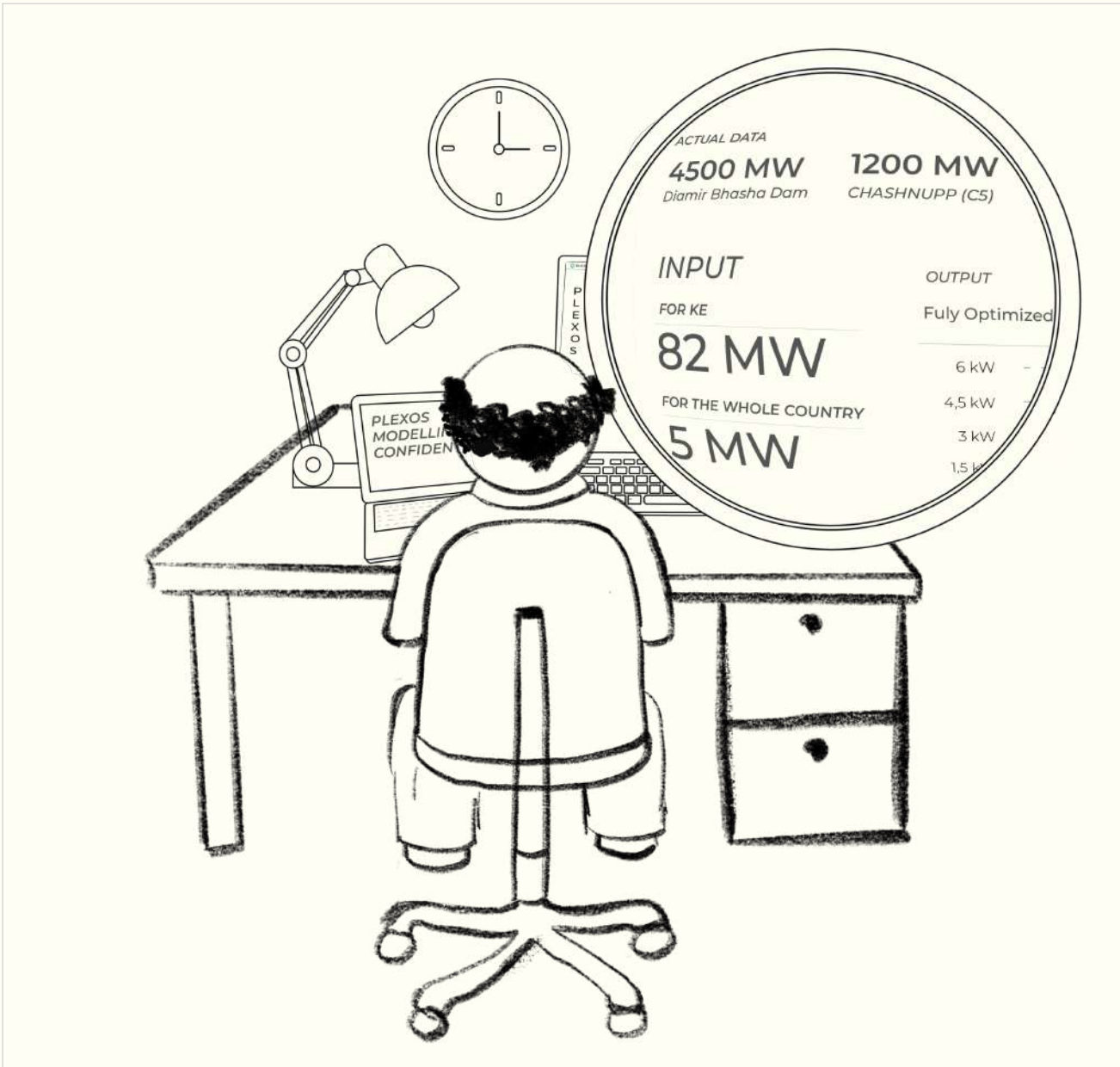
31,656 km of 132 kV lines
941 Grid Stations of 132 kV
2,267 Power Transformers
58,042 Distribution Transformers
358,219 Feeders of 11 kV

Share of electricity consumption - FY 2023



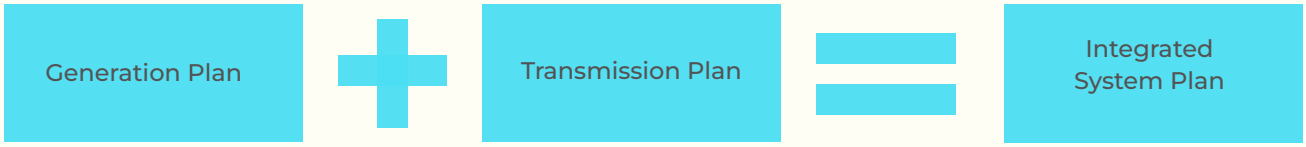

**In Pakistan, a two island approach is followed in which the power sector is split into two subsystems: NTDC and KE. Both sectors manage their own supply and demand which results in more expensive dispatch in the KE system. This approach also fails to benefit from demand-curve superposition.**




# Chapter 2:

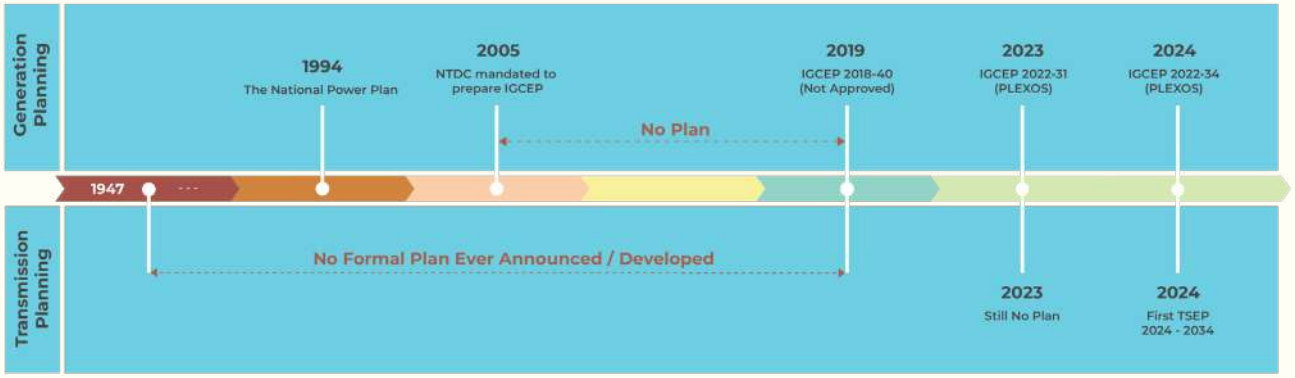
# Power Sector Planning Process

The planning process in the power sector can be divided into two major categories: generation planning (laid out in the IGCEP) and the transmission planning (laid out in the TSEP). The IGCEP is of at least 10 years, prepared every year with rolling horizon, whereas the Transmission System Expansion Plan (TSEP) may be prepared for one, three, five or ten years. The first IGCEP in the history of Pakistan was approved in 2021 whereas the first TSEP has come to light in 2024. Principally, both IGCEP and TSEP are required to be in place to form an Integrated System Plan, that lays out the viability of any generation project. This year, IGCEP 2024 has come together with the TSEP but has not been infused into an Integrated System Plan (ISP) document.




In early decades of post-independence, more emphasis was given to water resources management rather than energy concerns. WAPDA was established in 1958 mainly for managing water resources to boost the country's agricultural economy. Although various studies on power system planning were undertaken afterwards to supplement the regular five-year medium-term plans of the Government, no energy policy was formally announced by any government until 1994. It was only in 2005 that NTDC was mandated by the grid code to prepare a comprehensive capacity expansion plan, the IGCEP, which it failed to produce for the next 15 years. Power plants continued to be built on an ad-hoc and arbitrary basis, the disastrous economic implications of which have culminated in the economic crisis that we live in today.

When the NTDC finally submitted its first plan to NEPRA in 2019 and 2020, they were found inadequate and subsequently rejected. Only when in 2021 the World Bank made the disbursement of a $400 million loan contingent on the approval of IGCEP, was it hastily approved by the regulator. The subsequent plan, however, which was due in June next year, came more than three months late. Such institutional failures to deliver and arbitrary delays are a routine occurrence in the power sector of Pakistan and have cost the country dearly. While the grid code 2005 augmented by the latest approved grid code in 2023 also required NTDC to submit the TSEP along with the IGCEP, only recently in 2024, we have seen both planning documents coming into public together. Although the power sector planning is becoming continuous, now the challenge remains in making the planning process credible, transparent, and accessible to the relevant stakeholders.



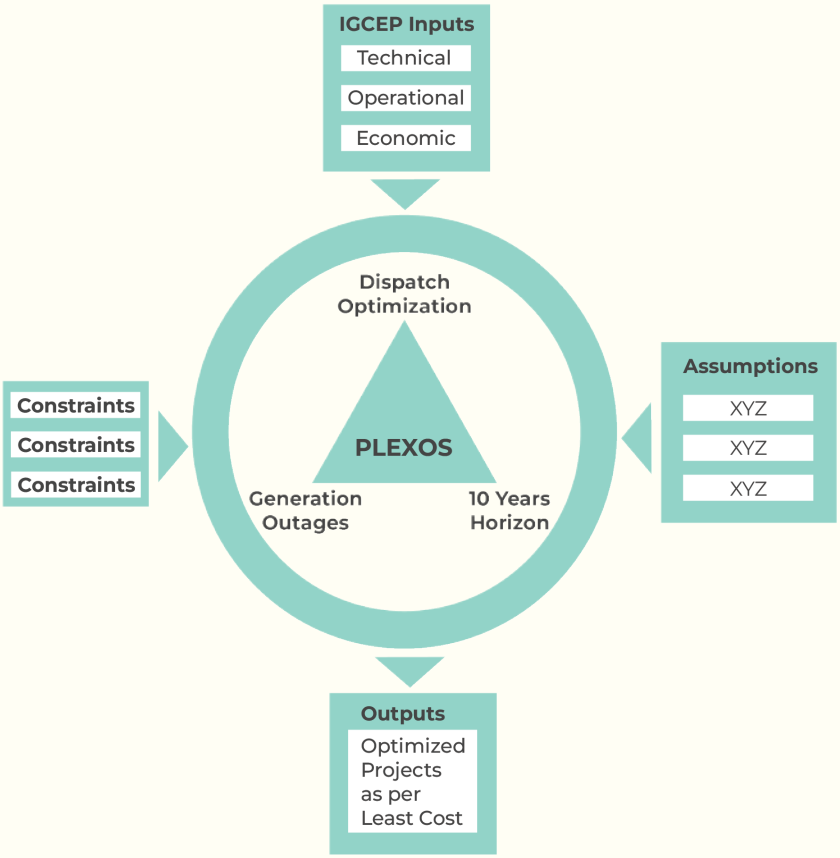
# Chapter 3:

# Assumptions & Methodology

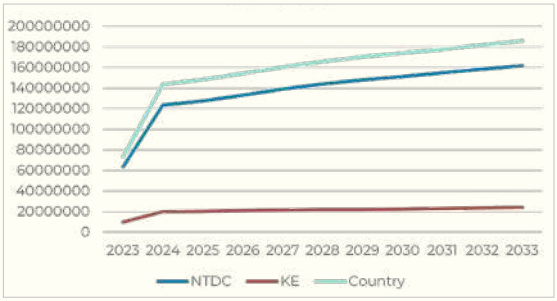
Our goal in this study is to assess the credibility of the IGCEP prepared by NTDC in terms of reliability and least cost criteria. To achieve this objective, we have developed a base case model that considers the same assumptions and constraints as provided in the IGCEP 2024. The planning horizon in this study is considered as 10 years as per the Grid Code 2023 — which stipulates a planning horizon of at least 10 years — under section PC 2.1 c, as per which we have shown the results until the FY33. The IGCEP 2024, however, extends it a year further and shows the results until the FY34. The base year is considered as Fiscal year 2024 (July 2023-June 2024). Where any data is found missing or incomplete, we have referred to data available on NEPRA's website to fill such gaps. We have also taken additional assumptions where required to get equivalent results as in the IGCEP 2024. Some of the key assumptions for our base case have been listed below for reference:

- Business as usual demand forecast has been considered for developing the base case which corresponds with low rate of GDP growth.
- For candidate local coal (330 MW coal fired Steam at Thar), the fixed fuel cost component of 113.205 $/kW-year and fuel price of 1.45 $/GJ has been considered respectively.
- For candidate local coal (660 MW coal fired Steam at Thar), the fixed fuel cost component of 107.72 $/kW-year and fuel price of 1.70 $/GJ has been considered respectively.
- The lead time of four years for candidate local coal projects and a 2-year for wind and solar technologies has been considered in the model.
- Only 1000 MW of wind can be added each year starting from July 2027 as candidate generation option. Only 1300 MWp of solar can be added each year starting from July 2027 as candidate generation option.
- A yearly block of 100MW of a new/disruptive/nascent technology has been considered from 2028 onwards until the end of study horizon. The cost parameters for the new technology have been added considering an average price lower than the basket price of electricity.
- Existing projects located near the load center have been considered as "Must Run", for summer months, i.e., May to September up till year 2027.

For the committed Hydel, we have used reference power purchase prices as issued by NEPRA in the Reference Power Purchase Prices for FY 2024 – 2025 dated June 14, 2024



# RF-PRIED PLEXOS Modelling Process



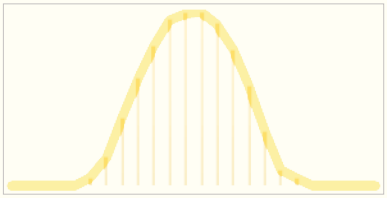

## Load Forecast

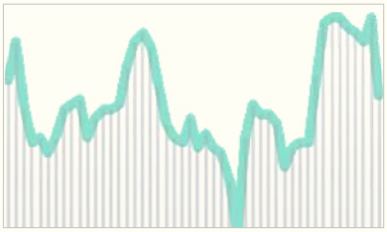
Based on projected GDP growth rate of 3.5% in the base case for a 10-year period, whereas CAGR stands at 3.1%.

## Feeding 300+ Generation Options

Technical data such as capaciity, min stable level, heat rates, ramp rates etc. from existing, committed, and candidate power is fed into PLEXOS for analysis.







## Wind and Solar Profiles

Hourly solar and wind data were fed into the model for 8,760 hours of the year. Separate data is entered for utility-scale, distributed generation, net metering, and existing plants.

## Hourly Load Profiles

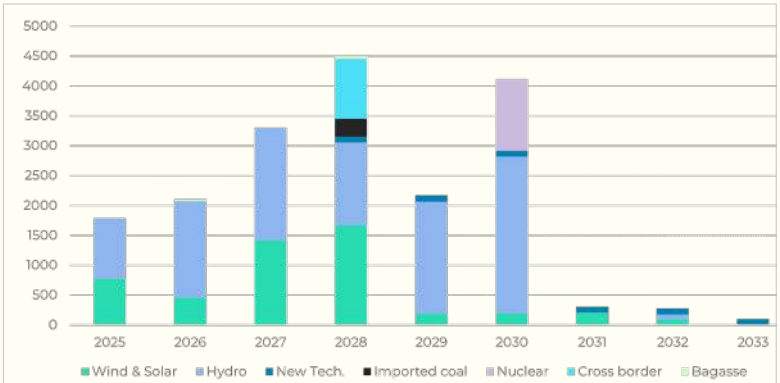
NTDC provided detailed hourly profiles for 175,200 hours, greatly improving the model's accuracy and reliability.

## Contractual Obligations & Constraints

Constraints such as minimum energy off take requirements, must run statuses, and retirements are added in the model in line with IGCEP 2024.

## Economic & Financial Parameters

Parameters such as build costs, operations and maintenance costs, economic life (e.g. 50 years for hydro and nuclear), fuel costs etc. are fed into the model based on EIA projections.




## Projects Optimized by PLEXOS

Considering all respective constraints, assumptions, build costs, dispatch capability, and financial parameters, PLEXOS recommends technology builds on a least-cost principle, year by year.

## Data Availability and Acquisition Challenge Still Exists

The best approach in power sector modeling is using practical and realistic data, which helps the model capture complex and rapidly changing ground conditions. The more accurate and practical data we procure the more accurately we predict future demand, optimal resource allocation and dispatch, future transmission and distribution planning, and prevent faults, losses, and blackouts.

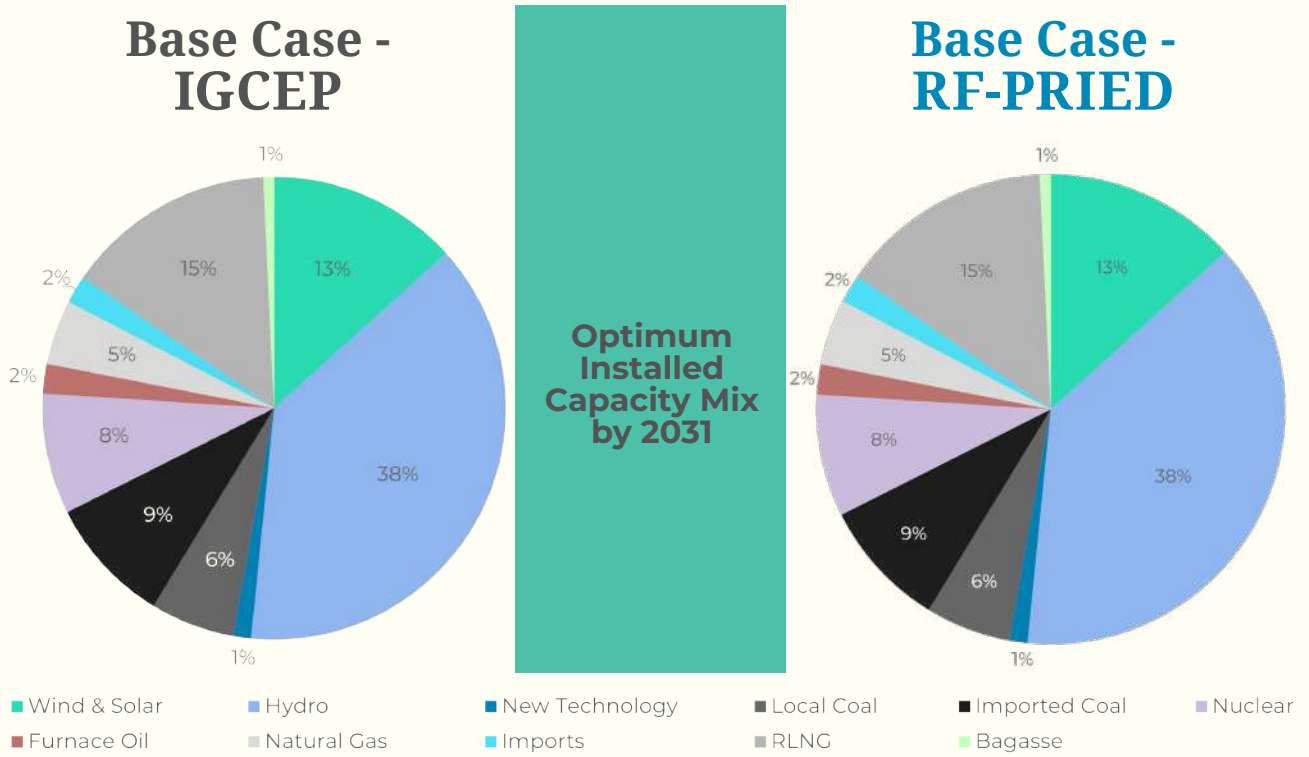
Data plays a crucial role in energy planning, but its availability, accessibility and usability, is often compromised. While some generation planning data can be found in the State of Industry Reports, tariff determinations, and generation licenses, the formats are not easily readable or understandable. While conducting this study, our major challenge has been data collection. Although we were provided load forecasting data by the NTDC, cooperation from other entities has been limited. Furthermore, data available on NEPRA's website is either not complete for power generators or is not updated. For example, tariff determinations at the stage of commercial operations start of recent coal-fired power plants are missing, and for nuclear no specific data is available on NEPRA's website or Pakistan Atomic Energy Commission.

Particularly for hydel committed projects, whose cost details are not available in IGCEP 2024, various websites were surfed to find relevant cost data and generation profiles. The ramp rates essential for increasing and decreasing energy generation as per variations in demand are missing; the ramp rate fluctuation may impact plant efficiency and indirectly the cost of generation, which is again missing in the document.

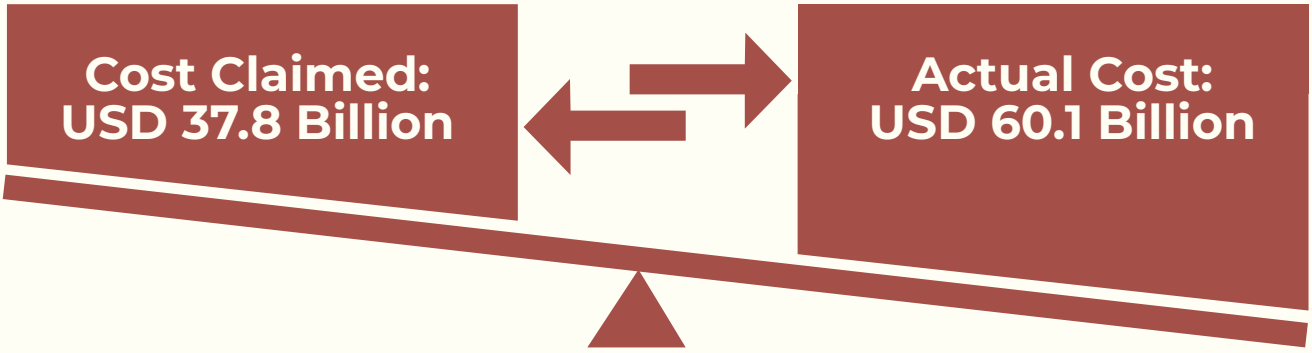
Pakistani citizens pay taxes for all their basic utilities, which helps the government in running the power sector; they have the right to know where their taxes have been spent. Therefore, data transparency is indispensable and its accessibility is crucial to ensure accountability, improving stakeholder participation, making informed decisions for generating revenue, reducing risks, potential research and planning, and attracting foreign investment in the power sector.

## Comparison of IGCEP & RF-PRIED Base Case

The figure below illustrates the differences between NTDC's IGCEP and RF-PRIED's base case, with the most significant disparity being the total system cost claimed by NTDC. This discrepancy primarily arises from the tariff assumptions for committed hydropower projects totaling approximately 10.47 GW. Notably, the IGCEP lacks clear tariff assumptions for these committed projects, and it appears that NTDC has significantly underestimated these tariffs. As a result, the cost projections do not accurately reflect the true financial implications of the committed hydropower projects.




| | 2033 | |
|---|---|---|
| 13% (7.5 GW) | Share of Wind and Solar | 13.4% (7.6 GW) |
| 56.8 GW | Total Installed Capacity | 56.8 GW |




*The total costs represent the NPV and are without inclusion of project costs of the committed projects. Since NTDC has not mentioned the projects costs of committed projects, we have also excluded the same from above, which is quite significant. A comprehensive comparison must take the cost of committed projects into account. The total scenario-wise investments are listed separately for each scenario in the results section.

## 3.1 Demand Forecasting

Demand forecasting is done by using PLEXOS and all the assumptions have been taken from IGCEP. While determining the demand, NEECA's energy efficiency targets have also been rationalized into the model. The demand forecast also considers the non-AT&C losses gradually in the system.

Table: Demand forecasting (NTDC System)

| FY | Low growth | | Medium Growth | | High Growth | |
|---|---|---|---|---|---|---|
| | Load | Peak Demand | Load | Peak Demand | Load | Peak Demand |
| | (GWh) | (MW) | (GWh) | (MW) | (GWh) | (MW) |
| 2024 | 122,743 | 24,453 | 128,240 | 24,812 | 139,039 | 26,902 |
| 2025 | 126,330 | 25,090 | 131,012 | 25,579 | 142,901 | 27,767 |
| 2026 | 130,991 | 25,936 | 135,536 | 26,706 | 144,672 | 28,231 |
| 2027 | 137,285 | 27,099 | 141,361 | 28,112 | 148,013 | 29,007 |
| 2028 | 141,619 | 27,870 | 148,822 | 29,872 | 152,829 | 30,080 |
| 2029 | 145,809 | 28,819 | 154,340 | 31,059 | 159,156 | 31,461 |
| 2030 | 150,200 | 29,817 | 160,690 | 32,420 | 166,852 | 33,125 |
| 2031 | 152,579 | 30,422 | 167,363 | 33,853 | 175,887 | 35,072 |
| 2032 | 156,799 | 31,401 | 175,103 | 35,510 | 186,304 | 37,312 |
| 2033 | 159,357 | 32,197 | 183,343 | 37,374 | 197,995 | 40,004 |

Table: Demand forecasting (K-Electric System)

| FY | Low growth | | Medium Growth | | High Growth | |
|---|---|---|---|---|---|---|
| | Load | Peak Demand | Load | Peak Demand | Load | Peak Demand |
| | (GWh) | (MW) | (GWh) | (MW) | (GWh) | (MW) |
| 2024 | 19,975 | 4,114 | 21,874 | 4,168 | 21,897 | 4,172 |
| 2025 | 20,257 | 4,234 | 21,969 | 4,290 | 22,271 | 4,349 |
| 2026 | 20,813 | 4,347 | 22,256 | 4,404 | 22,334 | 4,420 |
| 2027 | 21,271 | 4,463 | 22,545 | 4,522 | 22,861 | 4,586 |
| 2028 | 21,714 | 4,571 | 22,877 | 4,631 | 22,948 | 4,646 |
| 2029 | 22,032 | 4,706 | 23,100 | 4,768 | 23,747 | 4,901 |
| 2030 | 22,377 | 4,833 | 23,363 | 4,896 | 24,148 | 5,060 |
| 2031 | 22,768 | 4,963 | 23,682 | 5,028 | 24,891 | 5,285 |
| 2032 | 23,323 | 5,084 | 24,167 | 5,150 | 25,315 | 5,395 |
| 2033 | 23,852 | 5,235 | 24,622 | 5,303 | 26,007 | 5,602 |

### 3.2 New Generation Options

For the base case scenario, to keep the set of assumptions similar to what is mentioned in the IGCEP, we have kept the candidate generation options the same as given in the IGCEP. However, for the additional scenarios such as unconstrained renewable energy addition, the number of solar and wind power plants along with the new technologies such as Battery Energy Storage Systems (BESS) is increased and left for the PLEXOS to optimize as required. The CAPEX and tariff assumptions for additional solar and wind power projects are kept like the IGCEP and for the BESS, the CAPEX is estimated by consulting with independent consultants and research-based institutes like Bloomberg New Energy Finance and IRENA.

### 3.2.1. Candidate Generation Options

Keeping in line with the IGCEP, the following candidate options have been built in the NTDC and KE's system, for optimization by PLEXOS:

**NTDC System**

- Steam Turbine New Local Coal candidate (330 MW)
- Steam Turbine New Local Coal candidate (660 MW)
- Gas Turbine on RLNG (400 MW)
- Combined Cycle Gas Turbine on RLNG (1263 MW)
- Nuclear Steam PP on Uranium (1,200 MW)
- Wind Turbines 1000 MW each year starting from July 2027
- Utility-scale solar of 1300 MWp each year starting from 2027

**KE System**

- Steam Turbine New Local Coal candidate (330 MW)
- Steam Turbine New Local Coal candidate (660 MW)
- Wind Turbine (Block of ≤ 300 MW in July 2025 and Block of ≤ 100 MW each year till the end of study horizon)
- Solar PV (Block of ≤ 300 MWp in July 2025 and Block of ≤ 200 MWp each year till the end of study horizon)

### 3.2.2. Committed projects

Following is the list of projects which have been entered into the base case model as committed projects. When the projects are added in the model as committed, their cost is taken as 'zero', so that the modeling software (PLEXOS) can select them irrespective of their need in the system or affordability.

A review of the provided list below shows that most of the committed projects are based on hydropower technology.

| Technology | Project Name | Capacity (MW) | Expected Date |
|---|---|---|---|
| Hydro | Balakot | 300 | Dec-2027 |
| | Chamfall | 3.22 | June-2024 |
| | Dasu 1 | 360 | Mar-2027 |
| | Dasu 2 | 360 | Apr-2027 |
| | Dasu 3 | 360 | May-2027 |
| | Dasu 4 | 360 | Oct-2027 |
| | Dasu 5 | 360 | Nov-2027 |
| | Dasu 6 | 360 | Dec-2027 |
| | Diamer Bhasha U1 | 375 | Jan-2029 |
| | Diamer Bhasha U2 | 375 | Feb-2029 |
| | Diamer Bhasha U3 | 375 | Apr-2029 |
| | Diamer Bhasha U4 | 375 | May-2029 |
| | Diamer Bhasha U5 | 375 | June-2029 |
| | Diamer Bhasha U6 | 375 | Jul-2029 |
| | Diamer Bhasha U7 | 375 | Aug-2029 |
| | Diamer Bhasha U8 | 375 | Sep-2029 |
| | Diamer Bhasha U9 | 375 | Oct-2029 |

| | | | |
|---|---|---|---|
| Hydro | Diamer Bhasha U10 | 375 | Nov-2029 |
| | Diamer Bhasha U11 | 375 | Jan-2030 |
| | Diamer Bhasha U12 | 375 | Feb-2030 |
| | Gorkin Matiltan | 84 | Aug-2025 |
| | Jabori | 10.2 | May-2024 |
| | Jagran II | 48 | Dec-2024 |
| | Karora | 11.8 | Jul-2024 |
| | Kathai-II (U#1) | 4 | Jul-2026 |
| | Kathai-II (U#2) | 4 | Jul-2026 |
| | Koto | 40.8 | Jun-2024 |
| | Kurram Tangi | 18 | Jun-2024 |
| | Lawi | 69 | Dec-2024 |
| | Mohmand Dam U1 | 200 | Dec-2026 |
| | Mohmand Dam U2 | 200 | Jan-2027 |
| | Mohmand Dam U3 | 200 | Mar-2027 |
| | Mohmand Dam U4 | 200 | Apr-2027 |
| | Nardagian | 3.22 | Dec-2025 |
| | Riali-II (U#1) | 3.5 | Jun-2025 |
| | Riali-II (U#2) | 3.5 | Jun-2025 |
| | Suki Kinari (U#1) | 221 | Aug-2024 |
| | Suki Kinari (U#2) | 221 | Sep-2024 |
| | Suki Kinari (U#3) | 221 | Oct-2024 |
| | Suki Kinari (U#4) | 221 | Nov-2024 |
| | Tarbela_Ext_5 (U#1) | 510 | Nov-2025 |
| | Tarbela_Ext_5 (U#2) | 510 | Dec-2025 |
| | Tarbela_Ext_5 (U#3) | 510 | Jan-2026 |
| Imported Coal | Gwadar | 300 | Dec-2027 |
| | Jamshoro Coal | 660 | June-2024 |
| Bagasse | CSP | 30 | Jan-2028 |
| | Shahtaj | 32 | June-2024 |
| | TAY | 30 | Aug-2025 |
| Nuclear | C5 | 1200 | July-2029 |
| Import | CASA | 1000 | Aug-2027 |
| Wind | Trans-Atlantic | 50 | June-2028 |
| | Western | 50 | June-2028 |
| Solar | Access Electric | 10 | Dec-2023 |
| | Access Solar | 11.52 | Dec-2023 |
| | Helios | 50 | Dec-2023 |
| | HNDS | 50 | Dec-2023 |
| | Manjhand | 50 | June-2028 |
| | Meridian | 50 | Jan-2024 |
| | Safe | 10 | Nov-2024 |
| | Siachen | 100 | June-2028 |
| | Zorlu | 100 | Oct-2024 |
| | PV_Comm_A | 600 | Jan-2027 |
| | PV_Comm_B | 600 | Apr-2027 |
| | PV_Comm_C | 1200 | Nov-2027 |
| | Net Metering | 2107 | Throughout study horizon |

## 3.3. Fuel price indexation factors

For fuel prices modelling we have relied on the same mechanism as provided in the IGCEP. The IGCEP uses the Energy Information Administration (EIA)'s Annual Energy Outlook 2023 for fuel price indexations over the study horizon. The factors of indexations used are provided in the table below.

Table: Fuel Price Indexations

| FY | RFO | RLNG/NG | Imp. coal | Uranium | Thar coal | Bagasse |
|---|---|---|---|---|---|---|
| 2024 | 1 | 1 | 1 | 1 | 1 | 1 |
| 2025 | 0.992 | 0.797 | 0.998 | 1 | 0.99 | 1.02 |
| 2026 | 0.938 | 0.688 | 0.99 | 1 | 1.01 | 1.02 |
| 2027 | 0.936 | 0.602 | 0.976 | 1 | 0.94 | 1.04 |
| 2028 | 0.939 | 0.549 | 0.967 | 1 | 0.95 | 1.04 |
| 2029 | 0.939 | 0.528 | 0.955 | 1 | 0.95 | 1.061 |
| 2030 | 0.942 | 0.527 | 0.949 | 0.999 | 0.94 | 1.061 |
| 2031 | 0.946 | 0.534 | 0.95 | 0.999 | 0.95 | 1.082 |
| 2032 | 0.949 | 0.546 | 0.947 | 0.999 | 0.93 | 1.082 |
| 2033 | 0.953 | 0.561 | 0.932 | 0.999 | 0.7 | 1.104 |

## 3.4. Scenarios & Sensitivity Analysis

The IGCEP 2024 constitutes four scenarios on top of the base case. The base case has been modelled on a low-demand growth scenario. The other two scenarios are based on the sensitivities of demand forecasting such as medium growth and high demand growth. Two additional scenarios include the delay in the committed Diamer Bhasha HPP and considering two candidate HPPs Azad Pattan and Kohala as committed in the model.

In our study, we have replicated the base case as in the IGCEP 2024. In addition, we have developed following scenarios and sensitivities in our model:

### 1. The base case

The base case scenario is designed to keep the same assumptions as used in the IGCEP 2024 base case scenario so that the starting point of our study aligns with NTDC's IGCEP. All the constraints such as must-run obligations and committed projects are like the IGCEP base case, the only difference is the tariff assumptions of committed projects. Since the tariff assumptions of committed projects are missing in the IGCEP, therefore the tariff assumptions of such projects are considered as per the latest reference power purchase prices for the year 2023-2024 as available on NEPRA's website.

### 2. The True Load Scenario

In this scenario, we assess the impact of true load on the overall power sector generation planning. We do so by incorporating the industrial load of captive power plants (CPPs) in the model, which is not part of the base case. We estimate that by shifting the load that the industries are currently meeting through CPPs to the national grid it will improve the system performance and utilization of existing capacity. Moreover, the fuel (usually natural gas) provided to the industries at subsidized prices for operating their CPPs could also be directed to alternate needs and be used efficiently given the depletion of local natural gas reserves. In this scenario we shift the load that the industries are currently meeting through their captive power generators — fueled by either diesel or gas — onto the national grid. According to well-informed sources and industrial leaders, the estimated quantum of captive-based industrial load is around 5 to 7 Gigawatts. In our model, we considered three estimates for captives based on pessimistic, realistic, and optimistic considerations.

**Sensitivities:**

- **Pessimistic**: 4000 MW (2000 MW in KE's system and 2000 MW in NTDC's system)
- **Realistic:** 7,000 MW (3500 MW in KE's system and 3500 MW in NTDC's system)

◊ **Optimistic:** 10,000 MW (5000 MW in KE's system and 5000 MW in NTDC's system)

### 3. Electric Vehicles Impact on Grid

In this scenario, we assess the impact of an increase in electricity demand on generation expansion planning, owing to an increased quantum of electric vehicles (EVs) in the country's transportation fleet over the study horizon. Pakistan introduced its first national EV policy in 2019, which aimed at gradually increasing the share of EVs in the country, as well as creating a local market for EV development. However, this government direction, and public interest towards EVs, requires careful generation planning and enabling infrastructural developments such as the erection of charging stations. It will further set the stage for deliberation on required incentives to manage the load on the grid instead of straining the grid. An EVs' charging station typically ranges between 2kW— 43kW in the case of an AC charging station and 50kW—1 50kW and more in the case of a DC charging station. The projections of EVs and their charging station loads are taken from the technical brief "Scaling up Electric Mobility in Pakistan" by NEECA developed under the UNDP NDC Support Programme. The sensitivities for the load of EV charging stations in Pakistan are as follows:

**Sensitivities:**

◊ **Pessimistic Projection:** 2000 MW by FY 2033

◊ **Realistic Projection:** 3500 MW by FY 2033

◊ **Optimistic Projection:** 4500 MW by FY 2033

The demand as mentioned above is factored in the cumulative load profile of Pakistan and then the impact on utilization rates of the existing fleet and optimization of new generation mix is analyzed.

### 4. No Capacity Additions (NCA)

Since the installed power generation capacity is considerably higher than the utilized capacity due to which, Pakistan is facing the issues of surplus power capacity and capacity payments, a unique scenario of "No Capacity Additions (NCA)" is also simulated and analyzed. The objective of this scenario is to check whether the installed generation capacity would be enough to meet the future demand if no further power plants are added to the generation fleet. Another major assumption used in this scenario is solarization. One of the most critical oversights in the IGCEP 2024 is the net metering projections. The IGCEP 2024 assumes 2,107MW from net metering by FY2034. However, as of April 2024, the installed net metering capacity had already exceeded 2,000MW, and this trend is expected to always accelerate due to rising electricity prices and solar PV prices lowest in Pakistan. Additionally, the federal and provincial governments are aggressively promoting solarization to maximize the use of indigenous and renewable sources. It is believed that one reason for the decline in energy demand of the system is due to solarization whether net metered or non-net metered. According to *Bloomberg*, Pakistan imported around 13,000MW of solar modules in the first six months of 2024 alone, making it the third-largest destination for Chinese exporters. This massive import volume and the growing trend of rooftop solar installations necessitate a prudent consideration of solarization's impact on demand side. Incorporating rooftop solar, both net-metered and non-metered, across all consumer types—industrial, three-phase, and single-phase is crucial. In the NCA scenario, the committed generation capacity until FY 2027 is considered along with the net metering quantum of 10GW(around 1000 MW each year). The impact of net metering quantum is considered on the demand side.

### 5. Increased Electricity flow from NTDC to KE - Affordable Electricity for KE Consumers

In NTDC's current approach, both dispatch and expansion planning for KE and NTDC systems is carried out separately. A limited quantum of electricity — 1100 MW which has now been increased to 2,050 MW under a new agreement with the NTDC — is allowed to flow from NTDC to the KE system. This approach results in a cost discrepancy between the two systems. NTDC has a diverse energy generation pool comprising 64.4% thermal power, 27.4% hydro, 4.2% renewables, and 4% nuclear energy, the overall generation basket rate is thus lower than KE's. If the interconnection capacity between NTDC and KE is further increased in the future, more affordable power will flow from the NTDC network to the KE's network — as currently the per unit price is higher in the KE system due to the dominance of thermal power generation in their network.

For sensitivity analysis, we considered different scenarios: increasing the supply to K-Electric from 2,050 MW to 2,800 MW, 3,600 MW, and 4000 MW, respectively. The result of this analysis is explained in the Results Section.

**Sensitivities:**

- Supply to KE = 2800 MW
- Supply to KE = 3600 MW
- Supply to KE = 4000 MW

### 6. Collective impact of CPPs and EVs – Building Synergies

This scenario aims to see the impact of better utilization of the existing grid on long-term generation planning by a) shifting the captive power plants on the national grid and b) incorporating EVs' charging demand. Considering the envisaged drop in the energy demand as outlined in the IGCEP 2024 and surplus generation capacity, more benefits can be reaped by first utilizing the existing capacity. Doing this strategically, electricity from variable renewables can be optimally utilized further by aligning EV charging with their peak generation which may also reduce curtailments from the must-run plants. Overall, the synergy in increased energy utilization could alleviate the financial burden of idle capacity, improve the efficiency, reliability, and flexibility of the system, and increase the potential for renewable energy integration. It further paves the way for sustainable development by reducing carbon footprint in the transportation sector, contributing to a cleaner environment.

**Sensitivities:**

- Realistic only – 4000 MW captive load and 2000 MW EVs load by 2033

### 7. Collective impact of CPPs, EVs and enhanced electricity flow to KE – The Efficiency Trio

In continuation to building synergies, this scenario is designed to analyze the combined effect of captives, EVs and enhanced electricity flow to the KE system – "The efficiency trio". This integrated approach will not only enable grid operators and utilities to increase grid flexibility, improve reliability, and optimize energy efficiency but also create capacity for integrating more renewable energy sources. By harnessing the strengths of this trio, the peak demand can be managed efficiently and achieve maximum efficiency in energy generation and utilization.

**Sensitivities:**

- CPPs load of 4000MW (2000MW in the KE system and 2000MW in the NTDC System)
- Increase in demand by 2000 MW to facilitate EV charging
- Increase in transmission capacity (from NTDC to KE) of 4000 MW from 2050MW

### 8. Unconstrained VRE additions: The Greener Future

In the current IGCEP about 20GW of capacity expansion in the next 10 years is on a committed basis i.e., the model has not been allowed to assess their optimization on the least cost basis. In our unconstrained RE scenario, we are treating the committed hydro and nuclear projects as candidate generation options for the software to optimize if needed in the system. With respect to solar and wind, their yearly addition constraints have been removed. Their unlimited addition with a block size of 100MW has been allowed in the model for optimization on a least cost basis.

Committed hydro projects (7.76 GW) and nuclear projects modeled as candidate generation options are as follows:

- Balakot HPP (300 MW) as a candidate option from 1st December 2027
- Dasu (Units 1 to 6 total 2160 MW, each unit of 360 MW)
- Diamer Bhasha Dam (4500 MW)
- Mohmand Dam (800 MW, 4 units each of 200MW)
- C-5 (1200 MW) Nuclear as a candidate option in FY-30

## 9. Additional Cost of Diamer Bhasha Dam and CHASHNUPP-5 (C-5)

In the IGCEP 2024, a majority of power projects are classified as committed; however, the underlying justifications and rationales for their inclusion remain largely unexplained. Critical details such as CAPEX, tariff assumptions, and comprehensive cost evaluations are absent, raising concerns about whether these projects meet the least-cost selection criteria. Among these committed projects, the Diamer Bhasha Dam (4,500 MW) and CHASNUPP-5 (C-5, 1,200 MW) stand out due to their substantial capital costs. According to recent media reports and reliable sources, the estimated cost for C-5 is approximately $5 billion, while the Diamer Bhasha Dam is projected at $4.86 billion. Notably, IGCEP 2024 only accounts for their levelized tariffs, which appear to be significantly underestimated.

To accurately reflect the true costs and implications of these projects, a scenario analysis was conducted. In the first iteration, C-5 was treated as a candidate project with all other assumptions consistent with the base case scenario. In the second iteration, Diamer Bhasha was evaluated as a candidate project. Finally, in the third iteration, both Diamer Bhasha and C-5 were considered as candidate options to analyze cost differentials and determine whether these projects would be optimized by PLEXOS. This approach aims to provide a clearer understanding of the economic feasibility of these projects and whether they align with least-cost planning principles, ensuring that the generation expansion strategy remains both financially sound and aligned with policy goals.

# Chapter 4:
# The Study Outputs

## 4.1. Base Case

The base case scenario is built along the assumptions, constraints, and obligations as determined by the IGCEP. In case of any missing data inputs in the IGCEP and non-provision of data from relevant energy sector agencies on request, we have used the closest suitable assumptions. In the current IGCEP, the base case has been developed on a demand forecast based on low GDP growth rate. Additionally, we see that the 99.95% capacity which could have been added to the system based on least cost, is added as committed. As a result, the model optimizes only 87 MW of hydel capacity. For solar and wind, we see that except for the already committed capacity, no additional capacity is being optimized. For Hydropower, in addition to what has been added as committed, 87.2 MW gets optimized by the model.

**Key features**

- The tariffs for both existing and candidate projects align with those in the NTDC IGCEP 2024, ensuring consistency in cost assumptions across the board.
- Renewable energy projects totaling 640 MW within the K-Electric system are treated as committed projects, as their Request for Proposals (RFPs) have been issued and are included in K-Electric's approved Power Acquisition Program (PAP) for 2024.
- Total **installed capacity** reaches **56.8 GW** with **VRE share of 13 % and hydel share of 37%**.
- The **total system cost (NPV)** in our analysis is estimated at **$60.1 billion**, excluding the capital costs of committed projects. The required investments for these **committed projects** amount to approximately **$23.9 billion**.
- In contrast, **NTDC's projections** indicate a **total system cost** of **$38.1 billion**, excluding committed project costs. When these project costs are included, **NTDC** estimates the **total system cost (NPV)** to be **$63.31 billion**, highlighting a potential discrepancy in cost evaluations between the analyses.

### Capacity Addition and Investment Required

- New Capacity Built: 20.256 GW
- Total System Cost (NPV) = $60.1 billion

| | Solar | Wind | Nuclear | Hydro | Imported Coal |
|---|---|---|---|---|---|
| Capacity (MW) | 5608 | 232 | 1200 | 10564.2 | 960 |
| CAPEX ($ billion) | 3.86 | 0.325 | 5 | 14.08 | 0.714 |

Total Investments required = **$ 23.9 billion**

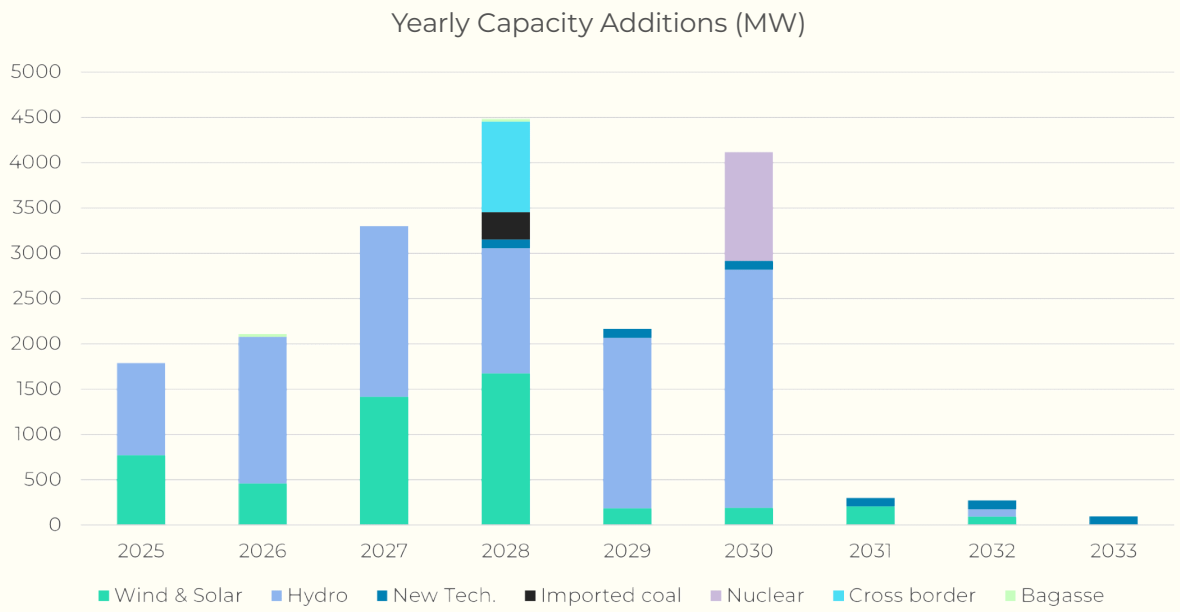
### Committed Projects (MW)

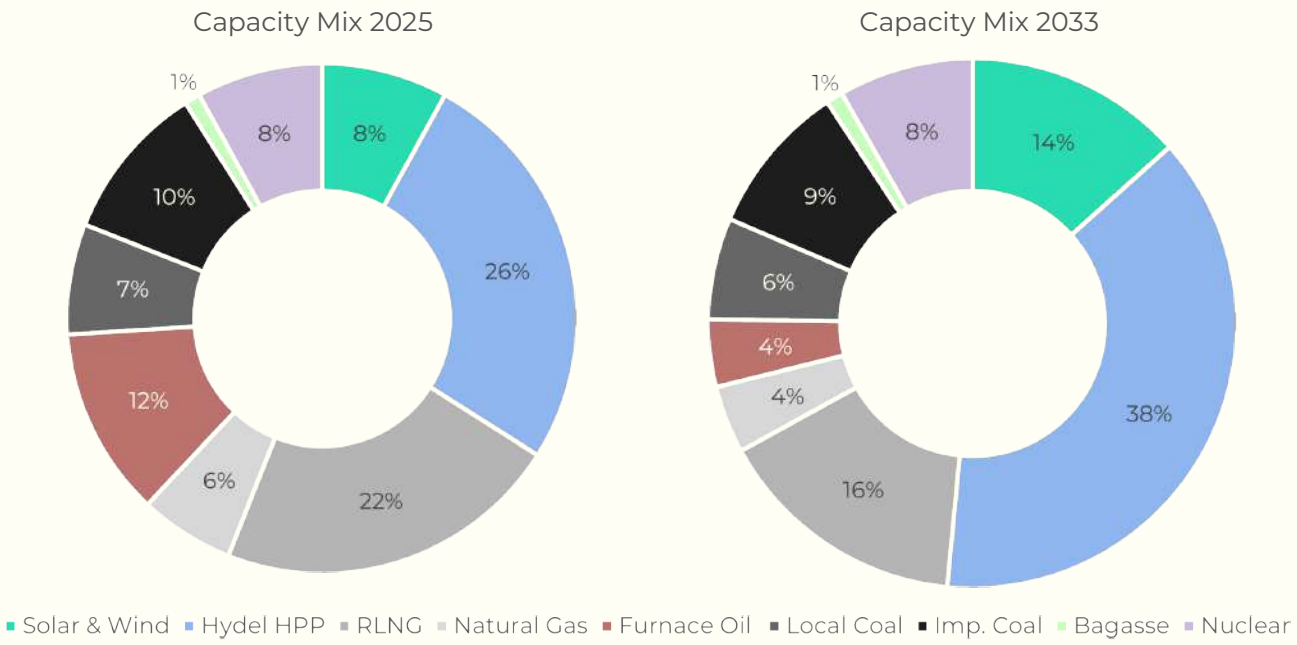
| FY | Hydro | Imp. Coal | Local Coal | Wind | Solar | Cross Border | Nuclear | New Tech | Bagasse | Yearly Addition |
|---|---|---|---|---|---|---|---|---|---|---|
| 2024 | 72 | 660 | 0 | 0 | 171.52 | 0 | 0 | 0 | 32 | 936 |
| 2025 | 357 | 0 | 0 | 0 | 770 | 0 | 0 | 0 | 0 | 1127 |
| 2026 | 818 | 0 | 0 | 132 | 329 | 0 | 0 | 0 | 30 | 1309 |
| 2027 | 1299 | 0 | 0 | 0 | 1465 | 0 | 0 | 0 | 0 | 2764 |
| 2028 | 1591 | 300 | 0 | 100 | 1645.4 | 1000 | 0 | 100 | 30 | 4766 |
| 2029 | 4310 | 0 | 0 | 0 | 236 | 0 | 0 | 100 | 0 | 4646 |
| 2030 | 1310 | 0 | 0 | 0 | 241 | 0 | 1200 | 100 | 0 | 2851 |
| 2031 | 360 | 0 | 0 | 0 | 255 | 0 | 0 | 100 | 0 | 715 |
| 2032 | 360 | 0 | 0 | 0 | 142 | 0 | 0 | 100 | 0 | 602 |
| 2033 | 0 | 0 | 0 | 0 | 353 | 0 | 0 | 100 | 0 | 453 |
| Total | 10477 | 960 | 0 | 232 | 5608 | 1000 | 1200 | 600 | 92 | 20,169 |

Inclusive of KE's committed projects and net-metering

## Optimized Candidate Projects (MW)

| FY | Hydro | Imp. Coal | Wind | Solar | Cross Border | Nuclear | New Tech | KE Hydro | KE Local Coal | KE Solar PV | KE Wind | Capacity Addition Per Year |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2024 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2025 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2026 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2027 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2028 | 5.2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 5.2 |
| 2029 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2030 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2031 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2032 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 82 | 0 | 0 | 0 | 82 |
| 2033 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Total | 5.2 | 0 | 0 | 0 | 0 | 0 | 0 | 82 | 0 | 0 | 0 | 87.2 |



## 4.2. Unconstrained VRE – The Green Future

In this scenario we evaluate the impact of treating committed projects as candidate generation options on long-term generation expansion planning. It is critical to evaluate committed projects with full transparency based on their costs, as about one third of our intended generation capacity by the end of 2033 is added in the IGCEP 2024 as committed. In the current IGCEP about 20GW of capacity expansion in the next 10 years is on committed basis i.e. the model has not been allowed to assess their optimization on the least cost basis.

### Key features

The unconstrained scenario suggests that the share of variable renewable energy sources can be increased if the hard constraints are removed from the model and committed projects are left open for optimization. Three sensitivities on demand growth are performed in unconstrained VRE scenario i.e., business -as-usual demand, medium demand and high demand.

### 4.2.1. Unconstrained VRE – Business as usual demand

- Total installed capacity reaches to 46.6 GW by 2033 with 6.2 GW VREs (13% VRE share).
- The share of hydel power generation increases from 10.75 GW to 13.6 GW (29% of total capacity) out which 2.7GW is committed quantum (until FY 2027) and only 216 MW is optimized by PLEXOS.
- The total cost savings amount to $5.79 billion (Total system cost NPV $54.31 billion) as compared to the Bas Case scenario.

### 4.2.2. Unconstrained VRE – Medium growth

- In this sensitivity, the total installed capacity reaches to 55.4 GW by 2033 with 14 GW VREs (25% VRE share).
- The share of hydel power generation increases from 10.75 GW to 14.3 GW (25% of total capacity) out which 2.7GW is committed quantum (until FY 2027) and only 956 MW is optimized by PLEXOS
- The total cost savings amount to $2 billion (Total system cost NPV $58 billion) as compared to the Bas Case scenario.

### 4.2.3. Unconstrained VRE – High demand

- In this sensitivity, the total installed capacity reaches to 59.3 GW by 2033 with 15.56 GW VREs (26% VRE share).
- The share of hydel power generation increases from 10.75 GW to 16.7 (28% of total capacity) GW out which 2.7GW is committed quantum (until FY 2027) and only 1173 MW is optimized by PLEXOS.
- The total system cost (NPV) in this sensitivity is $61.7 billion and this is due to new capacity optimization of 13.9 GW.

### Capacity Optimized by PLEXOS for 10-year Horizon

- New Capacity Built: 10.26 GW
- Total System Cost (NPV) = $58 billion

| | Solar | Wind | Nuclear | Hydro | Imported Coal |
|---|---|---|---|---|---|
| Capacity (MW) | 10836 | 1232 | 1200 | 3673.2 | 960 |
| CAPEX ($ billion) | 7.45 | 1.72 | 5 | 2.03 | 0.714 |

Total Investments required = **$ 11.93 billion**

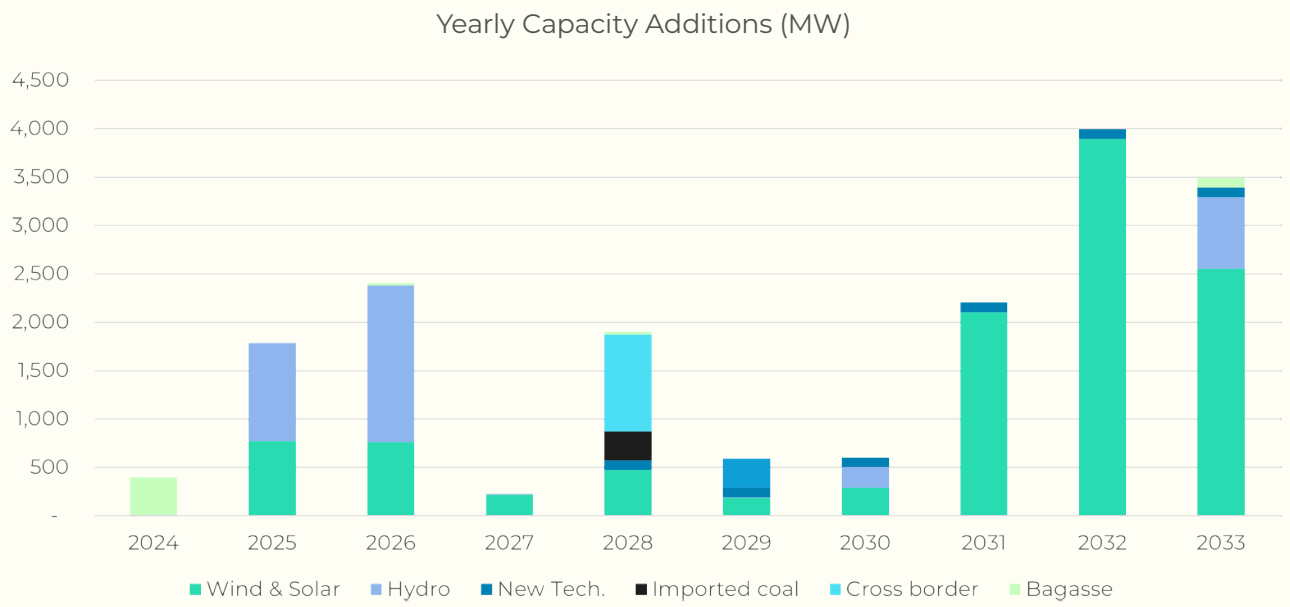
## Committed Projects (MW)

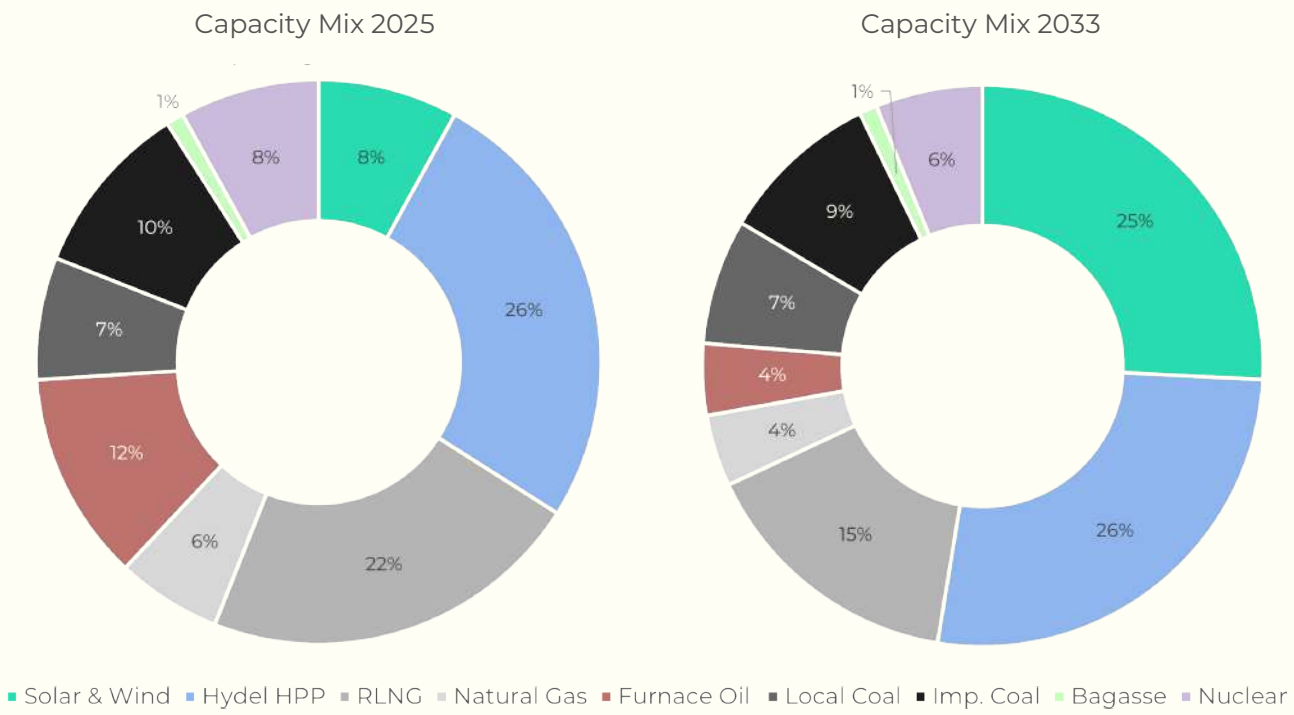
| FY | Hydro | Imp. Coal | Wind | Solar | Cross Border | Nuclear | New Tech | Bagasse | Yearly Addition |
|---|---|---|---|---|---|---|---|---|---|
| 2024 | 72 | 660 | 0 | 171.52 | 0 | 0 | 0 | 32 | 936 |
| 2025 | 1020 | 0 | 0 | 770 | 0 | 0 | 0 | 0 | 1790 |
| 2026 | 1617 | 0 | 132 | 329 | 0 | 0 | 0 | 30 | 2108 |
| 2027 | 8 | 0 | 0 | 265 | 0 | 0 | 0 | 0 | 273 |
| 2028 | 0 | 300 | 100 | 373 | 1000 | 0 | 100 | 30 | 1903 |
| 2029 | 0 | 0 | 0 | 236 | 0 | 0 | 100 | 0 | 336 |
| 2030 | 0 | 0 | 0 | 241 | 0 | 0 | 100 | 0 | 341 |
| 2031 | 0 | 0 | 0 | 255 | 0 | 0 | 100 | 0 | 355 |
| 2032 | 0 | 0 | 0 | 142 | 0 | 0 | 100 | 0 | 242 |
| 2033 | 0 | 0 | 0 | 353 | 0 | 0 | 100 | 0 | 453 |
| Total | 2717 | 960 | 232 | 3136 | 1000 | 0 | 600 | 92 | 8,737 |

Inclusive of KE's committed projects and net-metering

## Optimized Candidate Projects (MW)

| FY | Hydro | Imp. Coal | Wind | Solar | Cross Border | Nuclear | New Tech | KE Hydro | KE Local Coal | KE Solar PV | KE Wind | Capacity Addition Per Year |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2024 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2025 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2026 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 305 | 300 | 0 | 0 |
| 2027 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 605 |
| 2028 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2029 | 5.2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 5.2 |
| 2030 | 211 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 100 | 311 |
| 2031 | 0 | 0 | 0 | 1600 | 0 | 0 | 0 | 0 | 0 | 200 | 100 | 1900 |
| 2032 | 0 | 0 | 0 | 3600 | 0 | 0 | 0 | 0 | 0 | 0 | 200 | 3800 |
| 2033 | 740 | 0 | 0 | 2000 | 0 | 0 | 0 | 0 | 0 | 300 | 600 | 3640 |
| Total | 956.2 | 0 | 0 | 7200 | 0 | 0 | 0 | 0 | 0 | 500 | 1000 | 10261.2 |






## 4.3. No Capacity Additions

In the IGCEP 2024, the net metering projections are assumed to reach at 2107 MW by the year 2034 however, as of April 2024, the installed net metering capacity had already exceeded 2,000MW, and this trend is expected to always accelerate due to rising electricity prices and solar PV prices lowest in Pakistan. In the sensitivity of the NCA, the committed generation capacity until FY 2027 is also considered along with the net metering quantum of 10GW. The NCA scenarios suggest that no further capacity addition is required given the increased rooftop solar additions along with the adequacy of the existing generation fleet. The results show that the total system cost (NPV) is around 56.4 billion USD, and this increased system cost shows that an additional 19 billion USD would be needed on the generation side for the projects that are committed until FY 2027. Another important finding in this scenario is that with 10 GW of rooftop solar along with 4.7 GW of committed VREs, Pakistan can successfully meet and even exceed the ARE Policy 2019 targets of having at least a 30% share of renewables without making significant investments.

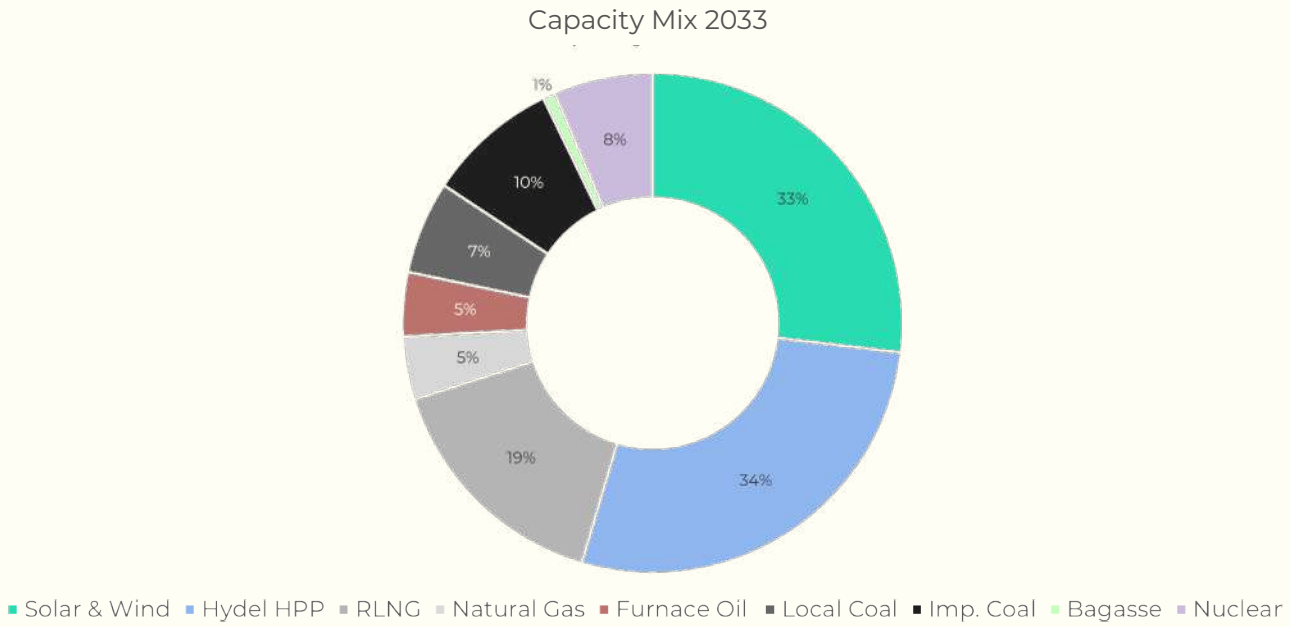
**Key features:**

- The total installed capacity reaches 44.7 GW by 2033 with 4.7 GW VREs (11 % VRE share), it is pertinent to mention here that 10GW of rooftop solar is also considered in the demand side, with this quantum the VRE share becomes 27%.
- The share of hydel power generation increases from 10.75 GW to 15.2 GW (34% of total capacity) and this increase only includes the committed quantum of hydel till FY 2027.
- The total cost savings amount to $3.7 billion (Total system cost NPV $56.4 billion) as compared to the Base Case scenario.

## Committed Projects (MW)

| FY | Hydro | Imp. Coal | Local Coal | Wind | Solar | Cross Border | Nuclear | New Tech | Bagasse | Yearly Addition |
|---|---|---|---|---|---|---|---|---|---|---|
| 2024 | 62 | 660 | 0 | 0 | 171.52 | 0 | 0 | 0 | 32 | 926 |
| 2025 | 1020 | 0 | 0 | 0 | 530 | 0 | 0 | 0 | 0 | 1550 |
| 2026 | 1617 | 0 | 0 | 132 | 88 | 0 | 0 | 0 | 30 | 1867 |
| 2027 | 1880 | 0 | 0 | 0 | 1200 | 0 | 0 | 0 | 0 | 3080 |
| 2028 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2029 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2030 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2031 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2032 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2033 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Total | 4579 | 600 | 0 | 132 | 1990 | 0 | 0 | 0 | 62 | 7423 |

Inclusive of KE's committed projects and net-metering




Share of solar & Wind includes 10 GW rooftop solar as well

## 4.4. True Load Scenario

For the captive's load to be connected to the grid, we have assumed that 40% of the captive load would be connected to the grid by FY 2027, 30% by FY 2029, and 30% by FY 2030. For the growth rate of the captive load, the GDP rate of 3.5 % is considered, and the projected captive load is calculated accordingly for the entire planning horizon. Moreover, the model is also given the option to optimize BESS. As a result, in addition to the committed capacity, the NTDC system only optimizes a micro hydro project to meet the increased load on the system implying better utilization of the existing fleet. Whereas the KE system needs additional capacity to serve increased industrial load which could be met by the energy mix comprising 2770 MW of solar and wind and 1320 MW of local coal along with BESS of 350 MW with up to 08 hours of backup. It is pertinent to mention that the KE system needs the coal power plant to meet the base load requirements, given the global sanctions on local coal along with other socio-environmental factors, the option of local coal remains no longer feasible. To meet the base load demand, the interconnection capacity of the KE system with NTDC must be enhanced so that cheaper electricity can flow from the NTDC system to the KE.

**Key features:**

- The total installed capacity reaches 60 GW by 2033 with 10.4 GW VREs (17% VRE share).
- The share of hydel power generation increases from 10.75 GW to 21.2 GW (35%) out of which 10.4GW is committed quantum and only 87.2MW is optimized by PLEXOS..
- The total cost in this case is $64.2 billion owing to new capacity additions and increased utilization of existing fleet (see annexures for utilization rates).
- The utilization rates of existing fleet of both KE and NTDC (RLNG and imported coal) have been increased as shown below:
  - **RLNG = 78%**
  - **Imported coal = 6.1%**

### Optimum Capacity Addition and Investment Required

- New Capacity Built: 24.6 GW
- Total System Cost (NPV) = $64.2 USD

| | Solar | Wind | Nuclear | Hydro | Imp. Coal | BESS | Local Coal |
|---|---|---|---|---|---|---|---|
| Capacity (MW) | 8258 | 352 | 1200 | 10564.2 | 960 | 350 | 1320 |
| CAPEX ($ billion) | 5.7 | 0.5 | 5 | 14.08 | 0.714 | 0.84 | 2.17 |

Total Investments required = **$ 29 billion**

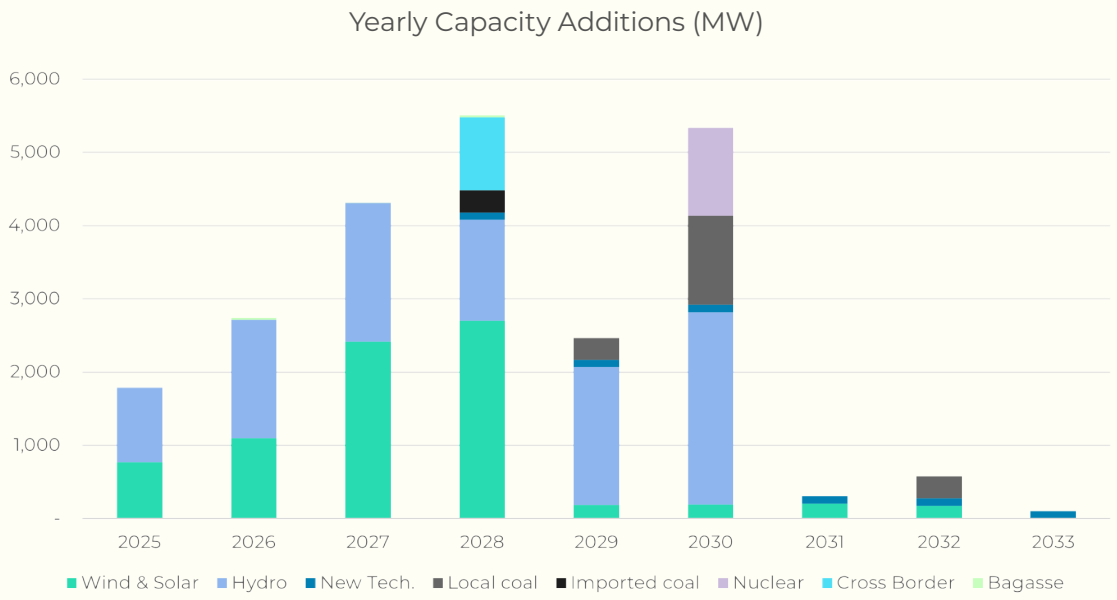
### Committed Projects (MW)

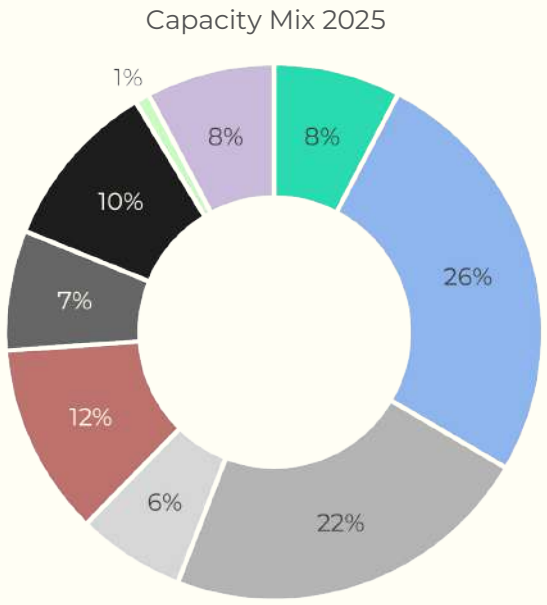
| FY | Hydro | Imp. Coal | Wind | Solar | Cross Border | Nuclear | New Tech | Bagasse | Yearly Addition |
|---|---|---|---|---|---|---|---|---|---|
| 2024 | 72 | 660 | 0 | 171.52 | 0 | 0 | 0 | 32 | 936 |
| 2025 | 357 | 0 | 0 | 770 | 0 | 0 | 0 | 0 | 1127 |
| 2026 | 818 | 0 | 132 | 329 | 0 | 0 | 0 | 30 | 1309 |
| 2027 | 1299 | 0 | 0 | 1465 | 0 | 0 | 0 | 0 | 2764 |
| 2028 | 1591 | 300 | 100 | 1645.42 | 1000 | 0 | 100 | 30 | 4766 |
| 2029 | 4310 | 0 | 0 | 236 | 0 | 0 | 100 | 0 | 4646 |
| 2030 | 1310 | 0 | 0 | 241 | 0 | 1200 | 100 | 0 | 2851 |
| 2031 | 360 | 0 | 0 | 255 | 0 | 0 | 100 | 0 | 715 |
| 2032 | 360 | 0 | 0 | 142 | 0 | 0 | 100 | 0 | 602 |
| 2033 | 0 | 0 | 0 | 353 | 0 | 0 | 100 | 0 | 453 |
| Total | 10477 | 960 | 232 | 5608 | 1000 | 1200 | 600 | 92 | 20,169 |

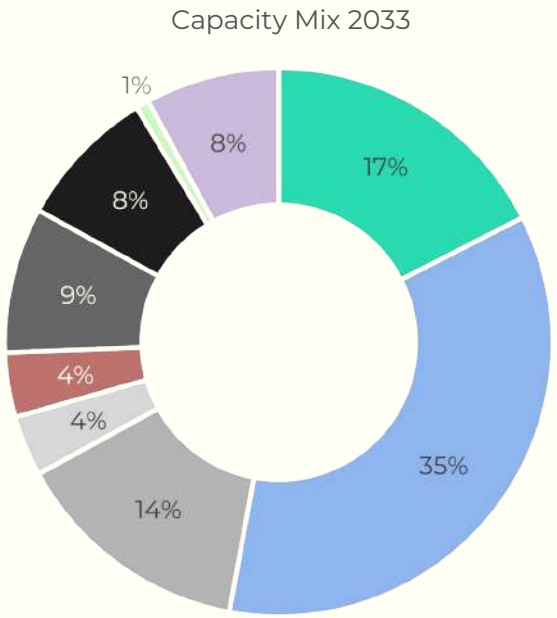
Inclusive of KE's committed projects and Net metering

## Optimized Candidate Projects (MW)

| FY | Hydro | Imp. Coal | Wind | Solar | Cross Border | Nuclear | New Tech | KE Hydro | KE Local Coal | KE Solar PV | KE Wind | Capacity Addition |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2024 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2025 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2026 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 300 | 0 | 300 |
| 2027 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 900 | 0 | 900 |
| 2028 | 5.2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1000 | 0 | 1000 |
| 2029 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 305 | 0 | 0 | 310.2 |
| 2030 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 915 | 0 | 0 | 915 |
| 2031 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2032 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 82 | 0 | 0 | 0 | 82 |
| 2033 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 450 | 120 | 570 |
| Total | 5.2 | 0 | 0 | 0 | 0 | 0 | 0 | 82 | 1220 | 2650 | 120 | 4077.2 |




Solar & Wind ▪ Hydel HPP ▪ RLNG ▪ Natural Gas ▪ Furnace Oil ▪ Local Coal ▪ Imp. Coal ▪ Bagasse ▪ Nuclear

## 4.5. Electric Vehicles (EVs) Impact on Grid

The quantum of the load of EV's charging stations is proportionated into KE's system and NTDC's system i.e., the 2000 MW load of EV's charging stations proportionated as 400 MW into KE's system by 2033 and 1600 MW into NTDC's system by 2033. The results of the pessimistic sensitivity i.e., 2000 MW are shown below:

**Key features:**

- The total installed capacity reaches 57.5 GW by 2033 with 8.4 GW VREs (15 % VRE share).
- The share of hydel power generation increases from 10.75 GW to 21.1 GW (36% of total capacity) out of which 10.4GW is committed quantum (until FY 2027) and only 5.2 MW is optimized by PLEXOS.
- The total cost in this case is $62 billion owing to new capacity additions and increased utilization of existing fleet (see annexures for utilization rates).
- The utilization rates of existing fleet of NTDC (RLNG) have been increased by 10% while utilization rates of KE's fleet (RLNG) reduced by 7% because of the addition of 1000MW wind.

### Capacity Optimized by PLEXOS for 10-year Horizon

- New Capacity Built: **21.56 GW**
- Total System Cost (NPV) = **$62 billion**

| | Solar | Wind | Nuclear | Hydro | Imp. Coal | Local Coal |
|---|---|---|---|---|---|---|
| Capacity (MW) | 5608 | 1232 | 1200 | 10482.2 | 960 | 330 |
| CAPEX ($ billion) | 3.8 | 0.5 | 5 | 13.91 | 0.714 | 0.54 |

Total Investments required = **$ 24.4 billion**

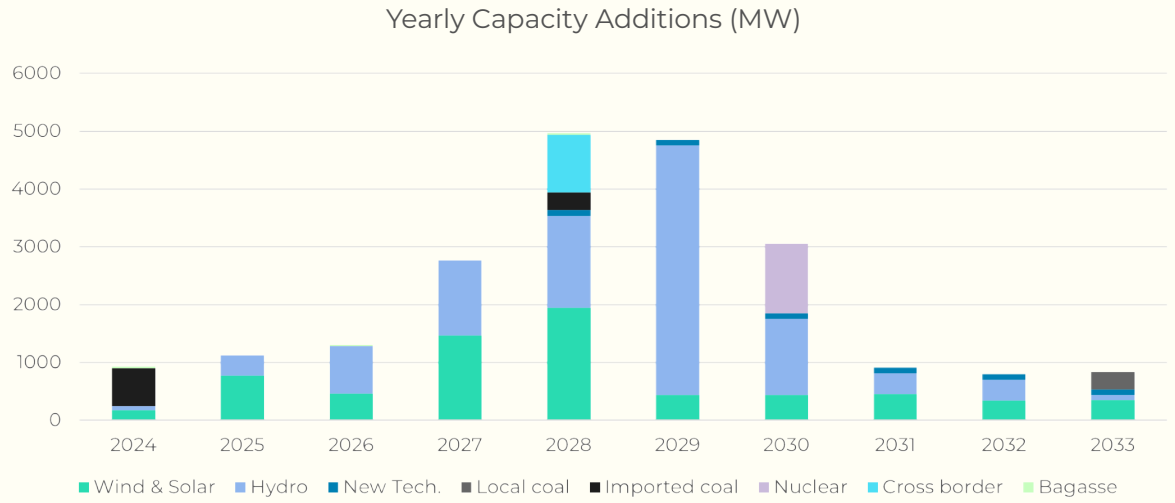
### Committed Projects (MW)

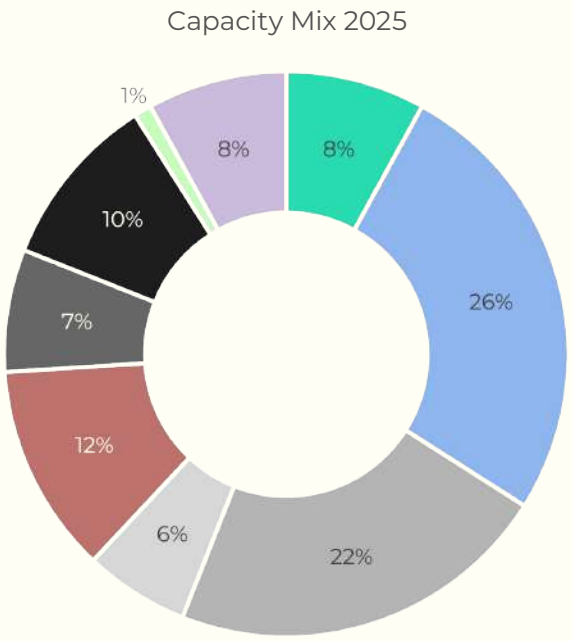
| FY | Hydro | Imp. Coal | Wind | Solar | Cross Border | Nuclear | New Tech | Bagasse | Yearly Addition |
|---|---|---|---|---|---|---|---|---|---|
| 2024 | 72 | 660 | 0 | 171.52 | 0 | 0 | 0 | 32 | 936 |
| 2025 | 357 | 0 | 0 | 770 | 0 | 0 | 0 | 0 | 1127 |
| 2026 | 818 | 0 | 132 | 329 | 0 | 0 | 0 | 30 | 1309 |
| 2027 | 1299 | 0 | 0 | 1465 | 0 | 0 | 0 | 0 | 2764 |
| 2028 | 1591 | 300 | 100 | 1645.42 | 1000 | 0 | 100 | 30 | 4766 |
| 2029 | 4310 | 0 | 0 | 236 | 0 | 0 | 100 | 0 | 4646 |
| 2030 | 1310 | 0 | 0 | 241 | 0 | 1200 | 100 | 0 | 2851 |
| 2031 | 360 | 0 | 0 | 255 | 0 | 0 | 100 | 0 | 715 |
| 2032 | 360 | 0 | 0 | 142 | 0 | 0 | 100 | 0 | 602 |
| 2033 | 0 | 0 | 0 | 353 | 0 | 0 | 100 | 0 | 453 |
| Total | 10477 | 960 | 232 | 5608 | 1000 | 1200 | 600 | 92 | 20,169 |

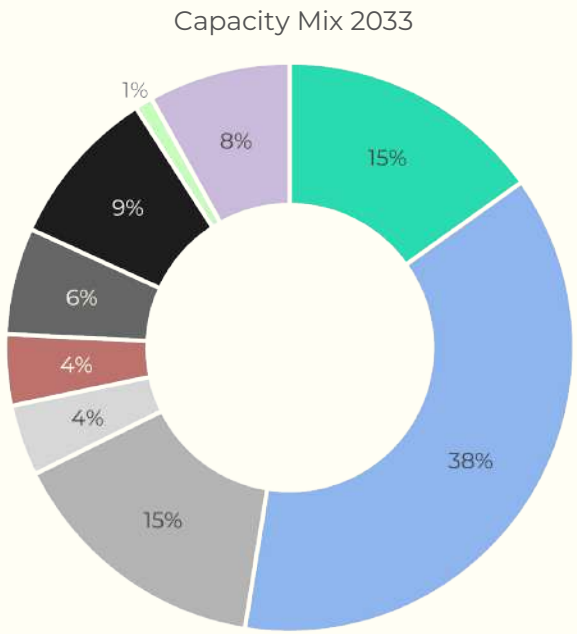
Inclusive of KE's committed projects and Net metering

## Optimized Candidate Projects (MW)

| FY | Hydro | Imp. Coal | Wind | Solar | Cross Border | Nuclear | New Tech | KE Hydro | KE Local Coal | KE Solar PV | KE Wind | Capacity Addition Per Year |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2024 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2025 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2026 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2027 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | | 0 |
| 2028 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 200 | 200 |
| 2029 | 5.2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 200 | 205.2 |
| 2030 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 200 | 200 |
| 2031 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 200 | 200 |
| 2032 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 200 | 200 |
| 2033 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 82 | 305 | 0 | 0 | 387 |
| Total | 5.2 | 0 | 0 | 0 | 0 | 0 | 0 | 82 | 305 | 0 | 1000 | 1,392.2 |




Solar & Wind · Hydel HPP · RLNG · Natural Gas · Furnace Oil · Local Coal · Imp. Coal · Bagasse · Nuclear

## 4.6. Collective impact of CPPs and EVs – Building Synergies

In this scenario we evaluate the collective impact of shifting industries on the main grid and increased demand from erection of charging stations for EVs. For this we taken realistic assumption of shifting 4000 MW load of CPPs and 2000MW EVs load on the main grid. As a result, in addition to the committed capacity, only a 5.2 MW hydro project is optimized in the NTDC system and 4.3 GW of energy mix comprising 990 MW local coal, 1600 MW wind and 1700 MW solar along with 01 unit of BESS (430 MW with up to 9 hours backup) are optimized in the KE system.

It is worth noting that KE system needs additional thermal power of 990 MW to meet the base load requirements, however given the global sanctions of coal financing along with its environmental impacts, coal is no longer a feasible option. This entails enhancement of tie line capacity from NTDC to KE as already mentioned in 4.6 section that will enable the flow of cheaper power from NTDC system to KE's system.

**Key features:**

- The total installed capacity reaches 61.1 GW by 2033 with 10.75 GW VREs (17.5 % VRE share).
- The share of hydel power generation increases from 10.75 GW to 21.2 GW (34% of total capacity) out of which 10.4GW is committed quantum (until FY 2027) and only 87.2 MW is optimized by PLEXOS.
- The total cost in this case is $64.5 billion owing to new capacity additions and increased utilization of existing fleet (see annexures for utilization rates).
- The utilization rates of existing fleet of both KE and NTDC (RLNG and imported coal) have been increased as shown below:
  - **RLNG = 90%**
  - **Imported coal = 7.1%**

### Capacity Optimized by PLEXOS for 10-year Horizon

- New Capacity Built: 24.47 GW
- Total System Cost (NPV) = $64.5 billion

| | Solar | Wind | Nuclear | Hydro | Imp. Coal | BESS | Local Coal |
|---|---|---|---|---|---|---|---|
| Capacity (MW) | 7308 | 1832 | 1200 | 10564.2 | 960 | 430 | 990 |
| CAPEX ($ billion) | 5.03 | 2.57 | 5 | 14.08 | 0.714 | 1.16 | 1.63 |

Total Investments required = **$ 30.17 billion**

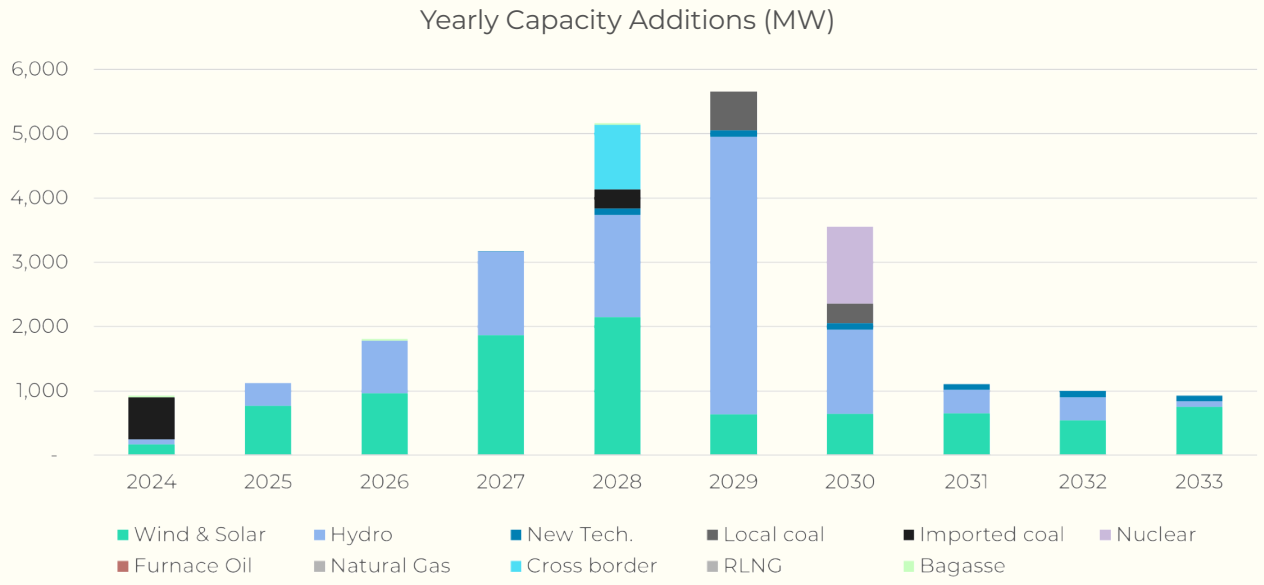
### Committed Projects (MW)

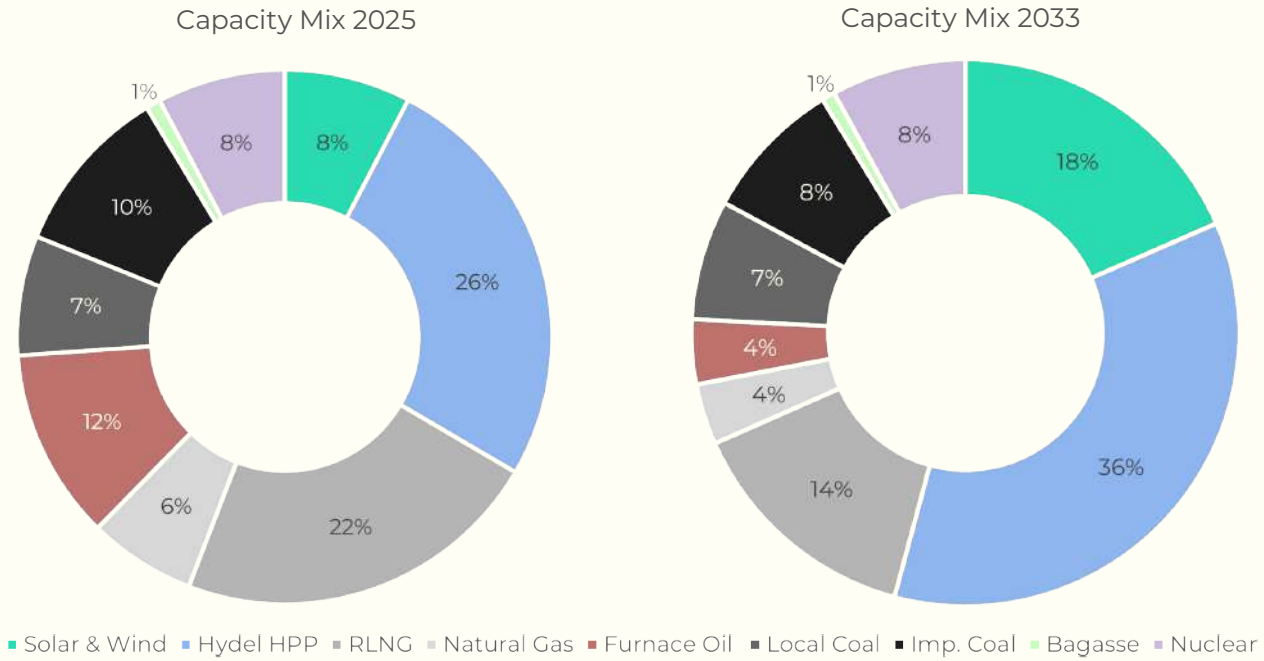
| FY | Hydro | Imp. Coal | Wind | Solar | Cross Border | Nuclear | New Tech | Bagasse | Yearly Addition |
|---|---|---|---|---|---|---|---|---|---|
| 2024 | 72 | 660 | 0 | 171.52 | 0 | 0 | 0 | 32 | 936 |
| 2025 | 357 | 0 | 0 | 770 | 0 | 0 | 0 | 0 | 1127 |
| 2026 | 818 | 0 | 132 | 329 | 0 | 0 | 0 | 30 | 1309 |
| 2027 | 1299 | 0 | 0 | 1465 | 0 | 0 | 0 | 0 | 2764 |
| 2028 | 1591 | 300 | 100 | 1645.42 | 1000 | 0 | 100 | 30 | 4766 |
| 2029 | 4310 | 0 | 0 | 236 | 0 | 0 | 100 | 0 | 4646 |
| 2030 | 1310 | 0 | 0 | 241 | 0 | 1200 | 100 | 0 | 2851 |
| 2031 | 360 | 0 | 0 | 255 | 0 | 0 | 100 | 0 | 715 |
| 2032 | 360 | 0 | 0 | 142 | 0 | 0 | 100 | 0 | 602 |
| 2033 | 0 | 0 | 0 | 353 | 0 | 0 | 100 | 0 | 453 |
| Total | 10477 | 960 | 232 | 5608 | 1000 | 1200 | 600 | 92 | 20,169 |

Inclusive of KE's committed projects and Net metering

## Optimized Candidate Projects (MW)

| FY | Hydro | Imp. Coal | Wind | Solar | Cross Border | Nuclear | New Tech | KE Hydro | KE Local Coal | KE Solar PV | KE Wind | Capacity Addition Per Year |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2024 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2025 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2026 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 300 | 200 | 500 |
| 2027 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 200 | 200 | 400 |
| 2028 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 200 | 200 | 400 |
| 2029 | 5.2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 607 | 200 | 200 | 1012.2 |
| 2030 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 305 | 200 | 200 | 705 |
| 2031 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 200 | 200 | 400 |
| 2032 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 200 | 200 | 400 |
| 2033 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 82 | 0 | 200 | 200 | 482 |
| Total | 5.2 | 0 | 0 | 0 | 0 | 0 | 0 | 82 | 912 | 1700 | 1600 | 4299.2 |



## 4.7. Enhanced Electricity Flow from NTDC to KE

In this scenario, we assess the impact on generation planning for the KE system by increasing the flow of electricity from NTDC's system to the KE system from 2050MW to 4000MW. The potential benefit of increased reliance on NTDC's system could be in the form of lesser utilization of expensive generation fleets in the KE system. As a result, the PLEXOS model does not optimize any generation option in the KE system for the next 10 years, and only a small hydel project of 5.2 capacity in the NTDC's system. Other than this, the majority of capacity additions are only those which have been treated as committed in the model. This also brings attention to the surplus capacity available in the main grid — sufficient for both the NTDC's and KE's systems — in the provision of affordable electricity to the end consumers. Another major observation in this scenario is that the need for a base load option i.e., local coal additions would no longer be needed even if the captive load along with the EV's load increases in the KE system because of the allowance of cheaper power to KE system via enhanced tie line capacity.

It is important to note that for the enhancement of tie-lines, the timelines have also been considered i.e., first augmentation in FY 27, the other one in FY 30 to make the scenario more realistic.

**Key features:**

- The total installed capacity reaches 56.7 GW by 2033 with 7.6 GW VREs (13 % VRE share).
- The share of hydel power generation increases from 10.75 GW to 21.1 GW (37 % of total capacity) out of which 10.4 GW is committed quantum (until FY 2027) and only 5.2 MW is optimized by PLEXOS.
- The total cost in this case is $60 billion as compared to the base case scenario.

### Capacity Optimized by PLEXOS for 10-year Horizon

- New Capacity Built: **20.175 GW**
- Total System Cost = **$60 billion**

| | Solar | Wind | Nuclear | Hydro | Imp. Coal | BESS |
|---|---|---|---|---|---|---|
| Capacity (MW) | 5608 | 232 | 1200 | 10482.2 | 960 | 0 |
| CAPEX ($ billion) | 3.86 | 0.325 | 5 | 13.91 | 0.714 | 0 |

Total Investments required = **$ 23.8 billion**

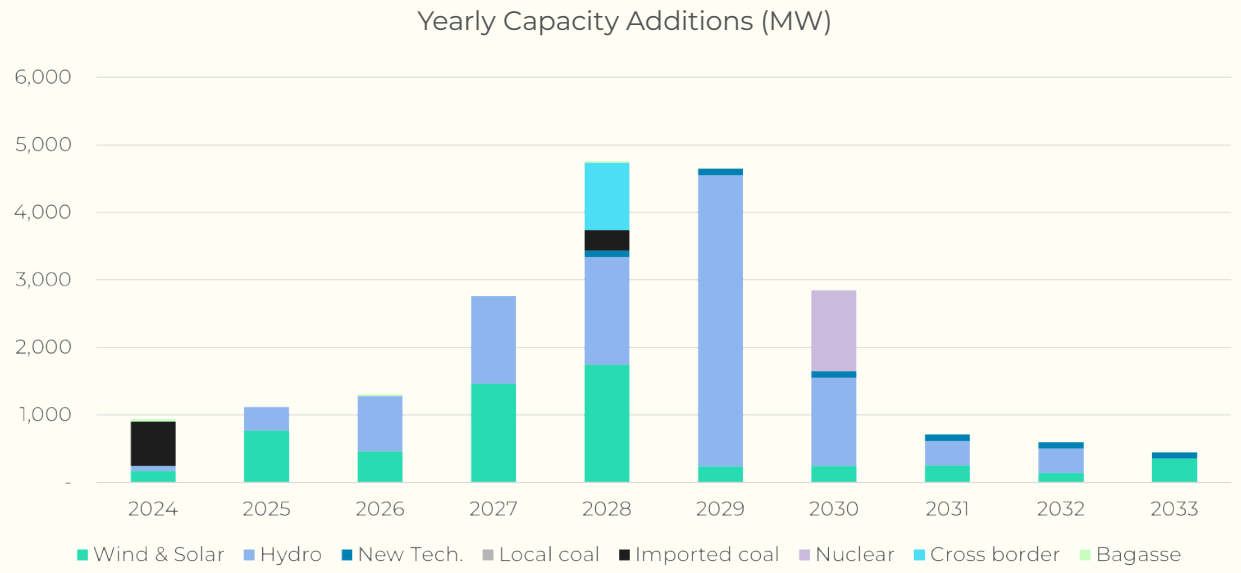
### Committed Projects (MW)

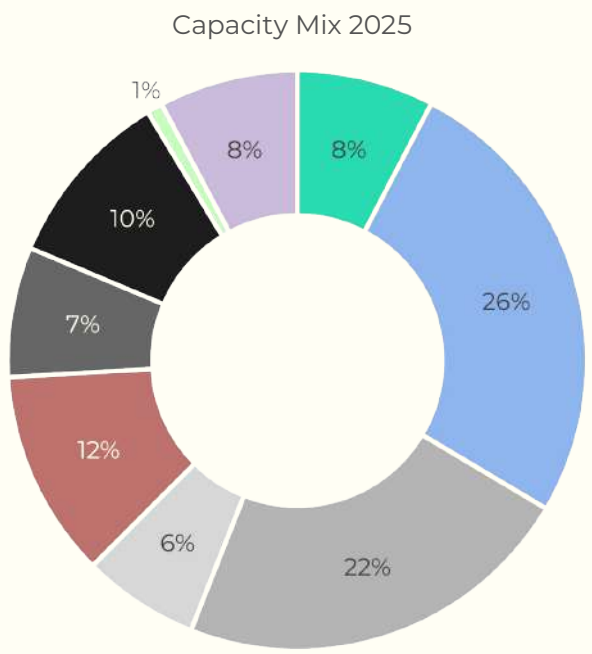
| FY | Hydro | Imp. Coal | Wind | Solar | Cross Border | Nuclear | New Tech | Bagasse | Yearly Addition |
|---|---|---|---|---|---|---|---|---|---|
| 2024 | 72 | 660 | 0 | 171.52 | 0 | 0 | 0 | 32 | 936 |
| 2025 | 357 | 0 | 0 | 770 | 0 | 0 | 0 | 0 | 1127 |
| 2026 | 818 | 0 | 132 | 329 | 0 | 0 | 0 | 30 | 1309 |
| 2027 | 1299 | 0 | 0 | 1465 | 0 | 0 | 0 | 0 | 2764 |
| 2028 | 1591 | 300 | 100 | 1645.42 | 1000 | 0 | 100 | 30 | 4766 |
| 2029 | 4310 | 0 | 0 | 236 | 0 | 0 | 100 | 0 | 4646 |
| 2030 | 1310 | 0 | 0 | 241 | 0 | 1200 | 100 | 0 | 2851 |
| 2031 | 360 | 0 | 0 | 255 | 0 | 0 | 100 | 0 | 715 |
| 2032 | 360 | 0 | 0 | 142 | 0 | 0 | 100 | 0 | 602 |
| 2033 | 0 | 0 | 0 | 353 | 0 | 0 | 100 | 0 | 453 |
| Total | 10477 | 960 | 232 | 5608 | 1000 | 1200 | 600 | 92 | 20,169 |

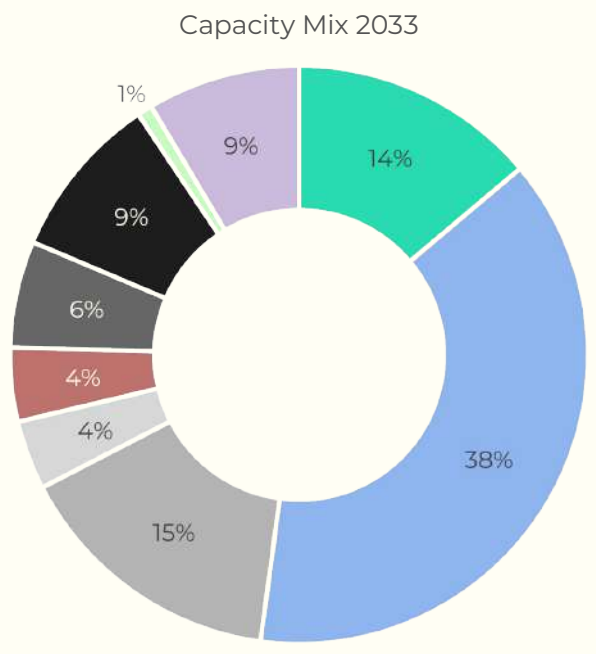
Inclusive of KE's committed projects and Net metering

## Optimized Candidate Projects (MW)

| FY | Hydro | Imp. Coal | Wind | Solar | Cross Border | Nuclear | New Tech | KE Hydro | KE Local Coal | KE Solar PV | KE Wind | Capacity Addition Per Year |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2024 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2025 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2026 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2027 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2028 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2029 | 5.2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 5.2 |
| 2030 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2031 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2032 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2033 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Total | 5.2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 5.2 |




Solar & Wind · Hydel HPP · RLNG · Natural Gas · Furnace Oil · Local Coal · Imp. Coal · Bagasse · Nuclear

## 4.8. Collective impact of CPPs, EVs and Enhanced Electricity Flow to KE – The Efficiency Trio

In this scenario, we evaluate the collective impact of shifting industries on the main grid, increased demand from the erection of charging stations for EVs and enhanced electricity flow to KE from the main NTDC system. For this, we assessed the realistic case only in which we have used the following three main assumptions: Realistic only – Captives load = 4000MW, EVs load = 2000MW, flow to KE = 4000MW

As a result, in addition to the committed capacity, the model optimizes 176 MW of hydropower in the NTDC system and 1400 MW of wind power in the KE system. The renewable energy optimization by PLEXOS strengthens the need for an affordable yet sustainable energy system while ensuring the maximum utilization of the existing system. The results of this scenario also back the findings of scenario 4.7 (Collective impact of CPPs and EVs) in which additional thermal power generation is optimized into KE's system as there were no other options for the base load. In this scenario, since the tie-line capacity from NTDC to KE has been enhanced, there is no need for additional thermal power generation in KE's system.

**Key features:**

- The total installed capacity will reach 58.1 GW by 2033 with 9.06 GW VREs (15 % VRE share).
- The share of hydel power generation increases from 10.75 GW to 21.1 GW (36% of total capacity) out of which 10.4GW is committed quantum (until FY 2027) and only 5.2 MW is optimized by PLEXOS.
- The total cost in this case is $63.4 billion owing to new capacity additions and increased utilization of existing fleet (see annexures for utilization rates).
- The utilization rates of existing fleet of both KE and NTDC (RLNG and imported coal) have been increased as shown below:
    - **RLNG = Utilization increased by 2.3 times** (118.3 TWh generation as compared to 51.2 TWh in the base case)
    - **Imported coal = Utilization increased by 10%**

### Capacity Optimized by PLEXOS for 10-year Horizon

- New Capacity Built: 21.57 GW
- Total system cost (NPV) = $63.4 billion

| | Solar | Wind | Nuclear | Hydro | Imp. Coal | BESS | Local Coal |
|---|---|---|---|---|---|---|---|
| Capacity (MW) | 5608 | 1632 | 1200 | 10482.2 | 960 | 0 | 0 |
| CAPEX ($ billion) | 3.85 | 2.28 | 5 | 13.91 | 0.714 | 0 | 0 |

Total Investments required = **$ 25.67 billion**

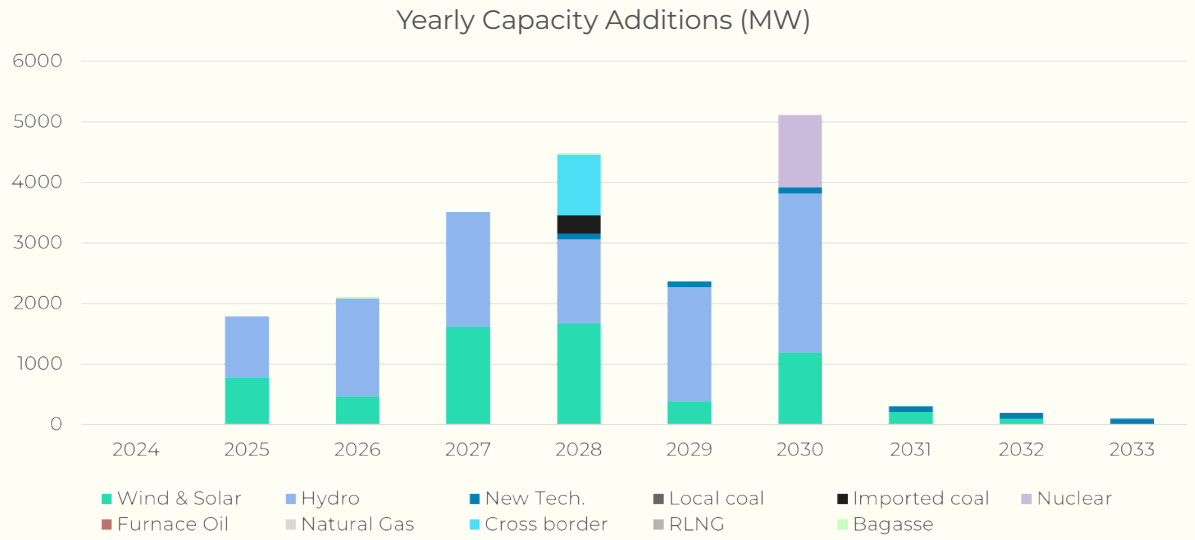
### Committed Projects (MW)

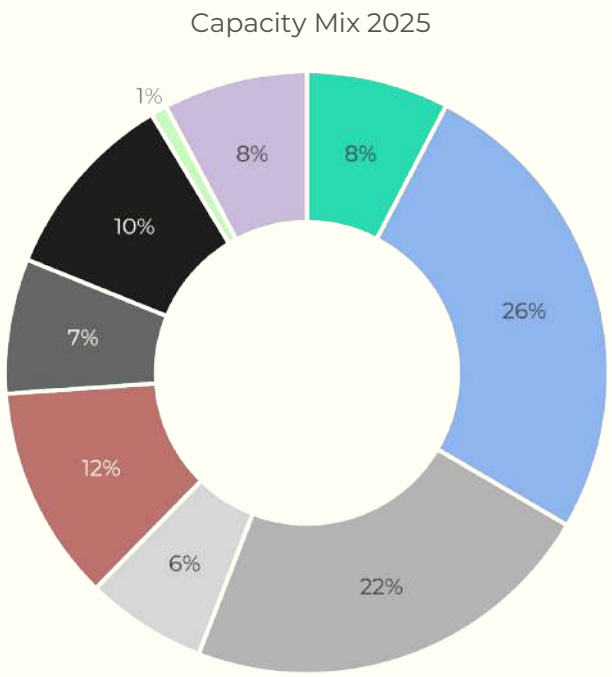
| FY | Hydro | Imp. Coal | Wind | Solar | Cross Border | Nuclear | New Tech | Bagasse | Yearly Addition |
|---|---|---|---|---|---|---|---|---|---|
| 2024 | 72 | 660 | 0 | 171.52 | 0 | 0 | 0 | 32 | 936 |
| 2025 | 357 | 0 | 0 | 770 | 0 | 0 | 0 | 0 | 1127 |
| 2026 | 818 | 0 | 132 | 329 | 0 | 0 | 0 | 30 | 1309 |
| 2027 | 1299 | 0 | 0 | 1465 | 0 | 0 | 0 | 0 | 2764 |
| 2028 | 1591 | 300 | 100 | 1645.42 | 1000 | 0 | 100 | 30 | 4766 |
| 2029 | 4310 | 0 | 0 | 236 | 0 | 0 | 100 | 0 | 4646 |
| 2030 | 1310 | 0 | 0 | 241 | 0 | 1200 | 100 | 0 | 2851 |
| 2031 | 360 | 0 | 0 | 255 | 0 | 0 | 100 | 0 | 715 |
| 2032 | 360 | 0 | 0 | 142 | 0 | 0 | 100 | 0 | 602 |
| 2033 | 0 | 0 | 0 | 353 | 0 | 0 | 100 | 0 | 453 |
| Total | 10477 | 960 | 232 | 5608 | 1000 | 1200 | 600 | 92 | 20,169 |

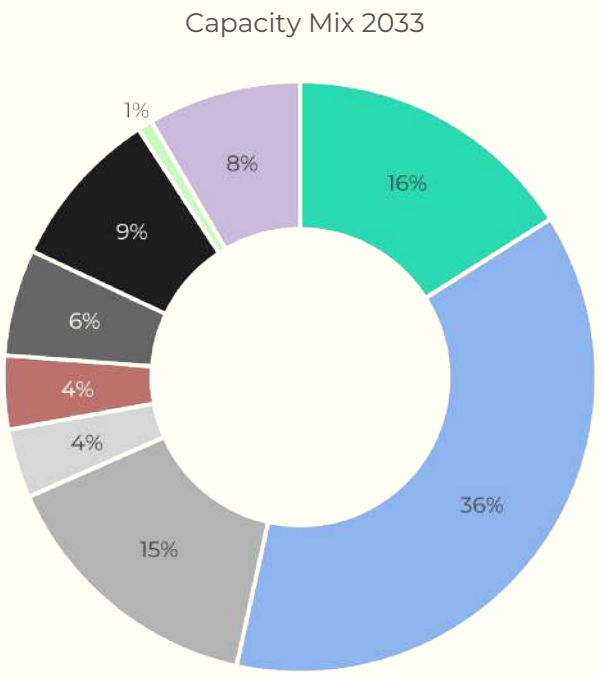
Inclusive of KE's committed projects and Net metering

## Optimized Candidate Projects (MW)

| FY | Hydro | Imp. Coal | Wind | Solar | Cross Border | Nuclear | New Tech | KE Hydro | KE Local Coal | KE Solar PV | KE Wind | Capacity Addition Per Year |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2024 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2025 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2026 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2027 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 200 | 200 |
| 2028 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2029 | 5.2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 200 | 205.2 |
| 2030 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2031 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1000 | 1000 |
| 2032 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2033 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Total | 5.2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1400 | 1405.2 |




Solar & Wind ▪ Hydel HPP ▪ RLNG ▪ Natural Gas ▪ Furnace Oil ▪ Local Coal ▪ Imp. Coal ▪ Bagasse ▪ Nuclear

## 4.9. Additional Cost of Diamer Bhasha Dam and CHASHNUPP-5 (C-5)

In this scenario, we assess the impact of incorporating the full costs of the Diamer Bhasha Dam (4,500 MW) and C-5 (1,200 MW) projects, beyond the levelized tariffs currently considered in the IGCEP 2024, where these projects are listed as committed. The current approach does not fully reflect the actual costs that will be incurred if these power projects come online, such as capital expenditures and increased capacity payments during the debt servicing periods.

**Key features (with C-5 as a candidate option):**

- The total installed capacity reaches 56.2 GW by 2033 with 8.2 GW VREs (15 % VRE share) and C-5 is not picked by PLEXOS.
- The share of hydel power generation increases from 10.75 GW to 21.2 GW (37% of total capacity) out of which 10.4GW is committed quantum (until FY 2027) and only 87.2 MW is optimized by PLEXOS.
- The total cost savings in this case amount to $5.52 billion (Project cost and running costs) as compared to the base case.

**Key features (with Diamer Bhasha Dam as a candidate option):**

- The total installed capacity reaches 52.9 GW by 2033 with 8.2 GW VREs (15 % VRE share) and Diamer Bhasha is not picked by PLEXOS.
- The share of hydel power generation increases from 10.75 GW to 16.7 GW (31% of total capacity) out of which 5.9 GW is committed quantum and only 87.2 MW is optimized by PLEXOS.
- The total cost savings in this case amount to $7.6 billion (Project cost and running costs) as compared to the base case.

In the first iteration, C-5 is evaluated as a candidate project, and results indicate that PLEXOS does not optimize C-5 due to its higher costs. This leads to a reduction in the total system cost (NPV) by 0.52 billion USD compared to the base case scenario where C-5 is committed. It is pertinent to mention that if this reduction does not include the project cost, and if the project cost is also included, the total reduction in total system cost amounts to $5.52 billion. Additionally, the share of renewable energy sources increases to 15%.

In the second iteration, Diamer Bhasha is treated as a candidate project, and similarly, PLEXOS does not optimize it due to higher costs. The analysis shows that the total system cost (NPV) is reduced by 2.74 billion USD compared to the base case scenario where Diamer Bhasha is committed. It is pertinent to mention that if this reduction does not include the project cost, and if the project cost is also included, the total reduction in total system cost amounts up to $7.6 billion with an increase in the renewable energy share to 16%.

These findings raise critical questions for the agencies concerned about whether these projects should be classified as committed, given that they do not meet the least-cost criteria and may displace more affordable energy sources. True costs of such projects must be transparently included in generation capacity expansion plans to ensure alignment with the least-cost criteria and policies promoting a higher share of renewable energy, especially since Pakistan is a signatory to the Paris Agreement.

### 4.9.1. Without C-5

- New Capacity Built: 19.6 GW
- Total system cost (NPV) = $59.5 billion

| | Solar | Wind | Nuclear | Hydro | Imp. Coal | BESS | Local Coal |
|---|---|---|---|---|---|---|---|
| Capacity (MW) | 5608 | 832 | 0 | 10564.2 | 960 | 0 | 0 |
| CAPEX ($ billion) | 3.85 | 1.165 | 0 | 14.08 | 0.714 | 0 | 0 |

Total Investments required = **$ 19.8 billion**

## Committed Projects (MW)

| FY | Hydro | Imp. Coal | Wind | Solar | Cross Border | Nuclear | New Tech | Bagasse | Yearly Addition |
|---|---|---|---|---|---|---|---|---|---|
| 2024 | 72 | 660 | 0 | 171.52 | 0 | 0 | 0 | 32 | 936 |
| 2025 | 357 | 0 | 0 | 770 | 0 | 0 | 0 | 0 | 1127 |
| 2026 | 818 | 0 | 132 | 329 | 0 | 0 | 0 | 30 | 1309 |
| 2027 | 1299 | 0 | 0 | 1465 | 0 | 0 | 0 | 0 | 2764 |
| 2028 | 1591 | 300 | 100 | 1645.42 | 1000 | 0 | 100 | 30 | 4766 |
| 2029 | 4310 | 0 | 0 | 236 | 0 | 0 | 100 | 0 | 4646 |
| 2030 | 1310 | 0 | 0 | 241 | 0 | 0 | 100 | 0 | 1651 |
| 2031 | 360 | 0 | 0 | 255 | 0 | 0 | 100 | 0 | 715 |
| 2032 | 360 | 0 | 0 | 142 | 0 | 0 | 100 | 0 | 602 |
| 2033 | 0 | 0 | 0 | 353 | 0 | 0 | 100 | 0 | 453 |
| Total | 10477 | 960 | 232 | 5608 | 1000 | 0 | 600 | 92 | 18969 |

Inclusive of KE's committed projects and Net metering

## Optimized Candidate Projects (MW)

| FY | Hydro | Imp. Coal | Wind | Solar | Cross Border | Nuclear | New Tech | KE Hydro | KE Local Coal | KE Solar PV | KE Wind | Capacity Addition Per Year |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2024 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2025 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2026 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2027 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2028 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 100 | 100 |
| 2029 | 87.2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 100 | 187.2 |
| 2030 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 100 | 100 |
| 2031 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 100 | 100 |
| 2032 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 100 | 100 |
| 2033 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 100 | 100 |
| Total | 5.2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 600 | 687.2 |

### 4.9.2. Without Diamer Bhasha Dam

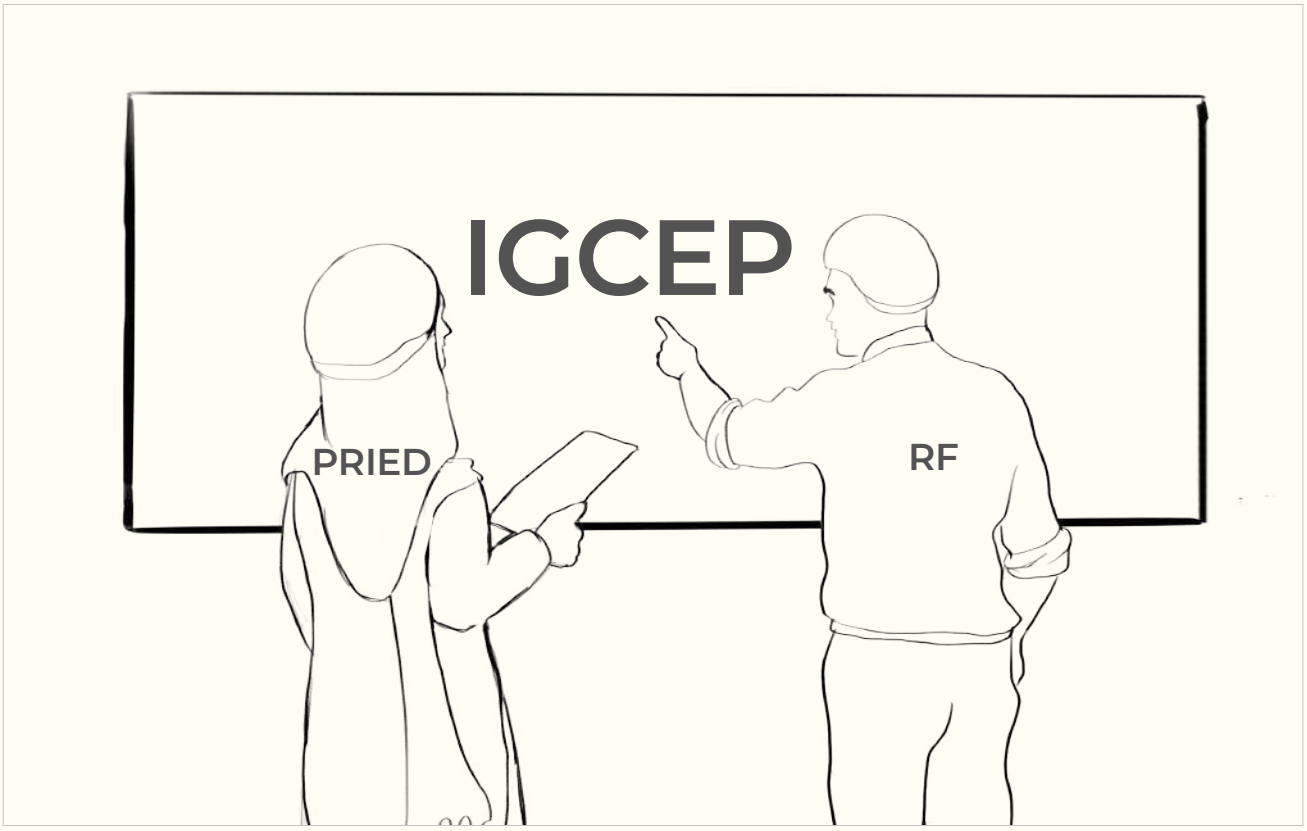
- New Capacity Built: **19.6 GW**
- Total system cost (NPV) = **$57.36 billion**

| | Solar | Wind | Nuclear | Hydro | Imp. Coal | BESS | Local Coal |
|---|---|---|---|---|---|---|---|
| Capacity (MW) | 5608 | 832 | 0 | 5977 | 960 | 0 | 0 |
| CAPEX ($ billion) | 3.85 | 1.165 | 0 | 9.215 | 0.714 | 0 | 0 |

Total Investments required = **$ 14.9 billion**

## Optimized Candidate Projects (MW)

| FY | Hydro | Imp. Coal | Wind | Solar | Cross Border | Nuclear | New Tech | KE Hydro | KE Local Coal | KE Solar PV | KE Wind | Capacity Addition Per Year |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 2024 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2025 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2026 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2027 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 2028 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 100 | 100 |
| 2029 | 87.2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 100 | 187.2 |
| 2030 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 100 | 100 |
| 2031 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 100 | 100 |
| 2032 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 100 | 100 |
| 2033 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 100 | 100 |
| Total | 5.2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 600 | 687.2 |




# Chapter 5:

# Generation Sequence & Recommendations

| | Scenario 4.1 | Scenario 4.2 | Scenario 4.3 | Scenario 4.4 | Scenario 4.5 | Scenario 4.7 | Scenario 4.6 | Scenario 4.8 | Scenario 4.9 |
|---|---|---|---|---|---|---|---|---|---|
| Solar | 0 | 7700 | 0 | 2650 | 0 | 1700 | | 0 | 0 |
| Wind | 0 | 1000 | 0 | 120 | 1000 | 1600 | | 1400 | 1000 |
| Local Coal | 0 | 330 | 0 | 1320 | 330 | 990 | | 0 | 0 |
| Hydro | 87.2 | 956.2 | 0 | 87.2 | 87.2 | 87.2 | 5.2 | 87.2 | 87.2 |
| BESS | 0 | 0 | 0 | 350 | 0 | 430 | | 0 | 0 |

The generation sequence analysis indicates that hydropower projects, particularly Batdara (5.2 MW) and Uzghor (82 MW), are optimized across all scenarios except Scenario 4.6, which involves the enhanced flow of power to KE. According to IGCEP 2024, the concerned agencies for Batdara and Uzghor are NTDC and KE respectively, with both projects optimized within their respective regions. Notably, since Uzghor is integrated into KE's system, applicable wheeling charges for power evacuation must be factored into the overall cost analysis. A thorough assessment is necessary to determine if the inclusion of these wheeling charges affects the economic feasibility of these projects for the responsible agencies. Local coal power plants are optimized within KE's system in scenarios where significant electric vehicle (EV) and captive loads are included on the demand side. To serve these captive loads, a reliable base load option is required, with current options limited to local coal or the conversion of existing imported coal plants to local coal within KE's system. However, due to global sanctions on coal financing and socio-environmental concerns, coal is no longer a sustainable option for KE.

Moreover, for captive users, particularly export-oriented industries, compliance with the Carbon Border Adjustment Mechanism (CBAM) and heightened accountability on Scope 2 and Scope 3 emissions make coal an impractical choice, especially given KE's already fossil fuel-dominated energy mix. Considering these constraints, it is imperative to explore cleaner and more reliable energy alternatives for KE, such as enhancing the tie-line capacity between NTDC and KE or integrating renewable energy sources like solar and wind with Battery Energy Storage Systems (BESS). It is important to highlight that while an agreement has been established between KE and NTDC to supply power to the KE system based on the tie-line capacity, there is a critical need for a separate agreement ensuring a firm dedicated to power supply to KE. This would always guarantee a consistent and reliable power flow TO KE, rather than relying on pro-rata-based power flow arrangements, which may not adequately meet the demand or operational stability required for KE's system. Establishing such an agreement would enhance predictability and ensure the continuous availability of power to KE, aligning with the long-term energy security goals for the region.

The declining cost of BESS, expected to decrease further, makes it a compelling option for peak shaving, mitigating the intermittency of renewables, and providing ancillary services. Notably, PLEXOS also optimizes BESS in KE's system when captive and EV loads are incorporated, underscoring its viability. The scenarios also show a significant increase in the share of renewable energy sources, particularly when large projects like CHASNUPP-5 (C-5) and Diamer Bhasha Dam are treated as candidates rather than committed options. These projects were not selected by PLEXOS due to their high costs and reduced demand, indicating that such expensive commitments would impose an undue burden on consumers.

New capacity additions should strictly adhere to the least-cost criteria, especially as the government encourages captives to connect with the national grid and KE's system. In a business-as-usual scenario, it is advisable to pursue a no-capacity-addition approach, considering Pakistan's substantial imports of solar PV modules—13 GW in the first half of 2024 alone. Current large-scale solar projects in Pakistan have an average power purchase price of PKR 37/kWh, while rooftop solar can deliver electricity to nearby loads at PKR 27/kWh due to the absence of additional system charges. Rooftop solar provides a stable price per kWh without the need for sovereign guarantees or tariff indexations, which simplifies financial obligations for the government. Given the underutilization of power plants and burdensome capacity payments, future investments should prioritize modernizing the distribution grid to better accommodate renewable energy, rather than expanding generation capacity. Grid modernization offers the dual benefits of improved theft prevention, reduced losses, and increased renewable energy absorption capabilities. This approach would enable Pakistan to meet and even exceed the Alternative and Renewable Energy (ARE) Policy 2019 target of achieving a 30% renewable energy share without substantial new investments. To address peak demand, particularly at night, encouraging the deployment of battery energy storage systems among consumers will be crucial. Furthermore, retiring power plants can be retained on a take-and-pay basis, providing ancillary services and improving system

stability by converting generators into synchronous condensers.

The primary objective of this study extends beyond presenting multiple scenarios; it emphasizes the critical importance of considering a range of scenarios and sensitivities in power generation capacity expansion planning. The final scenario selection must integrate all macroeconomic factors to ensure robust decision-making. Moving forward, we aim to integrate academic institutions into our generation planning processes, employing advanced decision-making methodologies from operations research, such as Multi-Criteria Decision Making (MCDM) and Fuzzy Failure Mode and Effect Analysis (FFMEA). This interdisciplinary approach will harness expertise from academia, research institutes, think tanks, and policy planning departments, ensuring that our strategies are comprehensive, cutting-edge, and aligned with the latest developments in the power sector. This collaborative effort will strengthen our strategic decision-making capabilities, enhancing the resilience and sustainability of Pakistan's energy future.

In summary, we strongly recommend the following to the relevant planning agencies, policymakers and the ministry:

- **Evaluate Committed Projects on Merit:**

  Treat committed projects as candidate options and evaluate their feasibility to save $5.79 billion in system costs and increase VRE share to 13% by 2033 (up to 26% in high-growth scenarios).

- **Avoid Capacity Additions Until 2027:**

  Given the rise in net metering, rapid solarization and declining solar PV prices, avoid capacity additions beyond committed projects until 2027, potentially saving $3.7 billion.

- **Incentivize & Facilitate Transfer of Captive Power Plant (CPP) Load to National Grid:**

  Shift CPP load to the national grid to improve utilization rates and optimize generation capacity without significant new additions. However, such initiative would require augmentation and rehabilitation of the transmission and distribution system. It is pertinent to mention that this will not only ensure the reliable and un-interrupted supply of power to CPPs but would also help avoid out of merit operations i.e., running costly fossil-fuel based power plants amid network constraints along with maximum RE absorption.

- **Expand NTDC-KE Transmission Capacity:**

  Increase tie-line transmission capacity between NTDC and KE based on a firm agreement to reduce the need for new generation in KE and decrease reliance on costly thermal power plants.

- **Avoid High-Cost Projects:**

  Do not forcibly add high-cost projects like Diamer Bhasha Dam and CHASNUPP-5, as they are not optimized and will lead to unnecessary costs ($7.6 billion for Diamer and $5.52 billion for Chashma).

- **Promote Rooftop Solar:**

  Encourage rooftop solar installations to meet demand without adding large-scale generation, benefiting from falling PV prices.

- **Optimize Grid for Renewables:**

  Focus on grid modernization to reduce losses, prevent theft, and integrate renewable energy, avoiding unnecessary generation capacity expansion.

- **Leverage BESS for Flexibility:**

  Utilize the declining cost of Battery Energy Storage Systems (BESS) to manage peak demand, intermittent renewables, and provide ancillary services.

- **Encourage EV and CPP Integration:**

  Strategically shift captive power plant loads and integrate EV charging demand to better utilize the existing generation fleet.

# APPENDIX (Utilization Rates)

## Scenario 4.1 – Utilization Rates (%)

| Fiscal Year | 2024 | 2025 | 2026 | 2027 | 2028 | 2029 | 2030 | 2031 | 2032 | 2033 |
|---|---|---|---|---|---|---|---|---|---|---|
| CASA | 0 | 0 | 0 | 0 | 42 | 42 | 42 | 42 | 42 | 42 |
| FPCL | 54.11 | 3.94 | 0 | 0 | 0 | 0 | 0.01 | 0.02 | 0.02 | 0.02 |
| Gul Ahmed | 51.52 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| SNPC 1 | 92 | 92 | 92 | 92 | 92 | 92 | 64.77 | 64.55 | 66.91 | 69.31 |
| SNPC 2 | 92 | 92 | 92 | 92 | 92 | 92 | 67.35 | 67.19 | 69.28 | 71.18 |
| Tapal | 46.21 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| BQPS 1 Unit 01 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| BQPS 1 Unit 02 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| BQPS 1 Unit 05 | 0.01 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| BQPS 1 Unit 06 | 0 | 0 | 0 | 0 | 0 | 0 | 0.02 | 0.02 | 0.02 | 0.02 |
| BQPS 2 | 20.71 | 0.88 | 1.22 | 1.75 | 2.38 | 3.22 | 4.18 | 4.19 | 6.33 | 9.64 |
| BQPS 3 Unit 01 | 67.68 | 13.2 | 14.09 | 17.9 | 39.14 | 24.09 | 28.52 | 49.63 | 51.43 | 39.93 |
| BQPS 3 Unit 02 | 86 | 39.9 | 39.32 | 42.59 | 27.48 | 48 | 48.66 | 26.59 | 31.13 | 49.54 |
| KGTPS | 0.01 | 0 | 0 | 0 | 0 | 0 | 0.02 | 0.02 | 0.02 | 0.14 |
| KPC | 0.68 | 0 | 0 | 0.01 | 0.04 | 0.1 | 0.13 | 0.1 | 0.26 | 0.42 |
| SGTPS | 0.01 | 0 | 0 | 0 | 0 | 0 | 0.02 | 0.02 | 0.02 | 0.02 |
| Engro | 77.3 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Foundation | 79.37 | 84.71 | 84.94 | 84.92 | 84.96 | 84.94 | 0 | 0.66 | 0.97 | 0.45 |
| Guddu II | 78.38 | 83.7 | 84.78 | 84.96 | 84.96 | 85 | 84.96 | 84.94 | 31.59 | 0.82 |
| Guddu V | 84.57 | 84.93 | 85 | 85 | 85 | 85 | 84.99 | 84.99 | 84.99 | 6.9 |
| Liberty | 2.96 | 2.96 | 2.96 | 2.96 | 85 | 0 | 0 | 0 | 0 | 0 |
| Uch | 2.95 | 2.96 | 2.96 | 2.96 | 2.95 | 2.96 | 2.96 | 85 | 0 | 0 |
| Uch II | 50 | 78.78 | 80.1 | 81.39 | 50 | 50.64 | 50 | 50 | 50 | 50 |
| AES Lalpir | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| AES Pakgen | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| AGL | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Atlas | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Engro 127MW | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Hub N | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| HUBCO | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Jamshoro I U1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Jamshoro II U4 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Kohinoor | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Liberty Tech | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Muzaffargarh I U1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Muzaffargarh I U2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Muzaffargarh I U3 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Muzaffargarh II U4 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Nishat C | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Nishat P | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Saba | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| C1 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 |
| C2 | 80 | 80 | 80 | 80 | 80 | 80 | 80 | 80 | 80 | 80 |
| C3 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 |
| C4 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| C5 | 0 | 0 | 0 | 0 | 0 | 0 | 90 | 90 | 90 | 90 |
| K2 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| K3 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| China HUBCO | 81.69 | 80.99 | 82.47 | 83.67 | 75.05 | 50 | 50 | 50 | 50 | 50 |
| Port Qasim | 61.81 | 75.19 | 77.64 | 50 | 50 | 50 | 50 | 50 | 50 | 50 |
| Sahiwal Coal | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 |
| Engro Thar | 85 | 85 | 85 | 85 | 85 | 84.98 | 21.5 | 8.71 | 84.91 | 84.99 |
| Lucky | 10.56 | 34.48 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Thal Nova | 85 | 85 | 85 | 85 | 85 | 84.96 | 21.39 | 8.98 | 84.96 | 84.96 |
| Thar I SSRL | 84.96 | 84.98 | 84.98 | 84.99 | 84.34 | 84.36 | 0 | 0.16 | 0.56 | 84.76 |
| Thar TEL | 85 | 85 | 85 | 85 | 85 | 85 | 84.93 | 84.96 | 84.96 | 85 |
| Altern | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Balloki | 0.48 | 2.28 | 3.97 | 1.94 | 0.93 | 0.59 | 0 | 0 | 0 | 0 |
| Bhikki | 0 | 0.16 | 0.5 | 0.38 | 0.04 | 0 | 0 | 0 | 0 | 0 |
| Davis | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| FKPCL | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Halmore | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Haveli | 5.32 | 11.25 | 39.89 | 86.64 | 3.55 | 1.55 | 0 | 0 | 0 | 0 |
| KAPCO 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Nandipur | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Orient | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Rousch | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Saif | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Sapphire | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Trimmu | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Gwadar | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0.19 |
| Jamshoro Coal | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| CSP | 0 | 0 | 0 | 0 | 43.77 | 43.35 | 0 | 0 | 0 | 0.2 |
| Shahtaj | 45.62 | 45.62 | 45.62 | 45.6 | 44.52 | 44.47 | 0 | 0 | 0 | 0.2 |
| TAY | 0 | 0 | 45.62 | 45.6 | 44.57 | 44.37 | 0 | 0 | 0 | 0.2 |

## Scenario 4.4 – Utilization Rates (%)

| Fiscal Year | 2024 | 2025 | 2026 | 2027 | 2028 | 2029 | 2030 | 2031 | 2032 | 2033 |
|---|---|---|---|---|---|---|---|---|---|---|
| CASA | 0 | 0 | 0 | 0 | 42 | 42 | 42 | 42 | 42 | 42 |
| FPCL | 54.23 | 3.68 | 0 | 0.08 | 0.94 | 0 | 0 | 0.53 | 0.56 | 1.39 |
| Gul Ahmed | 51.98 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| SNPC 1 | 92 | 92 | 92 | 92 | 92 | 92 | 92 | 92 | 92 | 92 |
| SNPC 2 | 92 | 92 | 92 | 92 | 92 | 92 | 92 | 92 | 92 | 92 |
| Tapal | 45.88 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| BQPS 1 Unit 01 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| BQPS 1 Unit 02 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| BQPS 1 Unit 05 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| BQPS 1 Unit 06 | 0 | 0 | 0 | 0 | 1.88 | 0.45 | 0.9 | 1.32 | 1.42 | 0.75 |
| BQPS 2 | 20.62 | 0.8 | 0.88 | 8.01 | 34.97 | 23.58 | 27.04 | 29.77 | 29.97 | 26.27 |
| BQPS 3 Unit 01 | 67.48 | 27.45 | 35.6 | 33.16 | 70.75 | 66.83 | 66.16 | 64.97 | 70.77 | 67.23 |
| BQPS 3 Unit 02 | 86.31 | 25.51 | 9.89 | 50.8 | 63.66 | 48.26 | 54.47 | 53.76 | 54 | 50.08 |
| KGTPS | 0 | 0 | 0 | 0.25 | 5.63 | 1.98 | 2.76 | 3.43 | 3.94 | 3.2 |
| KPC | 0.61 | 0.01 | 0 | 1.01 | 12.08 | 4.15 | 6.13 | 8.26 | 8.57 | 6.53 |
| SGTPS | 0 | 0 | 0 | 0 | 3.12 | 1.14 | 1.8 | 2.49 | 2.47 | 1.92 |
| New Coal 330 | 0 | 0 | 0 | 0 | 0 | 85 | 85 | 85 | 85 | 85 |
| New Coal 660 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Engro | 78.25 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Foundation | 79.3 | 84.81 | 84.95 | 85 | 85 | 85 | 85 | 85 | 85 | 84.98 |
| Guddu II | 78.66 | 83.93 | 84.83 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Guddu V | 84.65 | 84.94 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Liberty | 2.96 | 2.96 | 2.96 | 2.96 | 85 | 0 | 0 | 0 | 0 | 0 |
| Uch | 2.95 | 2.96 | 2.96 | 2.96 | 2.95 | 2.96 | 2.96 | 85 | 0 | 0 |
| Uch II | 50 | 78.99 | 79.89 | 84.58 | 83.74 | 84.92 | 53.47 | 50 | 50 | 82.73 |
| AES Lalpir | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| AES Pakgen | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| AGL | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Atlas | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Engro 127MW | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Hub N | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| HUBCO | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Jamshoro I U1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Jamshoro II U4 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Kohinoor | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Liberty Tech | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Muzaffargarh I U1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Muzaffargarh I U2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Muzaffargarh I U3 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Muzaffargarh II U4 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Nishat C | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Nishat P | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Saba | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| C1 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 |
| C2 | 80 | 80 | 80 | 80 | 80 | 80 | 80 | 80 | 80 | 80 |
| C3 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| C4 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 |
| C5 | 0 | 0 | 0 | 0 | 0 | 0 | 90 | 90 | 90 | 90 |
| K2 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| K3 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| China HUBCO | 81.67 | 80.89 | 82.53 | 84.91 | 84.87 | 84.46 | 50 | 50 | 72.65 | 84.52 |
| Port Qasim | 61.82 | 76.33 | 78.22 | 50 | 50 | 50 | 50 | 50 | 50 | 50 |
| Sahiwal Coal | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 |
| Engro Thar | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Lucky | 10.73 | 32.51 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1.69 |
| Thal Nova | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Thar I SSRL | 84.96 | 84.99 | 84.98 | 85 | 85 | 85 | 84.99 | 84.99 | 84.99 | 85 |
| Thar TEL | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Altern | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Balloki | 0.41 | 2.18 | 3.65 | 42.66 | 3.38 | 1.67 | 0 | 0.02 | 0.22 | 0.05 |
| Bhikki | 0 | 0.14 | 0.46 | 0.77 | 0.88 | 0.55 | 0 | 0 | 0 | 0 |
| Davis | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| FKPCL | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Halmore | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Haveli | 5.05 | 11.25 | 37.16 | 90 | 87.48 | 56.75 | 0.17 | 0.7 | 1.29 | 0.86 |
| KAPCO 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Nandipur | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Orient | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Rousch | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Saif | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Sapphire | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Trimmu | 0 | 0 | 0 | 0.04 | 0.08 | 0 | 0 | 0 | 0 | 0 |
| Gwadar | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 2.87 |
| Jamshoro Coal | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| New Bagasse | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| CSP | 0 | 0 | 0 | 0 | 44.5 | 44.5 | 44.45 | 21.67 | 44.42 | 44.47 |
| Shahtaj | 45.62 | 45.62 | 45.62 | 45.62 | 45.62 | 45.62 | 45.57 | 22.28 | 45.57 | 45.59 |
| TAY | 0 | 0 | 45.62 | 45.62 | 45.62 | 45.62 | 45.54 | 22.27 | 45.56 | 45.59 |

## Scenario 4.5 – Utilization Rates (%)

| Fiscal Year | 2024 | 2025 | 2026 | 2027 | 2028 | 2029 | 2030 | 2031 | 2032 | 2033 |
|---|---|---|---|---|---|---|---|---|---|---|
| CASA | 0 | 0 | 0 | 0 | 41.89 | 42 | 42 | 42 | 42 | 42 |
| FPCL | 53.79 | 2.51 | 0 | 0.01 | 0 | 0 | 0.01 | 0.01 | 0 | 0.01 |
| Gul Ahmed | 53.94 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| SNPC 1 | 92 | 92 | 92 | 92 | 92 | 92 | 65.81 | 64.01 | 63.32 | 63.58 |
| SNPC 2 | 92 | 92 | 92 | 92 | 92 | 92 | 67.62 | 67.21 | 65.5 | 64.29 |
| Tapal | 48.18 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| BQPS 1 Unit 01 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| BQPS 1 Unit 02 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| BQPS 1 Unit 05 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| BQPS 1 Unit 06 | 0 | 0 | 0 | 0 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 |
| BQPS 2 | 19.32 | 0.5 | 1.07 | 1.39 | 2.18 | 2.27 | 2.23 | 1.95 | 1.83 | 2.06 |
| BQPS 3 Unit 01 | 86.44 | 10.88 | 13.07 | 18.67 | 40.43 | 23.17 | 22.92 | 32.8 | 47.3 | 31.22 |
| BQPS 3 Unit 02 | 68.24 | 41.3 | 41.83 | 47.45 | 32.12 | 50.03 | 50.53 | 36.28 | 19.26 | 34.48 |
| KGTPS | 0 | 0 | 0 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 |
| KPC | 0.14 | 0 | 0 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.02 |
| SGTPS | 0 | 0 | 0 | 0 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 |
| New Coal 330 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| New Coal 660 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Engro | 77.19 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Foundation | 79.41 | 84.97 | 84.97 | 84.96 | 85 | 84.97 | 0.01 | 0.48 | 1.01 | 0.64 |
| Guddu II | 77.83 | 83.86 | 84.96 | 84.99 | 85 | 85 | 84.97 | 84.97 | 61.44 | 0.69 |
| Guddu V | 84.75 | 84.97 | 85 | 85 | 85 | 85 | 85 | 84.99 | 85 | 31.5 |
| Liberty | 2.96 | 2.96 | 2.96 | 2.96 | 148.75 | 0 | 0 | 0 | 0 | 0 |
| Uch | 2.95 | 2.96 | 2.96 | 2.96 | 2.95 | 2.96 | 2.96 | 99.17 | 0 | 0 |
| Uch II | 50 | 78.66 | 80.05 | 82.11 | 79.55 | 82.71 | 50 | 50 | 50 | 50 |
| AES Lalpir | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| AES Pakgen | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| AGL | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Atlas | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Engro 127MW | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Hub N | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| HUBCO | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Jamshoro I U1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Jamshoro II U4 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Kohinoor | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Liberty Tech | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Muzaffargarh I U1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Muzaffargarh I U2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Muzaffargarh I U3 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Muzaffargarh II U4 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Nishat C | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Nishat P | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Saba | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| C1 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 |
| C2 | 80 | 80 | 80 | 80 | 80 | 80 | 80 | 80 | 80 | 80 |
| C3 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 |
| C4 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| C5 | 0 | 0 | 0 | 0 | 0 | 0 | 90 | 90 | 90 | 90 |
| K2 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| K3 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| China HUBCO | 81.65 | 80.84 | 82.8 | 84.27 | 81.61 | 55.98 | 50 | 50 | 50 | 50 |
| Port Qasim | 62.89 | 75.38 | 77.94 | 50 | 50 | 50 | 50 | 50 | 50 | 50 |
| Sahiwal Coal | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 |
| Engro Thar | 85 | 85 | 85 | 85 | 85 | 84.98 | 41.44 | 27.34 | 84.96 | 85 |
| Lucky | 10.26 | 38.1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0.01 |
| Thal Nova | 85 | 85 | 85 | 85 | 85 | 85 | 41.04 | 27.14 | 84.96 | 84.98 |
| Thar I SSRL | 84.99 | 84.99 | 84.98 | 85 | 84.84 | 84.81 | 0 | 0.23 | 0.59 | 84.91 |
| Thar TEL | 85 | 85 | 85 | 85 | 85 | 85 | 84.97 | 84.97 | 84.99 | 85 |
| Altern | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Balloki | 0.42 | 1.67 | 3.7 | 11.91 | 0.86 | 0.53 | 0 | 0 | 0 | 0 |
| Bhikki | 0 | 0.18 | 0.54 | 0.49 | 0.05 | 0 | 0 | 0 | 0 | 0 |
| Davis | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| FKPCL | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Halmore | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Haveli | 4.55 | 10.29 | 44.91 | 90 | 4.35 | 1.09 | 0 | 0 | 0 | 0 |
| KAPCO 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Nandipur | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Orient | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Rousch | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Saif | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Sapphire | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Trimmu | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Gwadar | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0.37 |
| Jamshoro Coal | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| New Tech Y1 | 0 | 0 | 0 | 0 | 40 | 40 | 40 | 40 | 40 | 40 |
| New Tech Y2 | 0 | 0 | 0 | 0 | 0 | 40 | 40 | 40 | 40 | 40 |
| New Tech Y3 | 0 | 0 | 0 | 0 | 0 | 0 | 40 | 40 | 40 | 40 |
| New Tech Y4 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 40 | 40 | 40 |
| New Tech Y5 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 40 | 40 |
| New Tech Y6 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 40 |
| CSP | 0 | 0 | 0 | 0 | 44.09 | 43.81 | 0 | 0 | 0.01 | 0.25 |
| Shahtaj | 45.62 | 45.62 | 45.62 | 45.6 | 45.02 | 44.8 | 0 | 0.01 | 0.01 | 0.32 |
| TAY | 0 | 0 | 45.72 | 45.6 | 44.88 | 44.92 | 0 | 0 | 0.01 | 0.32 |

## Scenario 4.6 – Utilization Rates (%)

| Fiscal Year | 2024 | 2025 | 2026 | 2027 | 2028 | 2029 | 2030 | 2031 | 2032 | 2033 |
|---|---|---|---|---|---|---|---|---|---|---|
| CASA | 0 | 0 | 0 | 0 | 42 | 42 | 42 | 42 | 42 | 42 |
| FPCL | 54.23 | 3.68 | 0.04 | 0.14 | 1.02 | 0 | 0.07 | 0.73 | 0.7 | 1.37 |
| Gul Ahmed | 51.98 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| SNPC 1 | 92 | 92 | 92 | 92 | 92 | 92 | 92 | 92 | 92 | 92 |
| SNPC 2 | 92 | 92 | 92 | 92 | 92 | 92 | 92 | 92 | 92 | 92 |
| Tapal | 45.88 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| BQPS 1 Unit 01 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| BQPS 1 Unit 02 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| BQPS 1 Unit 05 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| BQPS 1 Unit 06 | 0 | 0 | 0 | 0 | 1.94 | 0.54 | 1.01 | 1.4 | 1.5 | 0.72 |
| BQPS 2 | 20.62 | 0.8 | 1.2 | 9.2 | 36 | 24.61 | 28.06 | 30.78 | 30.99 | 25.53 |
| BQPS 3 Unit 01 | 67.5 | 27.45 | 36.59 | 35.14 | 71.66 | 67.73 | 66.99 | 68.34 | 71.53 | 66.45 |
| BQPS 3 Unit 02 | 86.29 | 25.51 | 11.44 | 51.94 | 64.5 | 49.4 | 55.53 | 57.55 | 55.04 | 49.06 |
| KGTPS | 0 | 0 | 0 | 0.38 | 6.78 | 2.11 | 2.84 | 3.72 | 4.2 | 3.11 |
| KPC | 0.61 | 0.01 | 0 | 1.32 | 12.62 | 4.46 | 6.74 | 9 | 9.39 | 6.03 |
| SGTPS | 0 | 0 | 0 | 0.03 | 3.42 | 1.19 | 1.9 | 2.6 | 2.67 | 1.76 |
| New Coal 330 | 0 | 0 | 0 | 0 | 0 | 85 | 85 | 85 | 85 | 85 |
| New Coal 660 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Engro | 78.25 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Foundation | 79.3 | 84.81 | 84.95 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Guddu II | 78.66 | 83.93 | 84.83 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Guddu V | 84.65 | 84.94 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Liberty | 2.96 | 2.96 | 2.96 | 2.96 | 85 | 0 | 0 | 0 | 0 | 0 |
| Uch | 2.95 | 2.96 | 2.96 | 2.96 | 2.95 | 2.96 | 2.96 | 85 | 0 | 0 |
| Uch II | 50 | 78.99 | 80.2 | 84.77 | 84.48 | 84.98 | 84.88 | 72.06 | 65.21 | 83.44 |
| AES Lalpir | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| AES Pakgen | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| AGL | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Atlas | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Engro 127MW | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Hub N | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| HUBCO | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Jamshoro I U1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Jamshoro II U4 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Kohinoor | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Liberty Tech | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Muzaffargarh I U1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Muzaffargarh I U2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Muzaffargarh I U3 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Muzaffargarh II U4 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Nishat C | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Nishat P | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Saba | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| C1 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 |
| C2 | 80 | 80 | 80 | 80 | 80 | 80 | 80 | 80 | 80 | 80 |
| C3 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| C4 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 |
| C5 | 0 | 0 | 0 | 0 | 0 | 0 | 90 | 90 | 90 | 90 |
| K2 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| K3 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| China HUBCO | 81.67 | 80.89 | 82.79 | 84.93 | 84.91 | 84.73 | 55.57 | 50 | 84.71 | 84.82 |
| Port Qasim | 61.82 | 76.33 | 78.33 | 50 | 50 | 50 | 50 | 50 | 50 | 50 |
| Sahiwal Coal | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 |
| Engro Thar | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Lucky | 10.73 | 32.51 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 2.09 |
| Thal Nova | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Thar I SSRL | 84.96 | 84.99 | 84.98 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Thar TEL | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Altern | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Balloki | 0.41 | 2.18 | 4.09 | 55.23 | 16.77 | 2.14 | 0 | 0.07 | 0.38 | 0.14 |
| Bhikki | 0 | 0.14 | 0.56 | 0.96 | 1.07 | 0.7 | 0 | 0 | 0 | 0 |
| Davis | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| FKPCL | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Halmore | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Haveli | 5.05 | 11.25 | 41.07 | 90 | 90 | 72.23 | 0.27 | 0.87 | 1.59 | 1.12 |
| KAPCO 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Nandipur | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Orient | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Rousch | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Saif | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Sapphire | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Trimmu | 0 | 0 | 0.01 | 0.09 | 0.14 | 0.02 | 0 | 0 | 0 | 0 |
| CSP | 0 | 0 | 0 | 0 | 44.5 | 44.5 | 44.47 | 44.47 | 44.47 | 44.5 |
| Shahtaj | 45.62 | 45.62 | 45.62 | 45.62 | 45.62 | 45.62 | 45.59 | 45.59 | 45.59 | 45.62 |
| TAY | 0 | 0 | 45.62 | 45.62 | 45.62 | 45.62 | 45.6 | 45.59 | 45.59 | 45.62 |

## Scenario 4.8 – Utilization Rates (%)

| Fiscal Year | 2024 | 2025 | 2026 | 2027 | 2028 | 2029 | 2030 | 2031 | 2032 | 2033 |
|---|---|---|---|---|---|---|---|---|---|---|
| CASA | 0 | 0 | 0 | 0 | 41.89 | 42 | 42 | 42 | 42 | 42 |
| FPCL | 53.79 | 2.51 | 0 | 0.01 | 0 | 0 | 0 | 0 | 0 | 0 |
| Gul Ahmed | 53.94 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| SNPC 1 | 92 | 92 | 92 | 92 | 92 | 92 | 92 | 92 | 92 | 92 |
| SNPC 2 | 92 | 92 | 92 | 92 | 92 | 92 | 92 | 92 | 92 | 92 |
| Tapal | 48.18 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| BQPS 1 Unit 01 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| BQPS 1 Unit 02 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| BQPS 1 Unit 05 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| BQPS 1 Unit 06 | 0 | 0 | 0 | 0.01 | 0.01 | 0.01 | 0.09 | 0 | 0 | 0 |
| BQPS 2 | 19.32 | 0.5 | 0.75 | 12.13 | 4.04 | 3.35 | 25.83 | 0.02 | 0.29 | 0.73 |
| BQPS 3 Unit 01 | 86.71 | 10.88 | 11.04 | 45.04 | 40.21 | 28.6 | 55.19 | 13.15 | 26.37 | 19.19 |
| BQPS 3 Unit 02 | 67.98 | 41.3 | 41.07 | 63.99 | 46.56 | 53.81 | 71.68 | 12.01 | 4.71 | 18.51 |
| KGTPS | 0 | 0 | 0 | 0.01 | 0.01 | 0.01 | 0.95 | 0 | 0 | 0 |
| KPC | 0.14 | 0 | 0 | 0.32 | 0.02 | 0.04 | 4.43 | 0 | 0 | 0.01 |
| SGTPS | 0 | 0 | 0 | 0.01 | 0.01 | 0.01 | 0.26 | 0 | 0 | 0 |
| Engro | 77.19 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Foundation | 79.41 | 84.97 | 84.97 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Guddu II | 77.83 | 83.86 | 84.95 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Guddu V | 84.75 | 84.97 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Liberty | 2.96 | 2.96 | 2.96 | 2.96 | 148.75 | 0 | 0 | 0 | 0 | 0 |
| Uch | 2.95 | 2.96 | 2.96 | 2.96 | 2.95 | 2.96 | 2.96 | 99.17 | 0 | 0 |
| Uch II | 50 | 78.66 | 79.86 | 84.43 | 84.95 | 84.98 | 84.97 | 84.97 | 84.95 | 84.95 |
| AES Lalpir | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| AES Pakgen | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| AGL | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Atlas | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Engro 127MW | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Hub N | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| HUBCO | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Jamshoro I U1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Jamshoro II U4 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Kohinoor | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Liberty Tech | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Muzaffargarh I U1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Muzaffargarh I U2 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Muzaffargarh I U3 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Muzaffargarh II U4 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Nishat C | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Nishat P | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Saba | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| C1 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 |
| C2 | 80 | 80 | 80 | 80 | 80 | 80 | 80 | 80 | 80 | 80 |
| C3 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 |
| C4 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 | 82 |
| C5 | 0 | 0 | 0 | 0 | 0 | 0 | 90 | 90 | 90 | 90 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| K2 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| K3 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| China HUBCO | 81.65 | 80.84 | 82.51 | 84.97 | 84.99 | 84.95 | 84.93 | 84.92 | 84.97 | 84.98 |
| Port Qasim | 62.89 | 75.38 | 77.66 | 50 | 50 | 50 | 50 | 50 | 50 | 50 |
| Sahiwal Coal | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 |
| Engro Thar | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Lucky | 10.26 | 38.1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 84.76 |
| Thal Nova | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Thar I SSRL | 84.99 | 84.99 | 84.98 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Thar TEL | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 | 85 |
| Altern | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Balloki | 0.42 | 1.67 | 3.29 | 48 | 69.75 | 39.51 | 0.02 | 0.68 | 20.97 | 1.22 |
| Bhikki | 0 | 0.18 | 0.47 | 0.69 | 1.68 | 0.84 | 0 | 0.26 | 0.63 | 0.33 |
| Davis | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| FKPCL | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Halmore | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Haveli | 4.55 | 10.29 | 41.18 | 90 | 90 | 90 | 31.02 | 74.43 | 90 | 72.89 |
| KAPCO 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Nandipur | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Orient | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Rousch | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Saif | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Sapphire | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| Trimmu | 0 | 0 | 0 | 0 | 0.53 | 0.23 | 0 | 0 | 0.01 | 0 |
| Gwadar | 0 | 0 | 0 | 0 | 0.02 | 0 | 0 | 0 | 0 | 85 |
| Jamshoro Coal | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| CSP | 0 | 0 | 0 | 0 | 44.5 | 44.5 | 44.48 | 44.48 | 44.5 | 44.5 |
| Shahtaj | 45.62 | 45.62 | 45.62 | 45.62 | 45.62 | 45.62 | 45.6 | 45.6 | 45.62 | 45.62 |
| TAY | 0 | 0 | 45.72 | 45.62 | 45.62 | 45.62 | 45.6 | 45.6 | 45.62 | 45.62 |

# Glossary

### Base load:

Base load/demand represents the consistent and continuous demand of electricity over an extended period such as a day, a month or a year. Base demand is linked with the essential services that occur continuously such as streetlights, refrigeration and some industries.

### Candidate Power Plants:

These are potential projects that can be considered as an option in the future generation capacity expansion. These power plants are included in future capacity expansion models as possible additions, depending on the system's needs, economic viability, policy support, or market conditions. These plants serve as options for future capacity, and their inclusion depends on various scenarios or uncertainties, such as future demand growth, policy shifts, or technology changes.

### Captive power plants:

Captive power plants (CPPs) refer to small-scale power generation facilities that are set up by industries or businesses to meet their own electricity needs, rather than relying entirely on the national grid. These plants help industries avoid power outages and reduce costs, especially in areas where the grid supply is unreliable or expensive.

### Capacity factor:

Capacity factor is a term used in the context of power plants to describe the ratio of actual generation over a specified time interval to the maximum possible generation if the plant were operating at full capacity. It is calculated as the ratio of actual power output to the maximum power output.

### Committed Power Plants:

These are projects that have already been planned, approved, or are under construction with a high degree of certainty that they will be operational within the specified planning horizon. In other words, these projects have received the necessary approvals, financing, and other required permits, and construction may have already begun.

### Fixed Operations & Maintenance (FO&M) costs:

Expressed in USD per kilo-watt-year, FO&M costs refers to the expenses associated with ongoing operations and maintenance of a power plant or an energy storage facility that remain constant irrespective of the output of a power plant or an energy storage facility.

### IGCEP:

Indicative Generation Capacity Expansion Plan refers to the power generation planning which involves strategy for producing electricity to meet the demand of consumers. IGCEP includes determination of optimum mix of power generation sources that can meet the electricity demand based on least cost. In Pakistan, the System Operator (National Transmission & Despatch Company) is mandated to prepare the IGCEP every year with a ten-year rolling horizon.

### Net Present Value (NPV):

NPV is a financial metric used to evaluate the profitability and viability of future investments in power generation infrastructure, such as building new power plants or upgrading existing facilities. NPV calculates the difference between the present value of cash inflows (e.g., revenues from selling electricity) and the present value of cash outflows (e.g., initial capital investment, operating and maintenance costs) over a specified time horizon.

### Peak demand:

Peak demand refers to the instance when electricity demand is maximum. It represents the highest level of electricity consumption that occurs during the periods of high activity such as turning on an air conditioner

during heat wave, during a hot summer day

**Peaking plants:**

Refers to thermal power plants that are operated to meet the short duration demand spikes (peak demand) in a system. These power plants are usually HSD/RFO based power plants having fast ramp rates but significantly higher operating costs.

**Scope 2 emissions:**

Represents indirect greenhouse gas (GHG) emissions resulting from purchased electricity generation by an organization. These GHG emissions occur outside of the organization's direct control but are linked with the organization's electricity consumption.

**Scope 3 emissions:**

Include all the indirect greenhouse gas (GHG) emissions that arise throughout an organization's value chain, including activities such as procurement, transportation, waste disposal, and employee commuting. These emissions are not directly owned or controlled by the organization but are associated with its operations and activities.

**Total cost:**

Total cost of generation refers to the sum of generation costs (fuel costs, variable O&M costs) and fixed costs (FO&M costs).

**Variable Operations & Maintenance (VO&M) cost:**

Expressed in USD per Mega-Watthour, VO&M refers to the costs associated with ongoing operations and maintenance of a power plant or an energy storage system. The span of these costs varies and depends upon the nature of operations and the sort of maintenance that is required. Typically, VO&M costs include fuel costs, labor costs, lubricants costs and repairs & overhauling etc.

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RENEWABLES FIRST

PRIED
Policy Research Institute
for Equitable Development

# Integrating Battery Storage with Renewables: A Techno-economic Analysis

(On-grid and off-grid solutions)

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# List of Abbreviations

| | |
|---|---|
| AC | Alternate Current |
| BESS | Battery Energy Storage System |
| BMS | Battery Management System |
| C | C-Rate |
| CF | Capacity Factor |
| DC | Direct Current |
| DERs | Distributed Energy Resources |
| DOD | Depth of Discharge |
| EMS | Energy Management System |
| FO&M | Fixed Operations & Maintenance Cost |
| Hz | Hertz |
| IGCEP | Integrated Generation Capacity Expansion Plan |
| KE | K-Electric |
| kW | kilowatt |
| kWh | kilowatt-hour |
| LCOE | Levelized Cost of Energy |
| Lithium Ion | Li-ion |
| MoE | Ministry of Energy |
| MW | megawatt |
| NEP | National Electricity Plan |
| NEPRA | National Electric Power Regulatory Authority |
| NTDC | National Transmission & Despatch Company |
| Pmax | Max Power |
| PCS | Power Conversion System |
| RE | Renewable Energy |
| REs | Renewable Energy Sources (Wind and Solar) |
| ROE | Return on Equity |
| ROEDC | Return on Equity During Construction |
| SOC | State of Charge |
| SOH | State of Health |
| UNE | Universal National Electrification |
| VO&M | Variable Operations & Maintenance |

# Glossary

**Base demand:**

Base demand represents the consistent and continuous demand of electricity over an extended period such as a day, a month or a year. Base demand is linked with the essential services that occur continuously such as streetlights, refrigeration and some industries.

**CTBCM:**

CTBCM stands for Competitive Trading Bilateral Contracts Market. It is a market mechanism that facilitates the trading of electricity through bilateral contracts between buyers and sellers in a competitive environment. CTBCM allows market participants, such as generators, retailers, and large consumers, to negotiate and enter into contracts for the purchase and sale of electricity at mutually agreed prices and terms.

**IGCEP:**

Indicative Generation Capacity Expansion Plan refers to the power generation planning which involves strategy for producing electricity to meet the demand of consumers. IGCEP includes determination of optimum mix of power generation sources that can meet the electricity demand based on least cost. In Pakistan, the System Operator (National Transmission & Despatch Company) is mandated to prepare the IGCEP every year with a ten-year rolling horizon.

**Peak charge avoidance:**

Refers to the strategies employed to reduce electricity bills by shifting energy consumption to off-peak hours or using alternative energy sources and storage systems to minimize reliance on grid.

**Peak demand:**

Peak demand refers to the instance when electricity demand is maximum. It represents the highest level of electricity consumption that occurs during the periods of high activity such as turning on an air conditioner during heat wave, during a hot summer day.

**Spinning reserve:**

Refers to the portion of power generation capacity that is online and synchronized with the grid but not supplied to consumers. The generators providing spinning reserve are kept in standby mode and are ready to increase their output in the event of sudden load fluctuations. Spinning reserves play an important role in grid stability by providing response to unforeseen fluctuations in supply or demand.

**RE curtailments:**

Refers deliberate reduction or limitation of electricity generation from renewable energy sources such as wind or solar power plants due to over generation of electricity from renewable energy sources relative to demand or capacity of the grid. It usually occurs when the electricity demand is lower but electricity production from renewable energy sources is maximum. In Pakistan, another major reason behind RE curtailments is obsolete transmission infrastructure.

**Take or Pay Fuel contracts:**

In these types of agreements, a fixed amount of fuel (e.g., RLNG) is secured for the contract duration, and the supplier guarantees its availability at defined intervals. Consequently, a power plant is obligated to consume the contracted amount of fuel or incur charges for it, for instance, if a power plant shuts down due to low demand and does not require fuel.

**Energy density:**

Associated with batteries, energy density refers to the amount of energy in kilo watt-hours (kWh) that can be stored in battery per unit mass (kilograms). It indicates how much energy can be stored in the battery for every kilogram of battery's weight.

**State of Charge (SOC):**

Associated with batteries, SOC refers to the level of energy that is stored in a battery relative to its maximum capacity. SOC is expressed in percentage and is an essential parameter to avoid over charging and over discharging of a battery.

**State of Health (SOH):**

Associated with batteries, SOH refers to the overall condition of battery as compared to its original performance capability. It measures how well a battery can charge and discharge along with charge and storage duration. SOH is influenced by operating conditions.

**Peaking plants:**

Refers to thermal power plants that are operated to meet the short duration demand spikes (peak demand) in a system. These power plants are usually HSD/RFO based power plants having fast ramp rates but significantly higher operating costs.

**Capacity factor:**

Capacity factor is a term used in the context of power plants to describe the ratio of actual generation over a specified time interval to the maximum possible generation if the plant were operating at full capacity. It is calculated as the ratio of actual power output to the maximum power output.

**Variable Operations & Maintenance (VO&M) cost:**

Expressed in USD per Mega-Watthour, VO&M refers to the costs associated with ongoing operations and maintenance of a power plant or an energy storage system. The span of these costs varies and depends upon the nature of operations and the sort of maintenance that is required. Typically, VO&M costs include fuel costs, labor costs, lubricants costs and repairs & overhauling etc.

**Fixed Operations & Maintenance (FO&M) costs:**

Expressed in USD per kilo-watt-year, FO&M costs refers to the expenses associated with ongoing operations and maintenance of a power plant or an energy storage facility that remain constant irrespective of the output of a power plant or an energy storage facility.

**Scope 2 emissions:**

Represents indirect greenhouse gas (GHG) emissions resulting from purchased electricity generation by an organization. These GHG emissions occur outside of the organization's direct control but are linked with the organization's electricity consumption.

**Scope 3 emissions:**

Include all the indirect greenhouse gas (GHG) emissions that arise throughout an organization's value chain, including activities such as procurement, transportation, waste disposal, and employee commuting. These emissions are not directly owned or controlled by the organization but are associated with its operations and activities.

# Executive Summary

The electricity generation and consumption patterns vary significantly across sectors (industrial or commercial/ residential) and countries (developing or developed). While conducting the electricity generation planning exercises for the optimum electricity generation mix, the dynamics of that sector and country must be considered, including GDP growth rate, currency depreciation, inflation rate, industrial and residential demand projections, and projections of exogenous variables (Population, crude oil prices). These indicators significantly impact the electricity consumption in a country and are therefore crucial for crafting a well-thought-out and optimum electricity generation mix plan.

A critical examination of Pakistan's electricity consumption patterns reveals that electricity demand in Pakistan is highly variational, and while the peak demand has been increasing consistently over the past decade, the base demand has barely witnessed any significant increase. The negligence of this simple fact while drawing up electricity generation plans and policies has resulted in an installed generation capacity which is almost double the size of the utilized capacity. This concerning disparity, which has burdened ordinary consumers and business owners alike, must prompt reflection on the trajectory of decision-making and policy formulation in Pakistan. This current study is one such attempt of research-based reflection seeking to explore the potential of renewable energy technologies (REs) coupled with Battery Energy Storage Systems (BESS) in the context of variable demand patterns in Pakistan.

Energy storage technology using batteries has already demonstrated its commercial viability and estimates show that BESS market is expected to reach between 120 – 150 USD Billions by 2030 as per McKinsey's article on Battery energy systems[1].

The reason there has been so much inclination towards BESS installation is its compelling benefits. One of the key advantages of using BESS is that it offers improved frequency regulation as compared to traditional spinning reserves from thermal power plants and thereby also reduces RE curtailment. Likewise, BESS can provide peak charge avoidance to consumers via off-grid energy during on-grid peak consumption intervals. There are also instances where RE generation does not coincide with the electricity demand; the surplus power, in this case, can be stored in the BESS and used later when the demand is higher. Another significant benefit of integrating BESS with REs is promoting environmental sustainability by reducing dependence on fossil fuel power generation. The strategic integration of BESS with REs is a pivotal solution, offering peak support to avoid expensive Take or Pay fuel contracts and driving down consumer end tariffs.

The landscape of BESS is rapidly evolving, witnessing remarkable technological advancements. In the exciting world of emerging technologies in BESS, the most notable technologies include Lithium-ion (Li-ion), Flow batteries (Iron or Vanadium), Sodium-Sulfur (NaS), and nickel-cadmium. While the BESS technologies are still evolving, this study focuses on Li-ion technology for maturity, low cost, and high efficiency. With a compelling energy density (150-250 Watt-hour per Kilogram), extended life span, and eco-friendliness, Li-ion emerges as the forefront choice for innovative energy storage solutions [2].

For on-grid RE-BESS systems, the study proposes a hybrid model (wind and solar combined) to deal with yearly variation of peak and base demand in the national grid. Under this combination, solar and wind technologies complement one another across diurnal and annual cycles while the surplus energy generated is stored by the batteries during periods of low demand. Similarly, for off-grid RE-BESS systems or independent projects of wheeling/CTBCM regime, the study explores the "firm capacity" scenario during which reliable and uninterrupted power supply can be provided for selected intervals. On a broader level, this study aims to answer two questions. First, Can Renewable energy sources coupled with Battery energy storage systems serve the purpose of meeting variational demand patterns of Pakistan?" Second, can these systems be utilized by on-grid and off-grid consumers at an affordable and competitive cost? Both the questions have been answered in the study after extensive data collection, simulation, and analysis.

The use cases presented within this study are designed to provide off-grid and on-grid solutions by leveraging daily and yearly generation patterns of REs coupled with the BESS. This innovative synergy ensures optimal energy utilization for a sustainable and resilient power ecosystem. The study provides a comprehensive examination of these use cases involving the integration of solar with BESS, wind with BESS, and the synergy of both REs with BESS. The outcomes include a detailed techno-economic analysis of the integrated hybrid solution (REs coupled with BESS), revealing its potential to introduce flexibility into the power system. The use case on the Firm capacity reveals that optimal combination of solar and batteries is well suited for the firm capacity while peak support can be best provided by combination of hybrid power plant (wind and solar) with the BESS.

Beyond enhancing the system's flexibility, the study underscores significant economic advantages in avoiding costly fossil fuel-based power generation. Ultimately, adopting REs coupled with BESS emerges not only as a strategic choice for flexibility but also as a catalyst for financial savings.

# 1. Introduction

The electricity demand in Pakistan has seen a dynamic trajectory over the recent years due to macro and micro economic factors, including population growth, economic development, and changing electricity consumption patterns. During peak intervals, the demand-supply gap often strains the power grid, leading to power outages, blackouts, and challenges in meeting the country's electricity needs. The peak demand trend as per the NEPRA's state of industry report [3] is shown in the figure below:

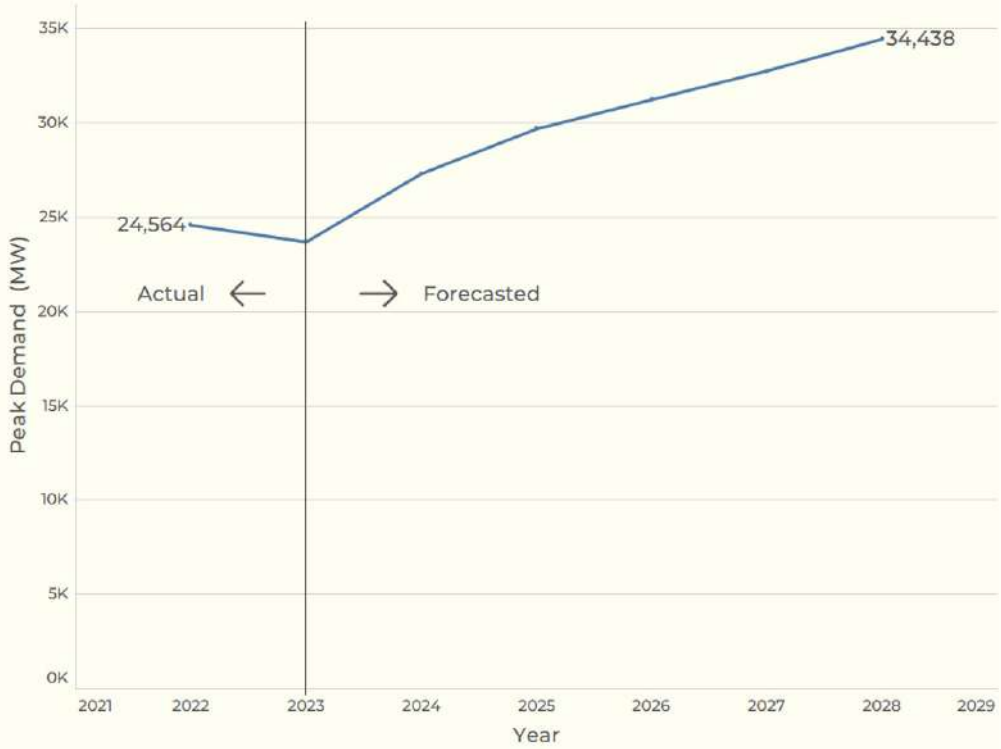


Figure 1 - Peak Demand Trend (Source: NEPRA State of Industries Report 2023)

The outlook of Pakistan's electricity demand profile exhibits two peaks during a typical day; one peak occurs during the day due to commercial and industrial loads, while the second peak occurs during the evening due to residential loads as shown in the figure 2 below [4]. The primary reasons behind these varying demand patterns include weather conditions, seasonal variations, spatial differences, and consumer behaviors. To ensure grid stability and reliability, efficiently meeting these fluctuations is imperative.

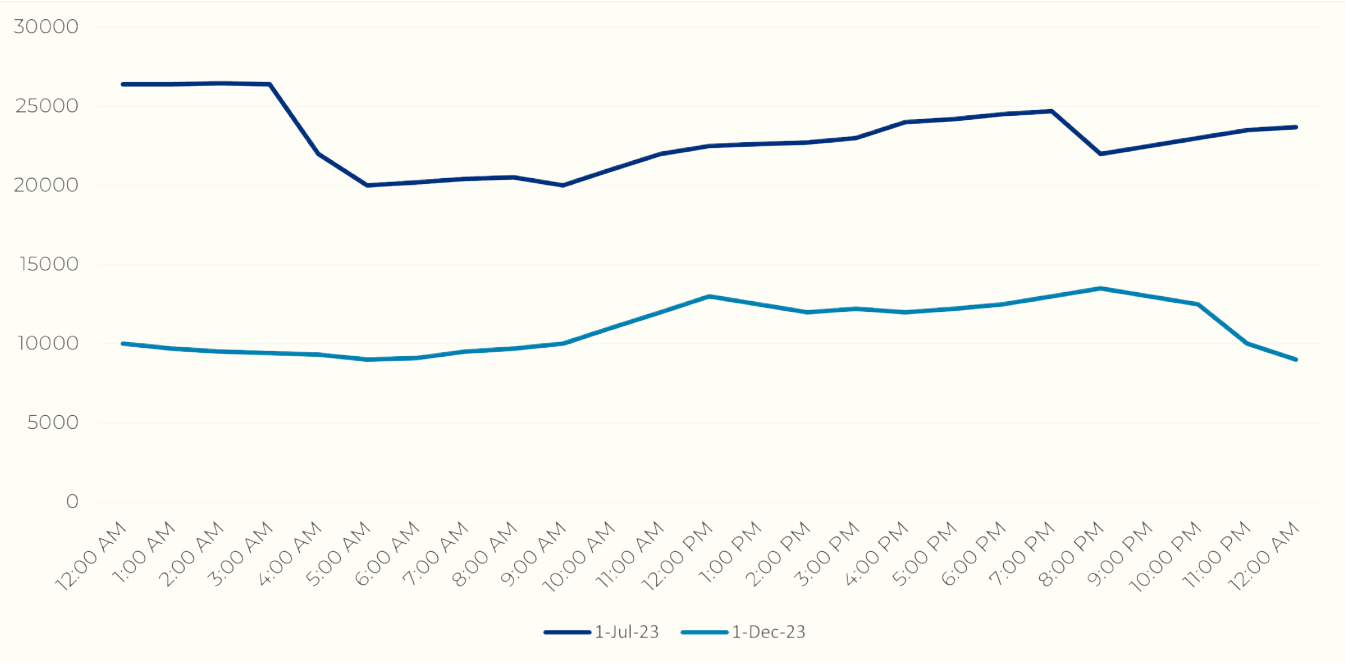


Figure 2 - Hourly Load Demand (Source: NTDC)

In Pakistan's Power sector planning, the key focus is on understanding and managing peak demand. Peak demand intervals are those where electricity demand is at its maximum, during the summer when the air conditioning load is at its maximum. The planning process involves forecasting peak demand to ensure sufficient electricity generation capacity to meet these high-demand periods, thereby averting shortages or blackouts.

Pakistan possesses significant potential for wind and solar energy due to its geographical positioning. Yet, there is only a 6% share of REs in the country's power generation mix. Due to intermittency issues and variability associated with REs like wind and solar, REs have been critiqued, particularly in Pakistan. While REs are acknowledged for their cost-effectiveness in electricity generation, there is a prevalent perception that the inherent variability of REs makes them costly. There is also an argument that baseload generation is imperative for a power system, and peak demand intervals can only be best met with thermal power plants, so-called peaking plants. Despite concerns about the variability of REs in power systems, there needs to be more discussion on leveraging the combined potential of REs and BESS to meet the fluctuating electricity demand in Pakistan effectively. This hybrid proposition serves the purpose of meeting the variational nature of demand in Pakistan. It will also replace the costly fossil fuel power generation and avoid expensive take or pay fuel contracts. Since BESS can provide better frequency regulation due to fast demand response time, fast ramp rates, accurate frequency regulation due to controlled output, and can be easily scaled up or down, integrating them with REs will complement the RE generation by effectively managing the intermittencies associated with REs.

Various technologies for large-scale renewable energy storage currently exist, each with unique advantages, limitations, potential, and applications. Among the notable options are lithium-ion batteries, sodium-sulfur batteries, vanadium-redox flow batteries, metal-air batteries, pumped hydro storage, flywheels, and compressed air energy storage. These technologies are either actively employed or under consideration for grid-scale energy storage. Battery technologies used in energy storage devices can be categorized based on energy density, charge, and discharge efficiency (round trip efficiency), lifespan (cycles), and eco-friendliness. Energy density refers to the energy a system can store per unit volume or weight. For instance, lithium secondary batteries have an energy storage capacity ranging from 150 to 250 watt-hours per kilogram. They can hold 1.5 to 02 times more energy than sodium-sulfur (Na–S) batteries, 02 to 03 times more than redox flow batteries (vanadium or iron), and around 05 times more than lead storage batteries. A comparison among various BESS technologies is provided in the table 1 below:

| Technologies | Lithium-Ion | | | | Sodium Sulfur | Redox Flow |
|---|---|---|---|---|---|---|
| | NMC | NCA | LFP | LTO | | |
| Characteristics | | | | | | |
| Operating Life (Yrs.) | 10 | 12 | 20 | 20 | >20 | >20 |
| Cycles | 4800 | 5000 | 12000 | 15000 | 20000 | 20000 |
| Energy Density (kWh/m3) | 200-250 | 250-300 | 95-200 | 50-80 | 150-350 | 30 |
| Efficiency (%) | 97 | 97 | 97 | 97 | 90 | 90 |
| Capital cost ($/kWh) | 170-210 | 250-300 | 150-190 | 400-900 | 300-600 | 240-390 |
| Operating cost ($/kWh) | 8 | 8 | 8 | 6 | 8 | 11 |
| Maturity | Mature | Mature | Mature | No | No | No |

Table – 1 Comparison Matrix-Storage technologies (Source: World Bank)

Therefore, Lithium-ion-based BESS is considered for this study due to its maturity, cost-effectiveness, and higher efficiency than other technologies. For its maturity, flexibility, and adaptability, Li-ion BESS has been actively deployed across the globe for large-scale energy storage. According to the International Energy Agency, China has emerged as the market leader in grid-scale battery storage additions, with yearly installations approaching approximately 5 Gigawatts. Next to China is the United States, which is also heavily investing in the grid-scale BESS, with annual installations reaching up to 4 GW [[5]. According to PV Magazine, Ark Energy secures approval for 275 MW and 2200 MWh Lithium-iron phosphate batteries that will be the world's largest 08-hour Lithium battery in the Australian state of New Wales [6].

One of the primary reasons behind this significant attention to Li-ion batteries is the reduction of Li-ion battery pack prices that hit the record low price of $139 per kWh in 2023, according to Bloomberg NEF. Further technological advancements and improvements in the manufacturing process would also decrease the Li-ion battery pack prices to $113 per kWh in 2025 and $80 per kWh in 2030, as per BNEF [7].

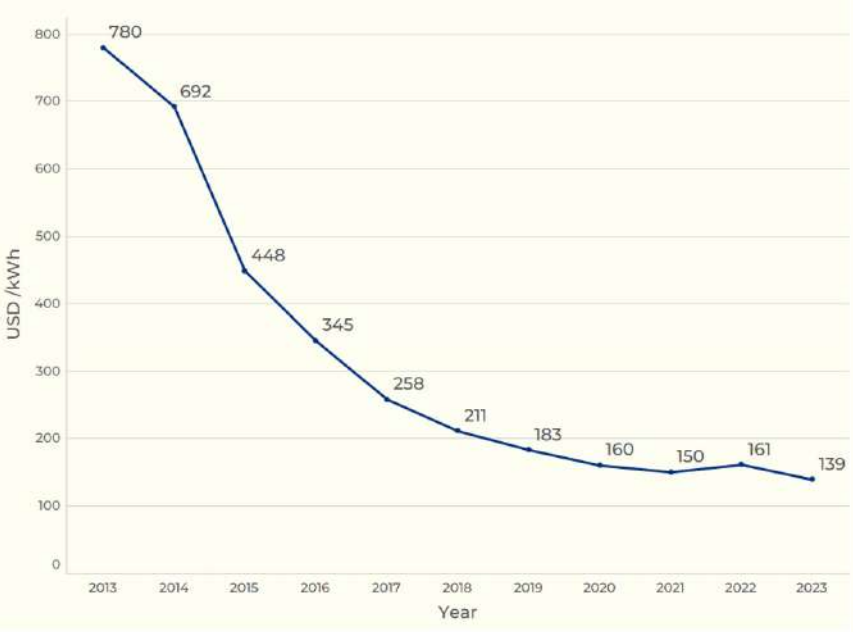


Figure 3 - Cost trend of the battery pack (Source: Bloomberg NEF 2023 Lithium-ion Battery Price Survey)

The techno-economic specifications highlighted above underscore the pivotal role of Li-ion Batteries, making them the cornerstone of this study. Hence, this study aims to investigate Li-ion batteries' economic and commercial feasibility through a comprehensive techno-economic analysis, focusing on specific use cases.

While applications of the BESS include frequency regulation, peak shaving, spinning, non-spinning and supplemental reserves, voltage support, and black start, this study underscores two compelling use cases: firm capacity and peak support. The firm capacity use case examines the practicality of utilizing this hybrid solution (BESS integrated with REs) to ensure a consistent and uninterrupted power supply at specified intervals. The firm capacity provided is essential for overcoming the inherent variability of REs. It is pivotal in maintaining grid stability, meeting energy demand, and ensuring a reliable and consistent power supply. Since it has been argued that large-scale integration of REs can pose severe problems to grid stability, integrating the BESS with REs can alleviate this problem by providing the firm capacity. By adequately scheduling the charging and discharging of BESS integrated with REs, they can ensure firm power during peak hours or specified intervals. Certain REs like wind and solar tend to have periods of higher generation during the off-peak hours that align with the valleys of net demand; for instance, wind generation can be maximum in the evening. Therefore, charging the BESS during the valleys of net demand and discharging them during peak hours can enhance the overall load factor and avoid the need for costly thermal peaking plants.

The electricity demand outlook of Pakistan indicates that demand in Pakistan is variational; the variations between the peak demand and base demand align with the generation of REs, which makes REs a cheaper source of meeting this variational demand. The adaptability of REs coupled with the BESS allows better management of the variational nature of demand by storing surplus power during off-peak hours and utilizing the stored power during peak demand intervals. This hybrid proposition would offer a flexible and sustainable approach that complements Pakistan's evolving energy needs in the coming years.

Regarding economic feasibility, the REs has seen a significant cost decline in Pakistan, with wind tariffs decreasing by 73% from 2011 to 2020 and solar tariffs falling by 78% from 2014 to 2020, the LCOE of wind and solar is given below as per NEPRA's tariff determinations [8].

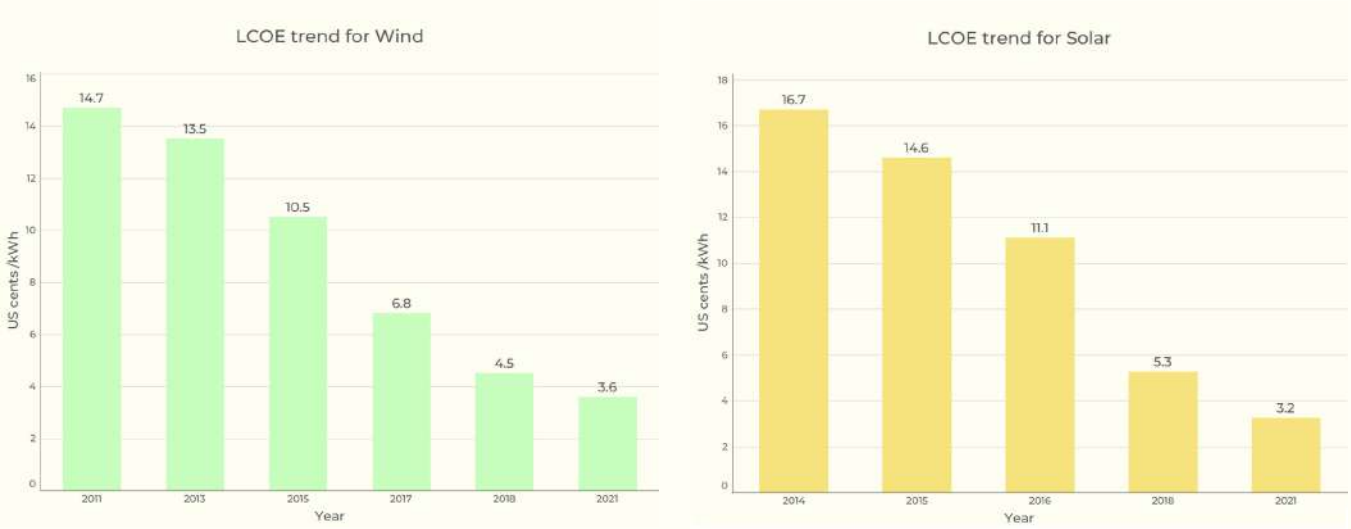


Figure 4 - Levelized Tariff for Wind & Solar Power plants (Source: NEPRA)

*Awarded to Moro Power Company Ltd., ** Awarded to Siachen Energy Limited

Hence, the topographic conditions, economic feasibility of REs, and their generation trends show that REs coupled with the BESS can address Pakistan's variations in electricity demand. This study's main objective is to demonstrate the techno-commercial viability of the hybrid proposition involving the coupling of REs with BESS.

This demonstration is accomplished through a comprehensive techno-economic analysis based on two critical use cases: firm capacity and peak support.

## 2. Overview of BESS

The BESS comprises four essential elements: a battery system, an inverter or power conversion system (PCS), a battery management system (BMS), and an energy management system (EMS).

- The battery system consists of multiple cells organized into modules, forming the battery pack. These cells convert chemical energy into electricity.
- The BMS ensures the safety of the battery system by monitoring cell conditions, including state-of-charge (SOC) and state-of-health (SOH). Battery Thermal Management System (B-TMS) safeguards against potential risks such as fires by controlling the temperature of cells in terms of temperature gradients in battery packs.
- The EMS monitors and controls power flow within the storage system, overseeing the operation of the BMS, PCS, and other components. It analyzes energy-related data to optimize the system's resources. The system monitoring and control is done by integration with overall supervisory control and data acquisition (SCADA).
- The inverter or PCS transforms direct current (DC) from the batteries into alternating current (AC) and vice versa. Equipped with bi-directional inverters, BESS systems facilitate both charging and discharging.

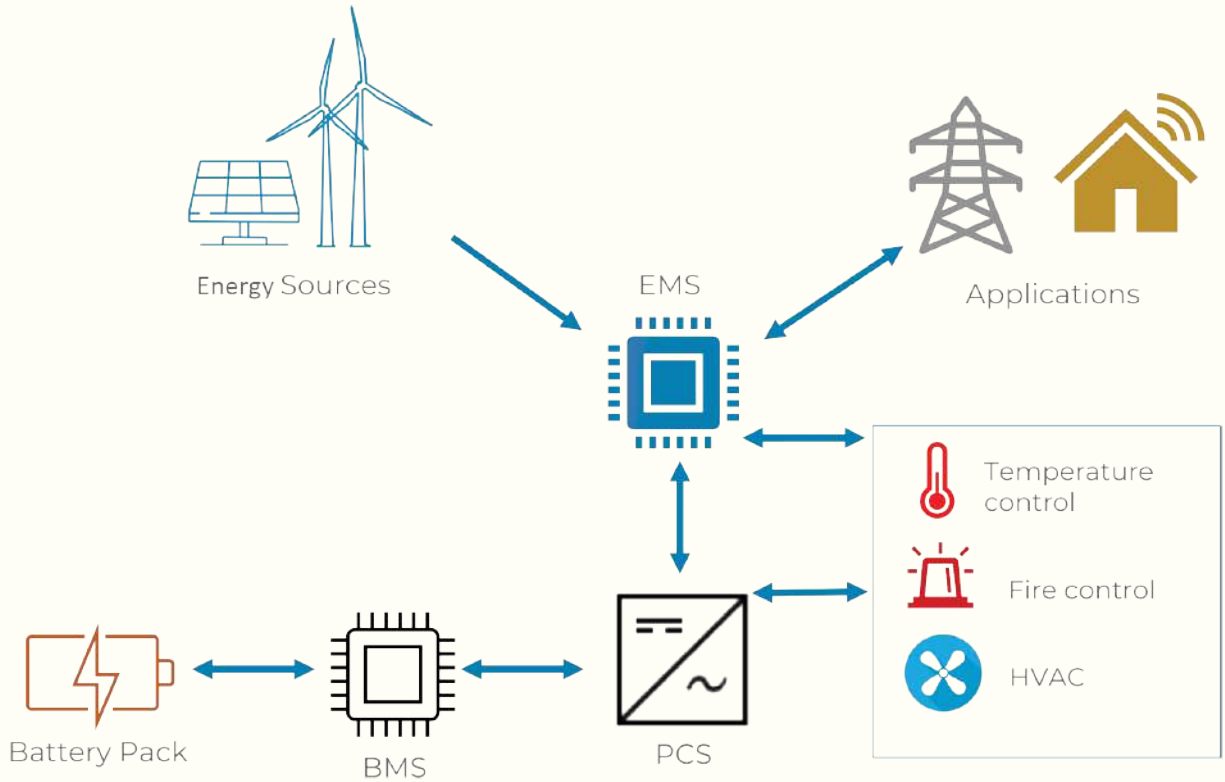


Figure 5 - Elements of BESS

Similarly, the BESS modeling design parameters are as follows.

**Energy capacity:** This is the maximum amount of energy that the battery can store, usually measured in megawatt-hours (MWhs) or gigawatt-hours (GWhs).

**Max Power (Pmax):** Max power is the power in kilowatts (kW) or Megawatts (MW) at full discharge measured at the connection point, i.e., excluding inverter losses.

**Round-trip efficiency:** The round-trip efficiency of the battery is the ratio of energy charged and discharged in the BESS. It is expressed as a percentage. The round-trip efficiency may represent the total DC to DC or AC to AC efficiency of the BESS incorporating all the losses, i.e., Self-discharges and other electrical losses.

**Maximum State of Charge (SoC):** Maximum SoC is a BESS's allowable minimum state of charge. This is expressed as a percentage of the Capacity defined.

**Minimum SoC:** Minimum SoC is the allowable minimum SoC of a BESS. This is expressed as a percentage of the Capacity defined.

**PCS:** Power conversion system comprised of inverters.

**Depth of Discharge (DOD):** The difference between the maximum and minimum state of charge is called depth of discharge.

**Battery Cycle:** A battery cycle refers to the complete process of charging a battery from its minimum SoC to its maximum SoC and then discharging it back to the minimum SoC again. Typically, a battery storage unit undergoes one cycle per day.

**Battery Life:** The expected lifespan of a battery is either expressed in years.

**Cost Parameters:** Various cost-related parameters such as capital, operating, maintenance, and replacement costs.

**C-Rate:** C-Rate is a unit which is used to express the rate at which a battery is charged or discharged. It represents the current relative to battery's capacity. 1C represents current will be fully charged or discharged in a battery in 01 hour.

Battery Energy Storage Systems (BESS) sizing varies depending on the intended applications. For frequency regulation, BESS sizes typically range from 10 to 40 MW, addressing short-term power needs with targeted discharge durations ranging from 15 minutes to 01 hour. In transmission upgrade deferral cases, where the goal is to delay utility investments in transmission system upgrades or avoid such investments, BESS sizes typically range between 10 to 100 MW, with targeted discharge durations of 2 to 8 hours. For providing electric supply capacity by deferring or avoiding new capacity additions, BESS can be sized up to 500 MW, with targeted discharge durations between 2 to 6 hours.

## 3. Use-Cases

BESS offers multiple applications, providing essential ancillary services such as Frequency Regulation, Peak Shaving, Grid Balancing and market arbitrage. Frequency regulation refers to a process where the alternate Current (AC) is maintained at a constant value of 50 Hz or 60 Hz, depending on the region. In Pakistan, the Normal operating range for frequency is 49.8Hz to 50.2Hz, and the Contingency Operating range is 49.3Hz to 50.5Hz as per the Grid Code 2023. Frequency regulation is essential to restore the balance between supply and demand. Failing to do so results in a malfunction that could cascade and cause widespread blackouts in the overall power system. There can be multiple reasons behind frequency deviations such as dynamic fluctuations in demand and generation that result in temporary imbalances and eventually cause frequency deviations. A significant contributor to frequency deviations is large scale integration of Renewable energy sources. Since REs are asynchronously integrated with power systems through power electronic devices, their inertial support to the power system is limited. This low inertia can lead to increased frequency oscillations that might exceed operating standards and eventually cause potential threat to uninterrupted power supply.

The primary responsibility of the System Operator is to ensure the system possesses adequate generation capacity to meet peak demand throughout the entire time horizon. Traditionally, costly thermal generators, known as peaking plants, have been employed to address this peak demand in accordance with the load curve. However, BESS also presents a cost-effective solution for meeting peak demand. Integrating REs with BESS empowers system operators to manage peak demand intervals efficiently. By strategically charging BESS during valleys of net demand and discharging them during peak demand intervals, the overall load factor of the system is enhanced, ensuring reliable demand management. This hybrid configuration of REs coupled with BESS has become increasingly economical, especially with the declining prices of BESS and REs. It can avoid costly peaking plants and fuel contracts, significantly advancing cost-effective and reliable peak demand management.

Market arbitrage using BESS involves taking advantage of price differentials between periods of low or high electricity demand or between regions with different electricity prices. BESS can be strategically charged when electricity prices are low and discharged when prices are higher to take market arbitrage opportunity. In this way, revenue can be maximized, and the costs of market participants can be minimized. In addition to this, stored energy in BESS during intervals of lower electricity prices can be used in ancillary service market for provision of ancillary services like voltage support and frequency regulation etc. However, this requires enactment of open market regulations entailed in the Competitive Trading Bilateral Contracts Markets (CTBCM) that are still pending in Pakistan. From the use cases of BESS described above, this study primarily focuses on two major applications: Firm Capacity and Peak Support, as these align well with the energy needs and dynamics of Pakistan's energy sector.

For the analysis of peak support case, hourly demand profiles from the National Transmission and Despatch Company (NTDC) and K-Electric (KE) during the fiscal year 2022 are utilized. Throughout the fiscal year 2022, the forecasted peak demand for NTDC was 24,755 MW, while KE's peak demand was 4,321 MW per the Indicative Generation Capacity Expansion Plan (IGCEP 2022) [9]. To facilitate comparison, both NTDC and KE demands are normalized, allowing for the analysis of relative trends. Normalization is achieved by dividing the respective hourly demands by the average demand of the entire fiscal year. Consequently, the peak normalized demand for NTDC reached 133 MW in quarter 3, while KE's demand peaked at 121 MW in quarter 2.

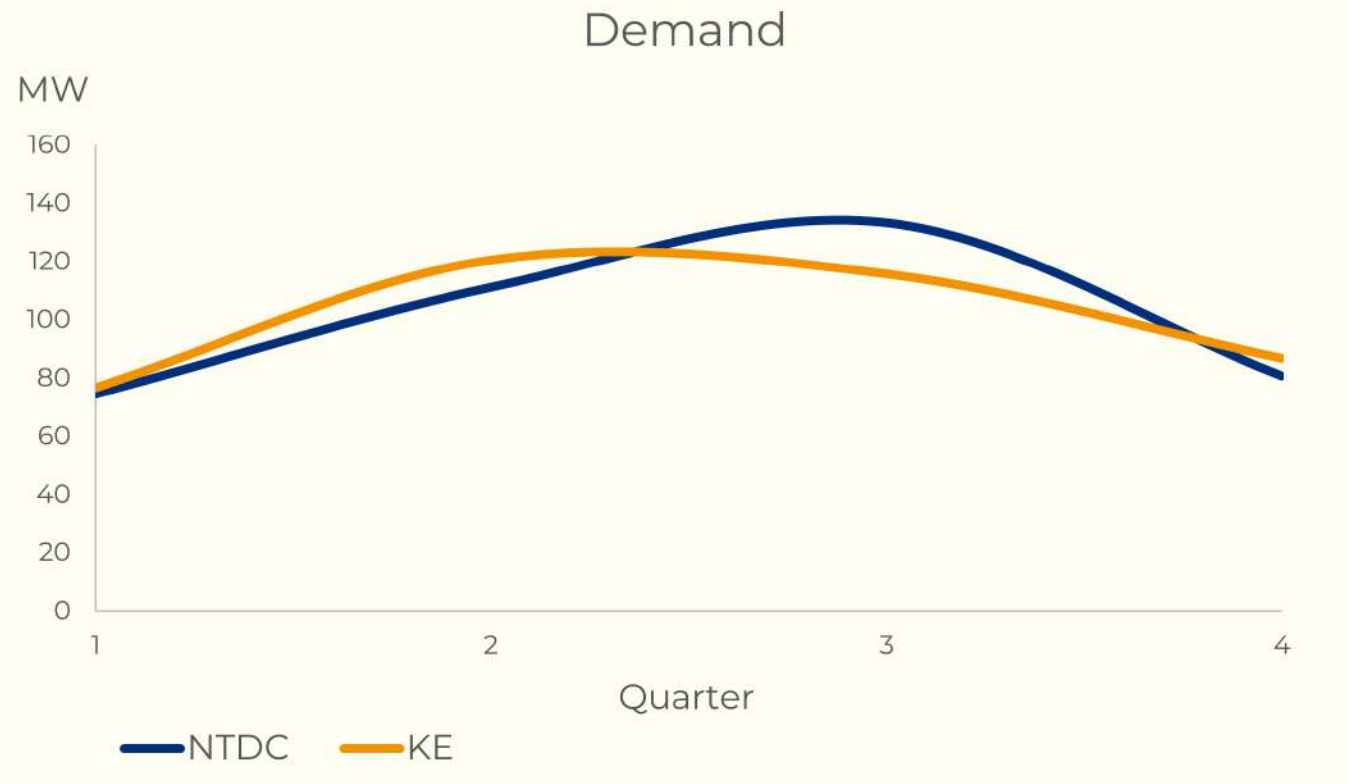


Figure 6 - Electricity Demand Pattern of NTDC & KE (Source: NTDC)

Note: these graphs have been normalized at 100 MW for the purpose of comparison

To meet this demand, an analysis of the generation profiles of REs is conducted quarterly. The analysis encompasses six REs i.e., net metering, utility-scale solar, and wind. Additionally, various hybrid profiles, incorporating specific combinations of wind and solar, are considered, as detailed in Table 2 below:

| Power Plants | Variations | Capacity Factor (CF) |
|---|---|---|
| Solar Net metering | Solar only | 17% |
| Utility-scale Solar | Solar only | 22% |
| Hybrid-27 | 1:3 (wind-to-solar ratio) | 27% |
| Hybrid-32 | 1:1 (wind-to-solar ratio) | 32% |
| Hybrid-37 | 3:1 (wind-to-solar ratio) | 36% |
| Utility-scale Wind | Wind only | 40% |
| All the power plants (solar, wind, and hybrid) are 100MW | | |

Table 2: Power Plants Considered for analysis.

It is pertinent to mention that among the three hybrid systems, the most efficient configuration is Hybrid-37 (3:1 wind-to-solar ratio) owing to its higher capacity factor and cost optimality, which is further explained in the cost section of this report. In this study, the Hybrid-37 configuration is chosen among other hybrid systems for its higher capacity factor and economic feasibility. The analysis extends to the generation profiles of REs, examined hourly and yearly. Results indicate that solar exhibits significant daily variation with lower annual variation, while wind displays higher annual variation and more downward daily fluctuation. Comprehensive figures 7 & 8 illustrating the yearly and quarterly generation of REs alongside the demand profiles of NTDC and KE systems are presented below for clarity:

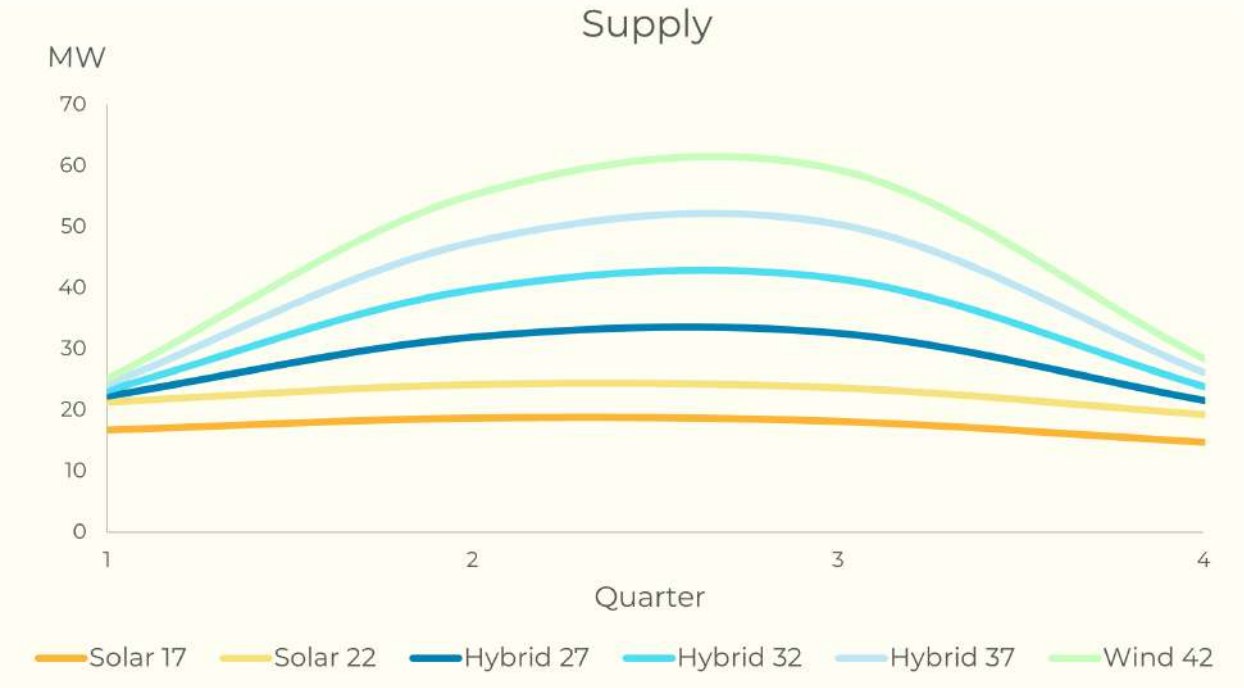


Figure 7 – Annual representative RE resource profiles

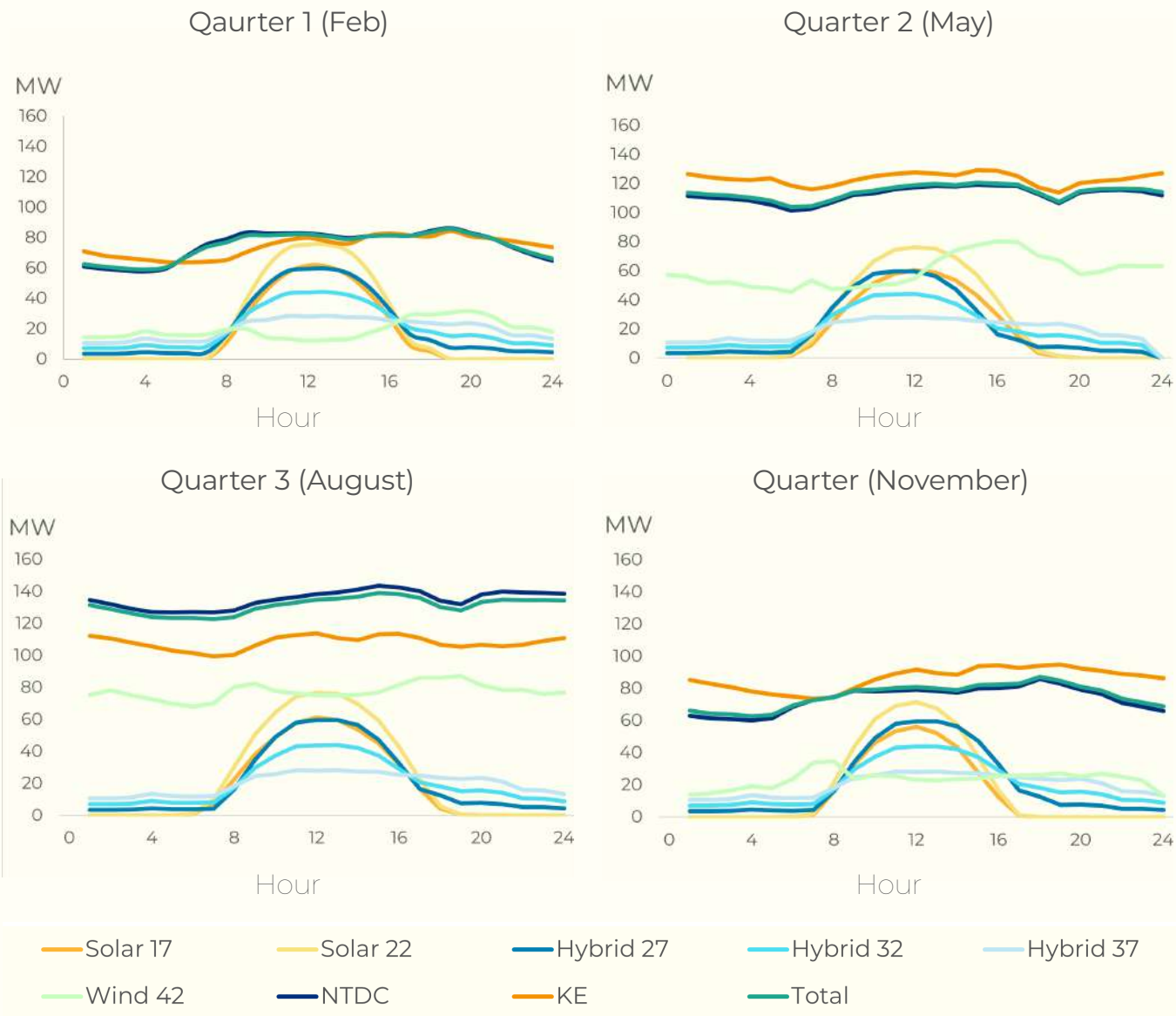


Figure 8 – Quarterly RE resource profiles against NTDC & KE demands

### 1. Firm Capacity

Expressed in kilowatts (kW) or Megawatts (MW), the firm capacity refers to the installed capacity that can reliably meet the demand during high-risk or peak intervals. The hybrid setting of REs paired with the BESS can contribute to the firm capacity. By adequately scheduling the charging and discharging of BESS, the energy stored in the BESS can be dispatched instantly when needed, ensuring consistent and robust capacity support. This configuration will provide a stable power supply even during variable generation intervals. In terms of applicability, this configuration of REs coupled with the BESS can be utilized to meet commercial load that usually lasts up to 08 hours, for instance, offices and markets. On a broader scale, this hybrid configuration can also be used as a customized package for industrial consumers having one or two working shifts in a day.

### 2. Peak Support

Peak support refers to the reliable and continuous supply of power from alternative sources to offset high demand from consumers, simultaneously helping ease out fluctuations in the grid and avoid strain on the system during peak usage times. Owing to fast response time, ramping rates, and flexibility in controlling the output, BESS can effectively meet spikes in demand and offer grid stability.

## 4. Modeling Parameters:

The primary goal of a power system planner is to meet the demand in a reliable and cost-efficient manner while maintaining the energy security of the power sector. Therefore, it is imperative to comprehend the modeling parameters for a system. This study utilizes the following modeling parameters to achieve its objectives:

### 4.1. Demand:

The key to crafting a well-thought-out and effective power generation plan is understanding and meeting the electricity demand cost-efficiently. Since a power generation plan aims to meet the demand, it is imperative to analyze and understand the demand and devise strategies accordingly. The constant demand can be called the baseline heart of power grids. Continuous demand involves consistent and steady electricity demand over a longer time horizon, and it can be predicted more accurately because of its stability. The continuous demand/load usually includes residential loads like lighting, basic electronics, public infrastructures, and some industries. The continuous demand exhibits gradual changes that can be predicted more efficiently, for instance, commercial and industrial loads, owing to their shift patterns.

In this study, the continuous demand includes 10 MW, 15 MW, 20 MW, and 30 MW of continuous demand.

### 4.2. Battery Size:

Measured in units like kilowatt-hours (kWh) or Ampere-hour (Ah), battery size is the total amount of energy that a battery can store, like the dam storage system in the case of hydel power plants where water is held in the reserve which can be used later. For optimizing the BESS, the sizing of the battery plays a pivotal role. The size of the battery depends on several factors like expected energy needs, i.e., the consumption patterns of a region or the system that needs to be powered up, the duration for which battery backup is needed, and the depth of discharge. Careful consideration of these factors is essential for optimizing a BESS. In this study, the BESS size varies as per the demand profiles and power plant configurations and described in the findings of the study.

## 5. Findings of study

This study explores the two use cases of the BESS coupled with REs, the firm capacity, and the peak support. For this, four distinct demand profile scenarios are analyzed, encompassing 10 MW, 15 MW, 20 MW, and 30 MW of continuous demand. As mentioned earlier, various combinations of REs coupled with BESS are considered and analyzed to find the best match for fulfilling the demand profiles effectively and efficiently.

The detailed findings of the study are mentioned below scenario-wise:

### 5.1. Firm Capacity

For the “firm capacity” use case, the scenarios include the integration of a standalone solar power plant with the BESS, a standalone wind power plant with the BESS, and integrating a hybrid power plant (Solar and wind) with the BESS. The scenario-wise analysis is presented below:

#### 5.1.1. Solar with BESS

In the first scenario focusing on firm capacity, the analysis considers three solar power plants with capacities of 100

MW, 70 MW, and 50 MW in conjunction with demand profiles. Given that solar power plants typically operate with capacity factor of approximately 21% to 22 % and generate electricity solely during daylight hours, these specific plant sizes were selected to ensure firm power supply. Various Battery Energy Storage System (BESS) sizes were evaluated to create firm support packages. Let's delve into the analysis of the 100 MW solar power plant:

#### 5.1.1.a. Solar 100 MW against 10 MW continuous demand

A solar power plant rated at 100 MW, paired with a 137 MWh BESS, can reliably provide a continuous power supply of 10 MW throughout the day. Considering that the solar plant is inactive for an average of 13 hours daily, the BESS capacity needed to sustain a consistent 10 MW output is determined to be 130 MWh. We assumed a round-trip efficiency of 95% for the BESS, based on industry standards. As depicted in Figure 9 below:

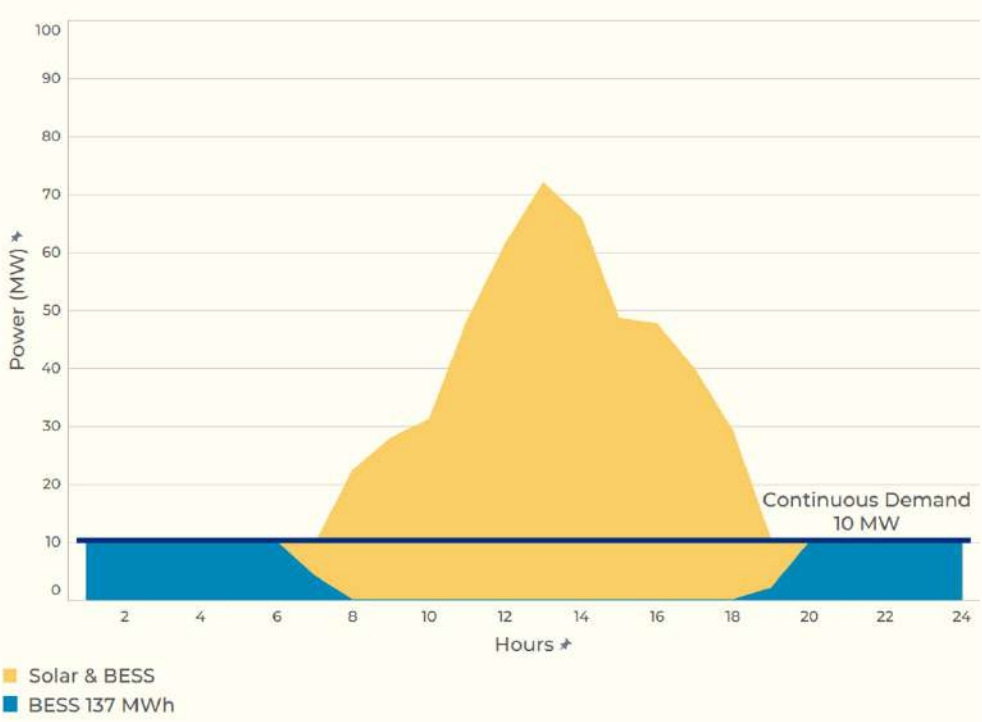


Figure 09 – 10 MW Firm support – All day

The analysis shows that a 137 MWh BESS, combined with a 100 MW solar power plant, effectively meets the continuous 10 MW demand. Notably, there are intervals during which solar generation exceeds demand by a factor of five to six. This surplus energy presents opportunities for utilization, such as selling to the market or supplying multiple consumers. However, capitalizing on these opportunities would require regulatory frameworks conducive to competitive trading, such as the implementation of Competitive Trading Bilateral Contracts Markets (CTBCM) in Pakistan.

For detailed information on Levelized Cost of Energy and other cost parameters, please refer to the methodology section of this report under section 6.

#### 5.1.1.b. Solar 100 MW against 15 MW continuous demand

In the next iteration, the continuous demand of 15 MW needs to be served. For this demand, a solar power plant of 100 MW is paired with 206 MWh BESS to ensure a consistent supply of 15 MW all day. The information is presented in the figure below:

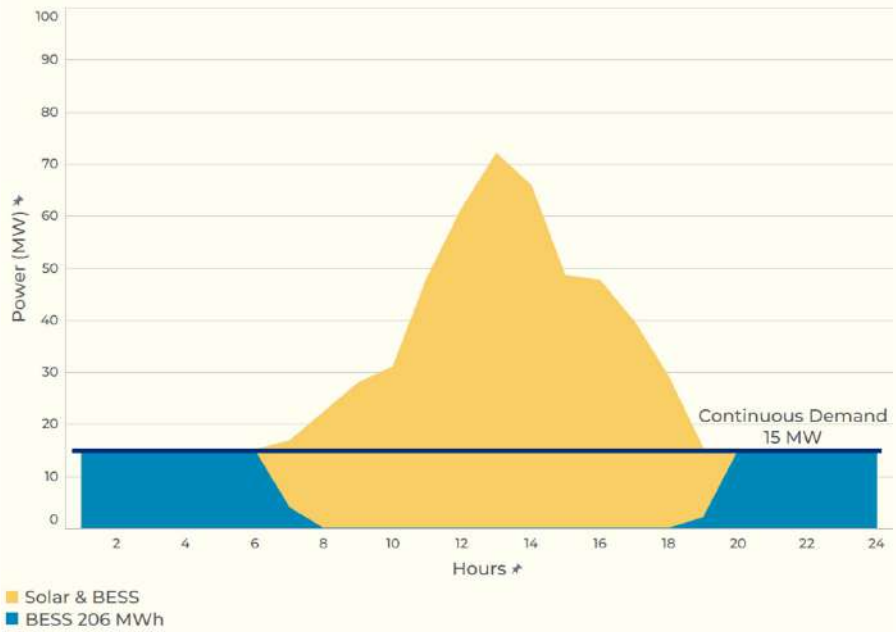


Figure 10 – 15 MW Firm support – All day

### 5.1.1.c. Solar 100 MW against 20 MW continuous demand

To ensure a consistent supply of 20 MW, a solar power plant with a capacity of 100 MW is integrated with battery energy storage systems (BESS) totaling 276 MWh. The sizing of BESS is determined by identifying periods of solar generation downtime and optimizing BESS charging schedules accordingly. It's worth noting the significance of BESS charging and discharging strategies, which can vary based on control algorithms such as fast or slow charging, depending on the C-Rate. For the purposes of this report, standard charging and discharging procedures are assumed. The configuration is shown in the figure below:

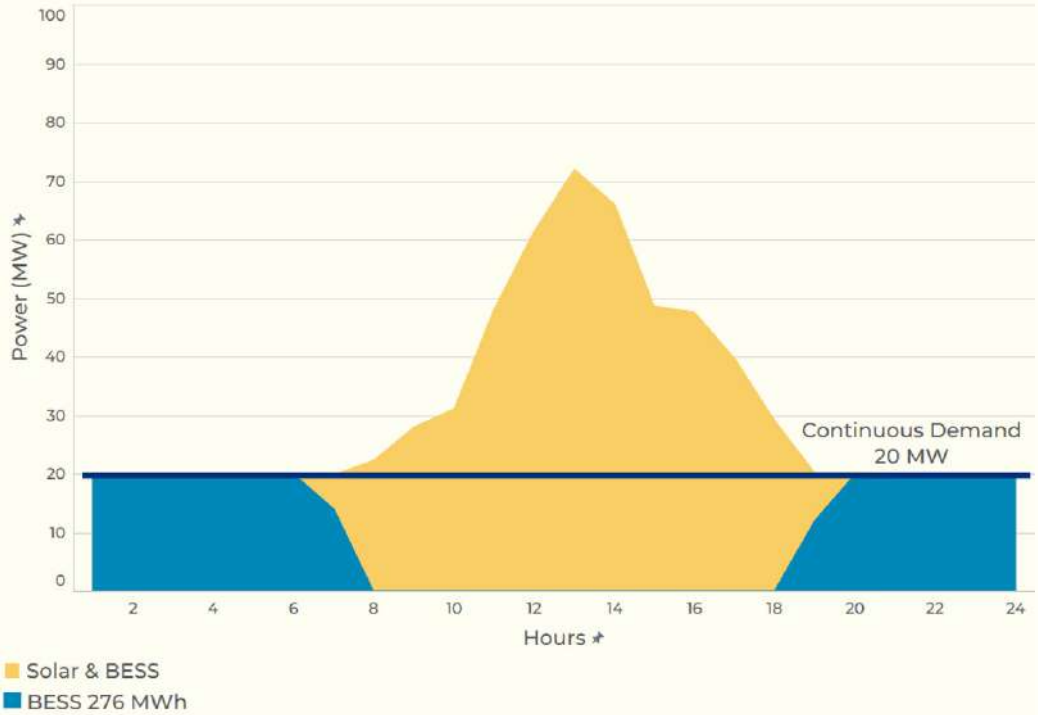


Figure 11 – 20 MW Firm support – All day

### 5.1.1.d. Solar 100 MW against 30 MW continuous demand

In the last scenario of 100 MW solar power plant, the continuous demand of 30 MW is considered. The analysis shows that a solar power plant of 100 MW is not capable of meeting consistent demand of 30 MW with any size of BESS as there is not enough generation to charge the BESS after serving the demand of 30 MW during daytime. However, a 100 MW solar power plant coupled with BESS of size ~127 MWh can provide firm support of 30 MW up to 14 hours. For this setting, 04 hours of backup is assumed from the BESS having capacity of ~31.7 MW (with 95 % round-trip efficiency) the configuration is shown in the figure below:

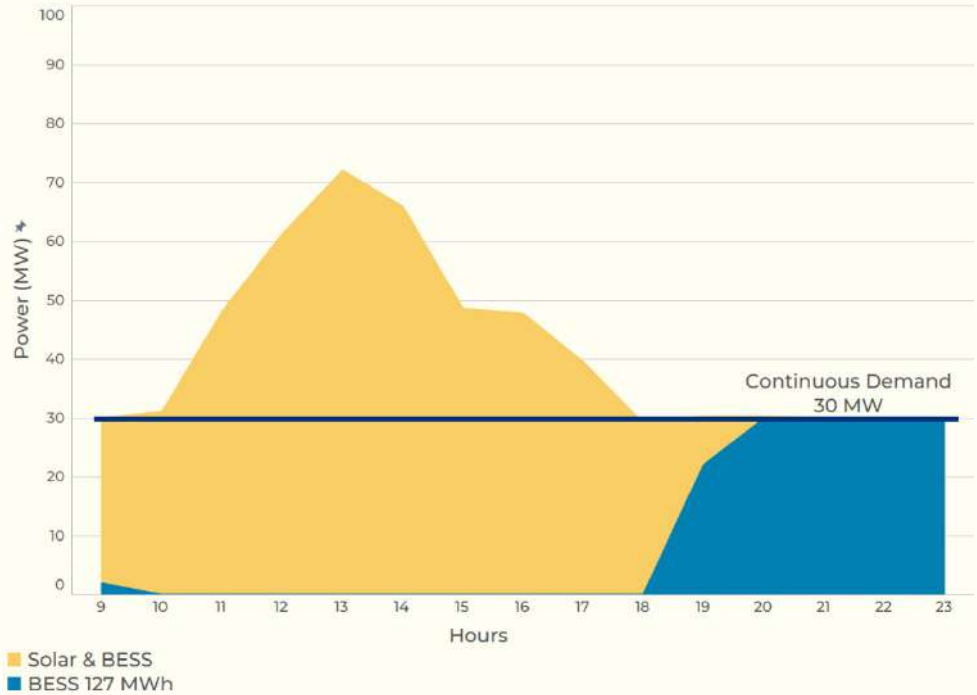


Figure 12 – 30 MW Firm support – 14 hours

### 5.1.1.e. Solar 70 MW against 10 MW continuous demand

In the second iteration, we explore the feasibility of a 70 MW solar power plant in optimizing combinations with Battery Energy Storage Systems (BESS) to effectively meet the specified demand profiles. Initially, we analyze the performance of a 70 MW solar plant coupled with a 137 MWh BESS to provide a consistent supply of 10 MW, as depicted in Figure 15 below:

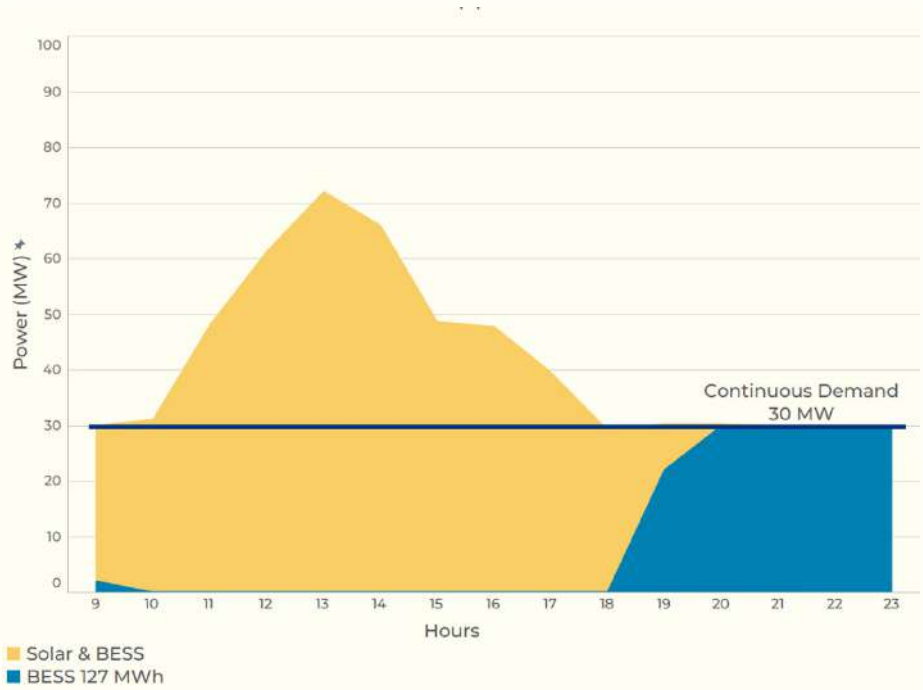


Figure 13 – 10 MW Firm support – All day

In contrast to solar power plant of 100 MW, a solar power plant of 70 MW can also meet the continuous demand of 10 MW when coupled with 137 MWh BESS and the surplus energy has also reduced as compared to 100 MW solar power plant. The comparison between LCOEs for both the settings is detailed in the methodology part under section 6.

### 5.1.1.f. Solar 70 MW against 15 MW continuous demand

In the scenario where a continuous demand of 15 MW is required, a 70 MW solar power plant paired with a Battery Energy Storage System (BESS) of approximately 206 MWh capacity proves capable of supplying firm support for the entire day. Since there are intervals when solar generation is twice or thrice than the demand to be served the battery sizing can vary however for this report, BESS size of 206 MWh is considered. The configuration is shown in the figure below:

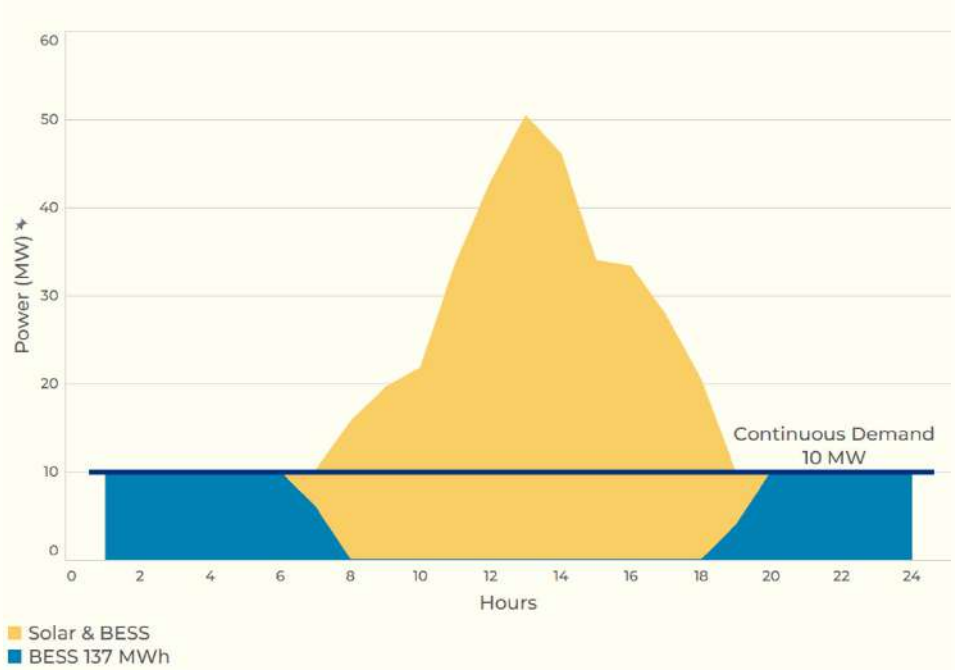


Figure 14 – 15 MW Firm support – All day

### 5.1.1.g. Solar 70 MW against 20 MW continuous demand

In the next iteration, the generation profile of 70 MW solar power plant is analyzed against the continuous demand of 20 MW, in this case the firm support of 20 MW can be supplied up to 14 hours by pairing BESS of ~85 MWh. The analysis is illustrated in the figure below:

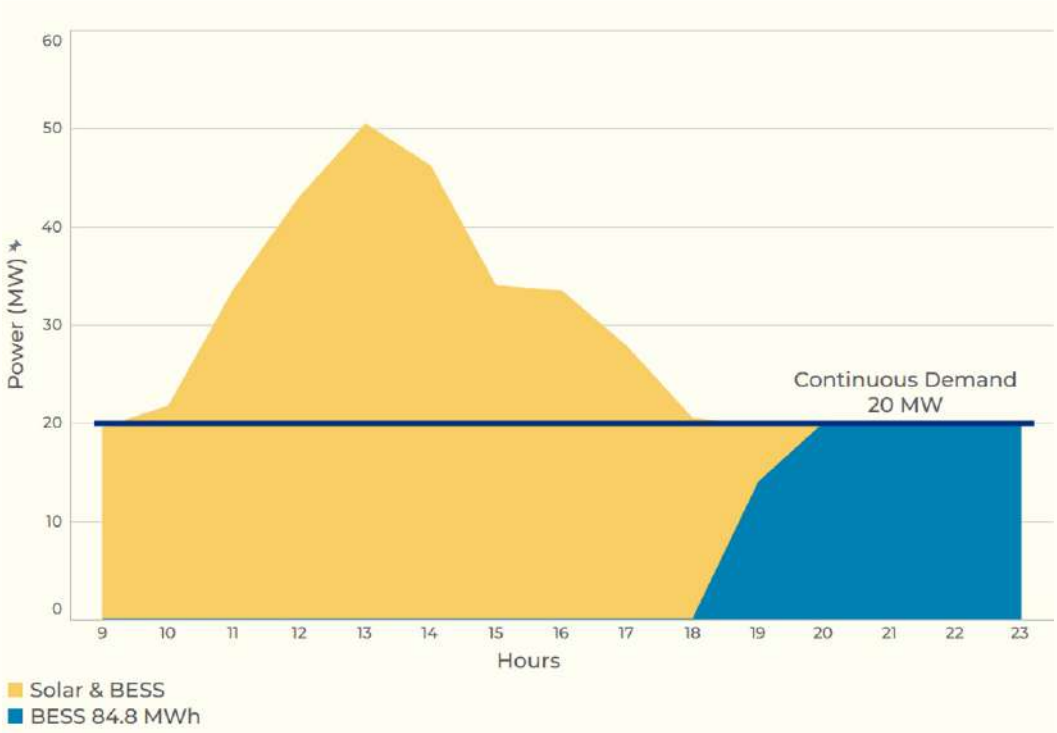


Figure 15 – 20 MW Firm support – 14 hours

### 5.1.1.h. Solar 50 MW against 10 MW continuous demand

In the final iteration, a 50 MW solar power plant in conjunction with demand profiles to find the optimal configuration for a hybrid power plant is examined. To meet a continuous demand of 10 MW, pairing the 50 MW solar power plant with a Battery Energy Storage System (BESS) of approximately 137.5 MWh ensures a consistent supply throughout the day. The analysis is illustrated in the figure below:

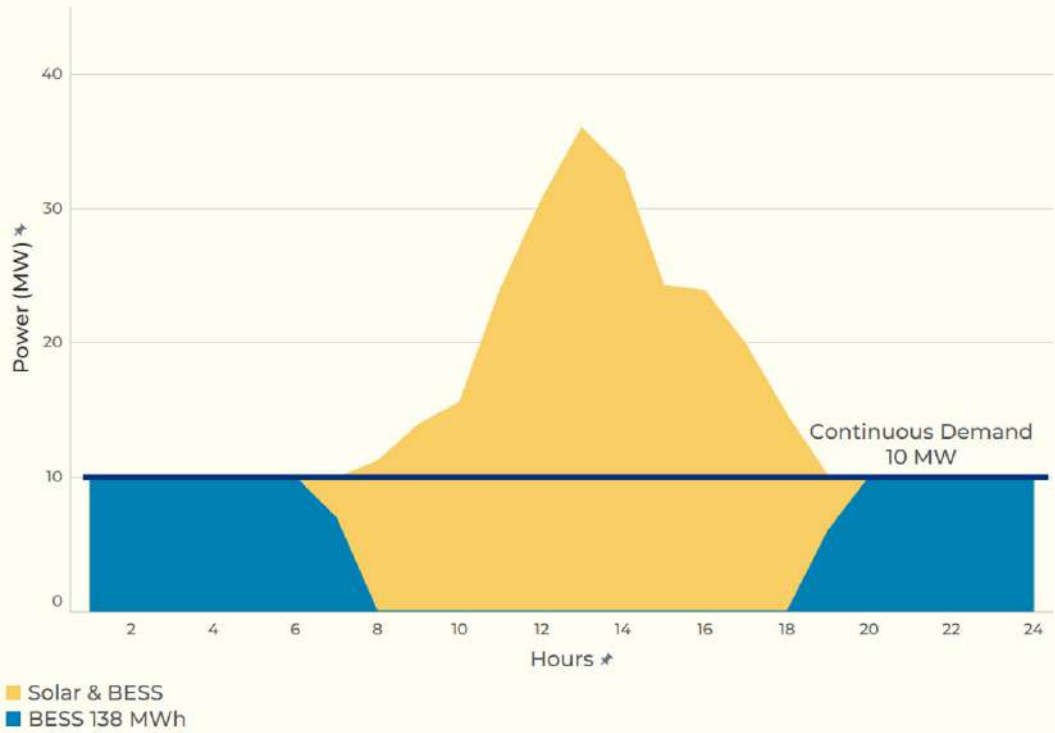


Figure 16 – 10 MW Firm support – All day

### 5.1.1.i. Solar 50 MW against 15 MW continuous demand

For the continuous demand of 15 MW, solar power plant of 50 MW paired with the BESS of 64 MWh can provide the firm support of 15 MW for up to 13 hours reason being not enough generation from solar power plant. This is illustrated in the figure below:

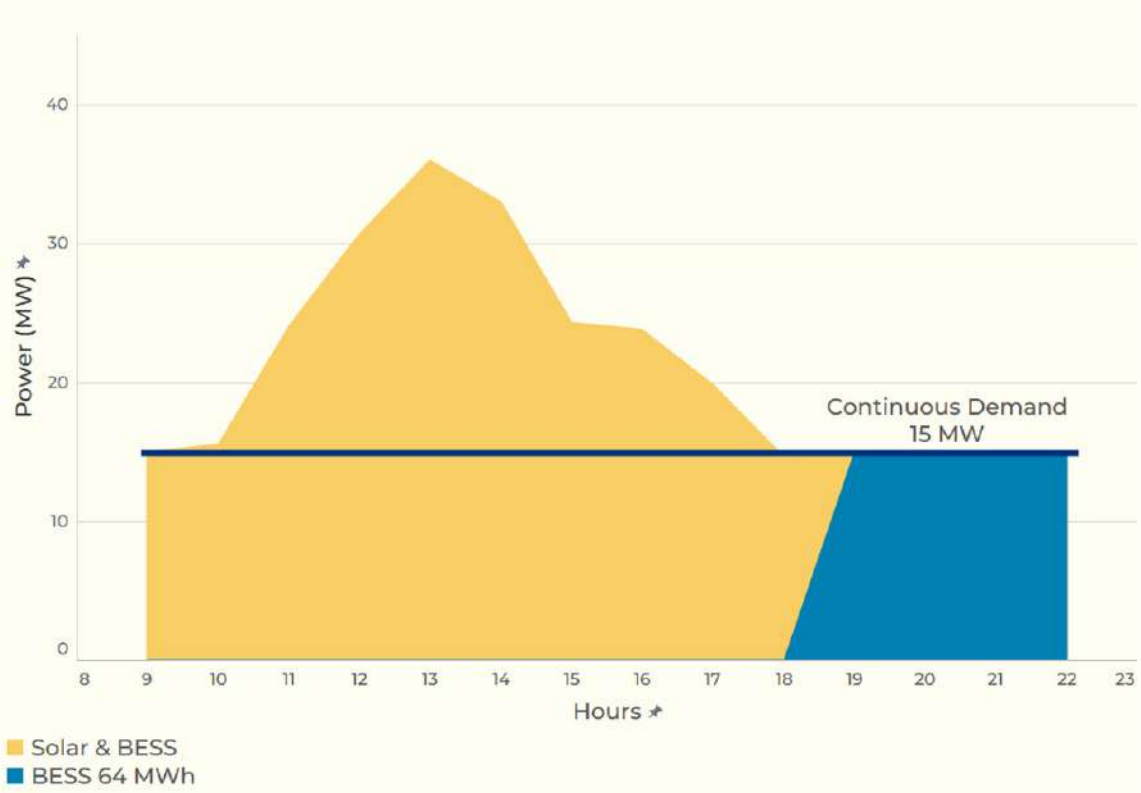


Figure 17 – 10 MW Firm support – 14 hours

The analysis reveals that combinations of solar power plants ranging from 100 MW to 50 MW, paired with varying sizes of Battery Energy Storage Systems (BESS), show the potential for hybrid setups to provide firm support. While all combinations highlight the feasibility of solar-BESS integration, some configurations offer firm support for limited durations. This hybrid proposition presents an opportunity for Renewable Energy (RE) developers to tailor offerings to bulk power consumers, particularly export-based industrial sectors. As industries face mounting pressure to reduce Scope 2 and Scope 3 emissions and meet science-based targets (SBTs) to remain competitive in global markets, securing access to clean energy becomes imperative especially for export-based industry. The firm support scenarios outlined in this report serve as a valuable reference for the development of customized products, positioning them as critical assets in Pakistan's electricity market, particularly following the implementation of Competitive Trading Bilateral Contracts Markets (CTBCM).

### 5.1.2. Wind with BESS

In contrast to solar power plants, wind power plants show low diurnal variations and significant yearly variations in their generation. The reason behind this varying generation pattern of wind power plants is that during windy seasons, the variations are minimal as per the topography of the region in which they exist, while in low wind periods, the generation is minimal. Therefore, integrating the BESS with a standalone wind power plant in an area with such variations is not technically feasible. This is evident from the results of the scenario in which a wind power plant of 100 MW is coupled with the BESS units of 20 MW and 60 MW, as shown in Figure below:

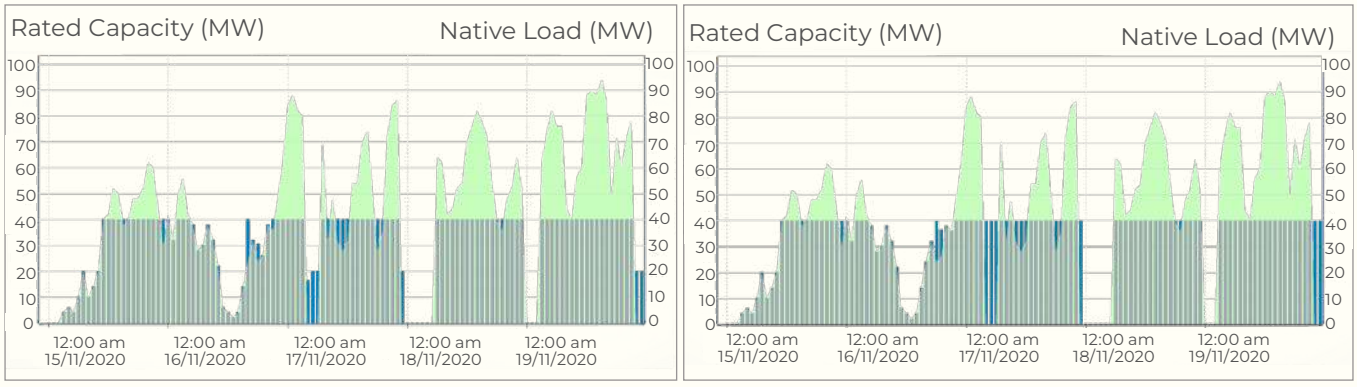


Figure 18 - Comparison of 40 MW demand served by 20 MW 4h and 60 MW 4h BESS with Wind

The extent to which a wind power plant can meet demand depends on seasons, i.e., wind seasons. For instance, a wind power plant of 100 MW coupled with 80 MWh BESS can meet the continuous demand of 10 MW and 20 MW during May and August when the wind speed is higher. However, this configuration cannot meet the demand during February and November when the wind speed is minimal, as depicted in Figures 19 & 20 below:

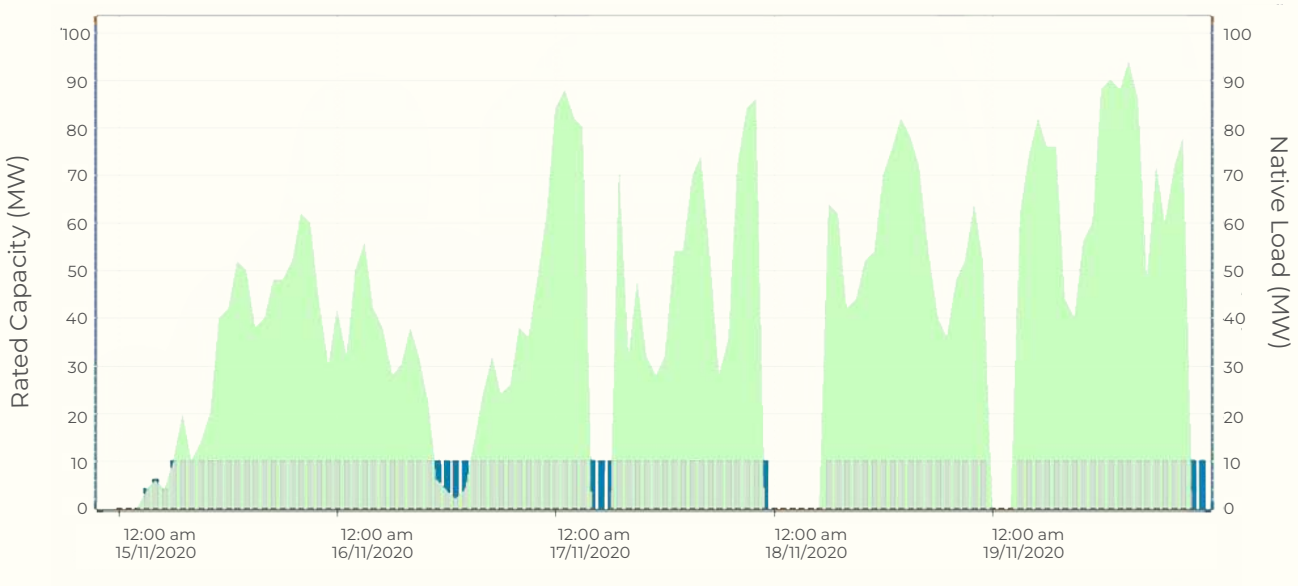


Figure 19 - 10 MW demand served by 20 MW, 40 MW 4h BESS with Wind

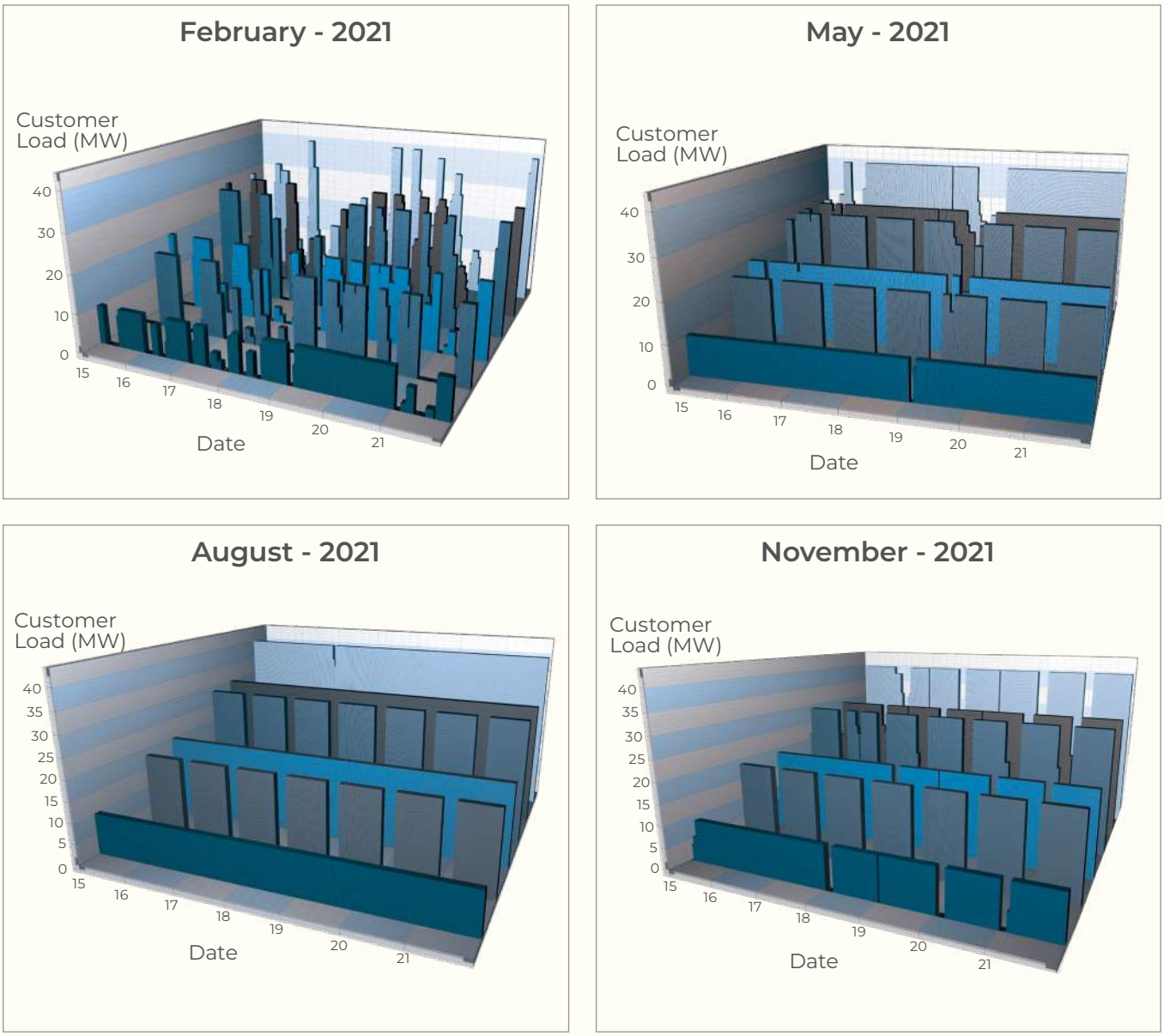


Figure 20- Demand served with wind and 20 MW & 4h BESS system

Similarly, a wind power plant of 100 MW coupled with the three BESS units of 20 MW, each with a capacity of 240 MWh, can meet the continuous demands of 10, 20, 30, and 40 MW during the high wind months, i.e., August and May. However, this system is not able to meet the demand requirements during low wind periods, as shown in Figure below:

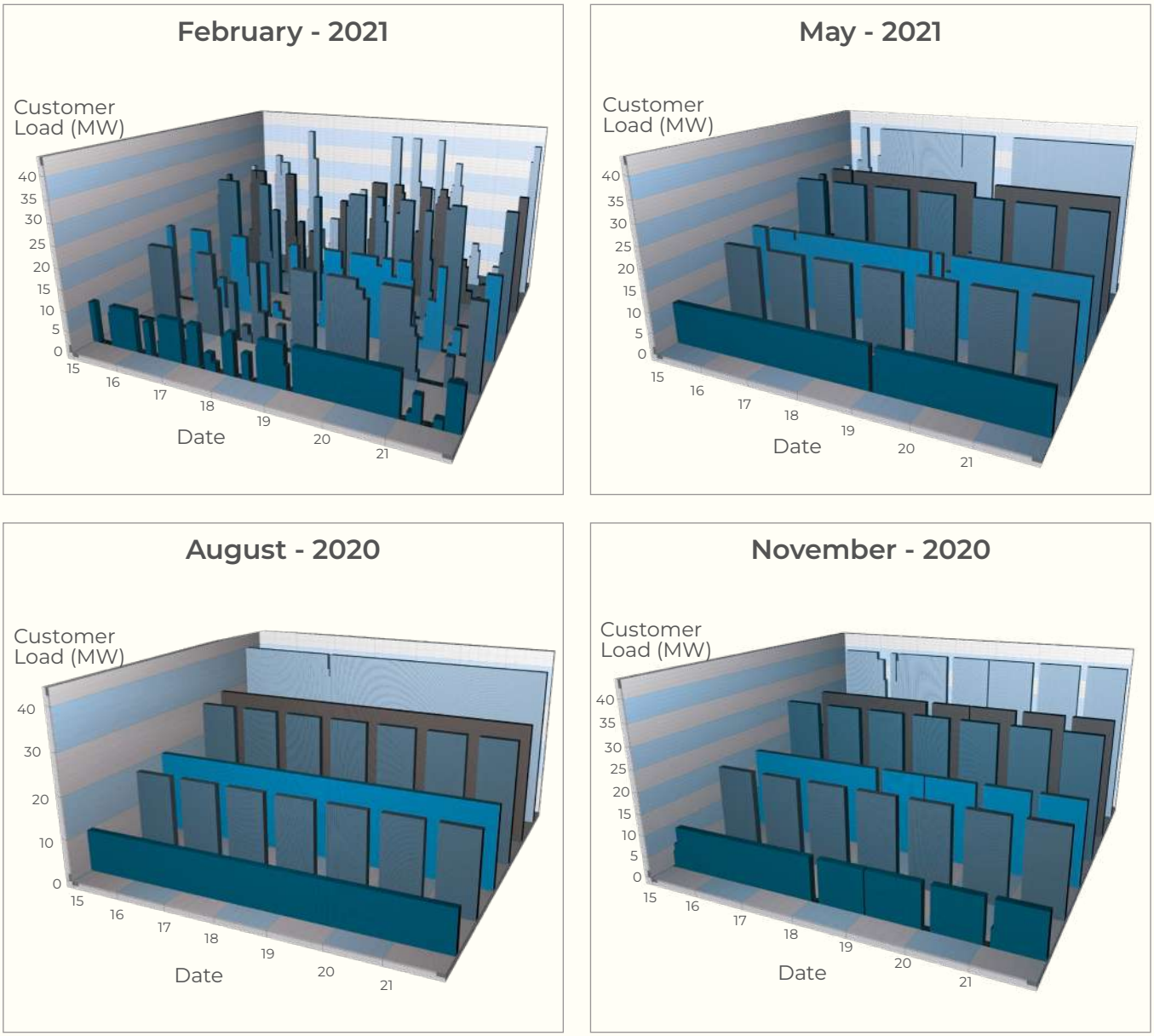


Figure 21 - Demand served with wind and 60 MW & 4h BESS system

### 5.1.3. Solar and Wind (Hybrid-37) with BESS

The hybrid model of solar power plants coupled with wind power plants has appeared as an influential contributor to sustainable power generation. In a wind farm, where turbines are strategically spaced at a considerable distance from each other (approximately 8 to 15 times the rotor diameter), incorporating solar panels within these gaps yields synergies. This approach maximizes energy yield from the land, providing a reliable, enhanced, and consistent power output compared to standalone wind or solar power plants.

As per the analysis, the optimum configuration for this hybrid setting is 3:1 (wind to solar), which results in a capacity factor of around 36 percent. However, the capacity factor and hybrid configuration may vary depending on the spatial differences and topography of the region under consideration. Therefore, in this study, a hybrid power plant of 100 MW is considered with a wind-to-solar ratio of 3:1, having a capacity factor of around 36 percent. To analyze how this hybrid power plant, when coupled with the BESS, will meet the demands, a single unit of BESS with 80 MWh capacity is considered. The results show that this configuration successfully met the continuous demand of 10 MW. However, the continuous demand for 15 MW was only fulfilled during the third and fourth quarters of the year because of the unavailability of wind during the first quarter. The results are shown in figure as follows:

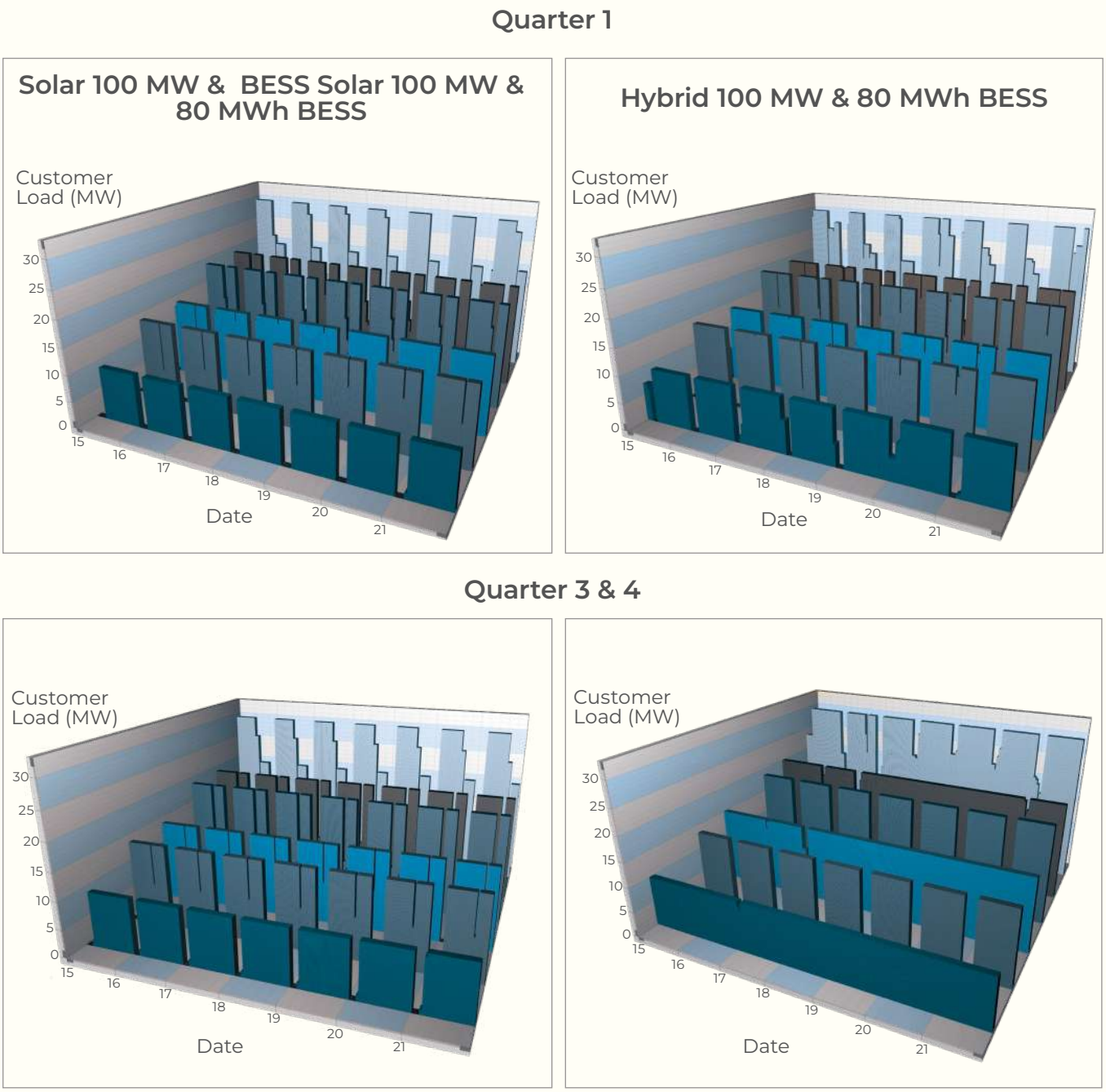


Figure 22- Comparison of solar and hybrid systems for quarterly firm support

### 5.2. Peak Support

For the use case on Peak Support, the demand of overall Pakistan is analyzed. The annual demand curves for NTDC and KE are plotted hourly throughout FY 2022-23. To compare NTDC and KE's demand, the demand profiles of both NTDC and KE are normalized. During the first quarter, NTDC experiences a peak demand that is relatively higher than KE but subsequently undergoes a steep decrease compared to KE. Likewise, in the third quarter, NTDC and KE exhibit low demand until the fourth quarter, when KE's peak demand surpasses that of NTDC as shown in the figure below:

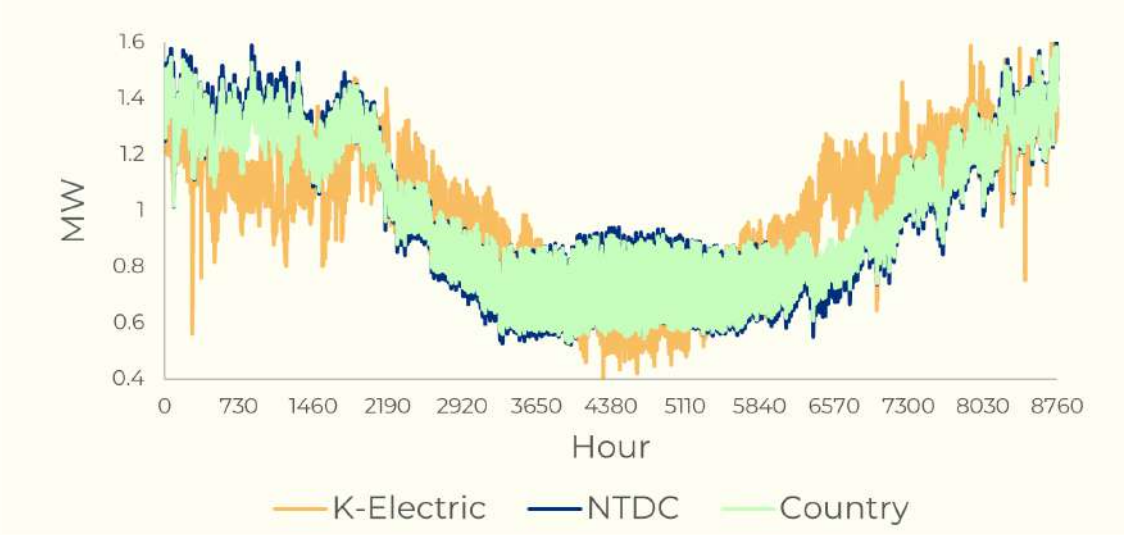


Figure 23 – Annual Demand Profiles

Note: this graph has been normalized at 100 MW for the purpose of comparison

The quarterly demand profiles of NTDC and KE are also plotted as shown in figure below:

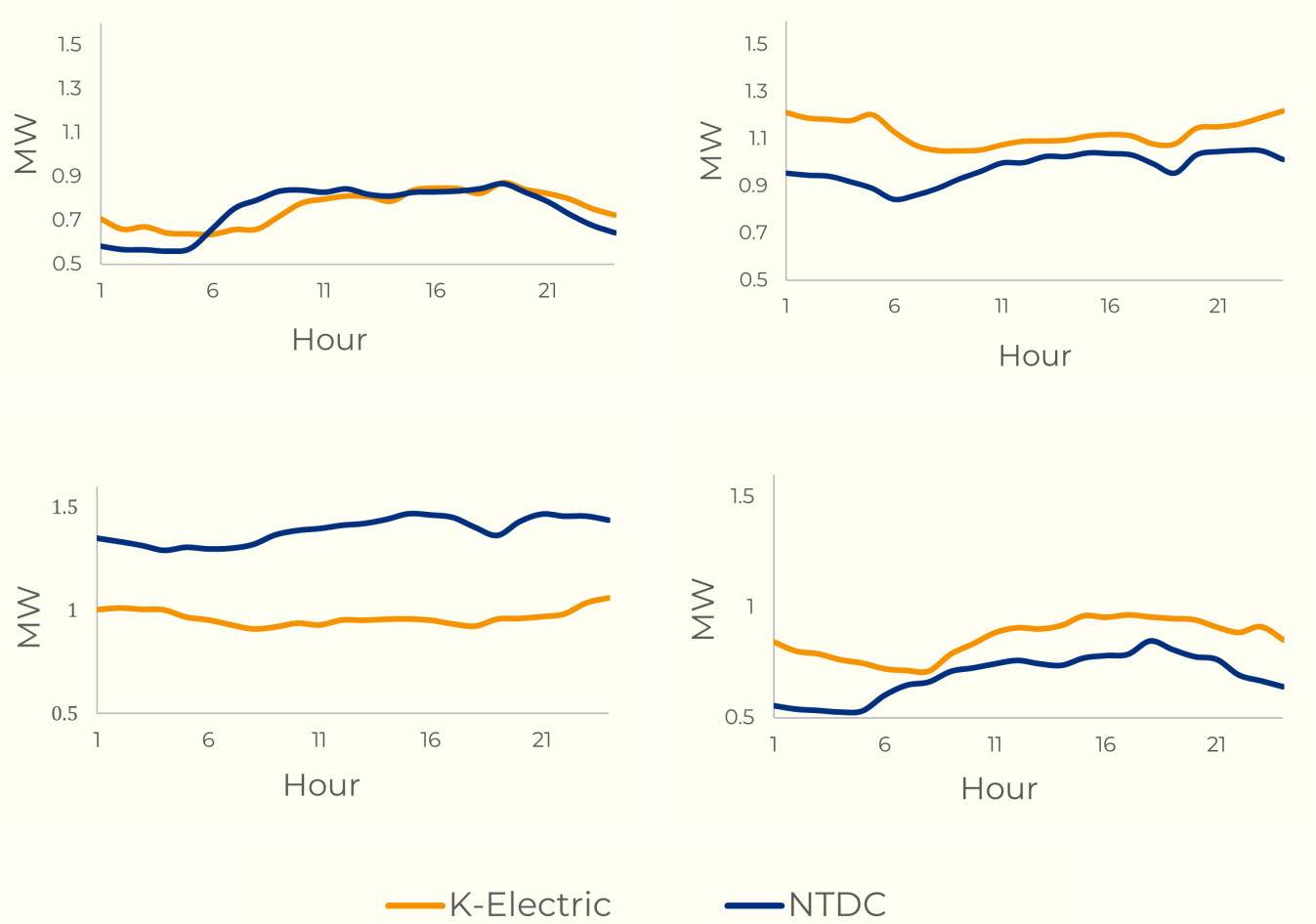


Figure 24 – Quarterly Demand Profiles

Based on the demand profile of Pakistan, six demand profiles (10 MW continuous, 15 MW step and continuous, 20 MW step and continuous, and 30 MW continuous) are created to analyze how the hybrid proposition of BESS coupled with REs fulfill these demand profiles as shown in figure below:

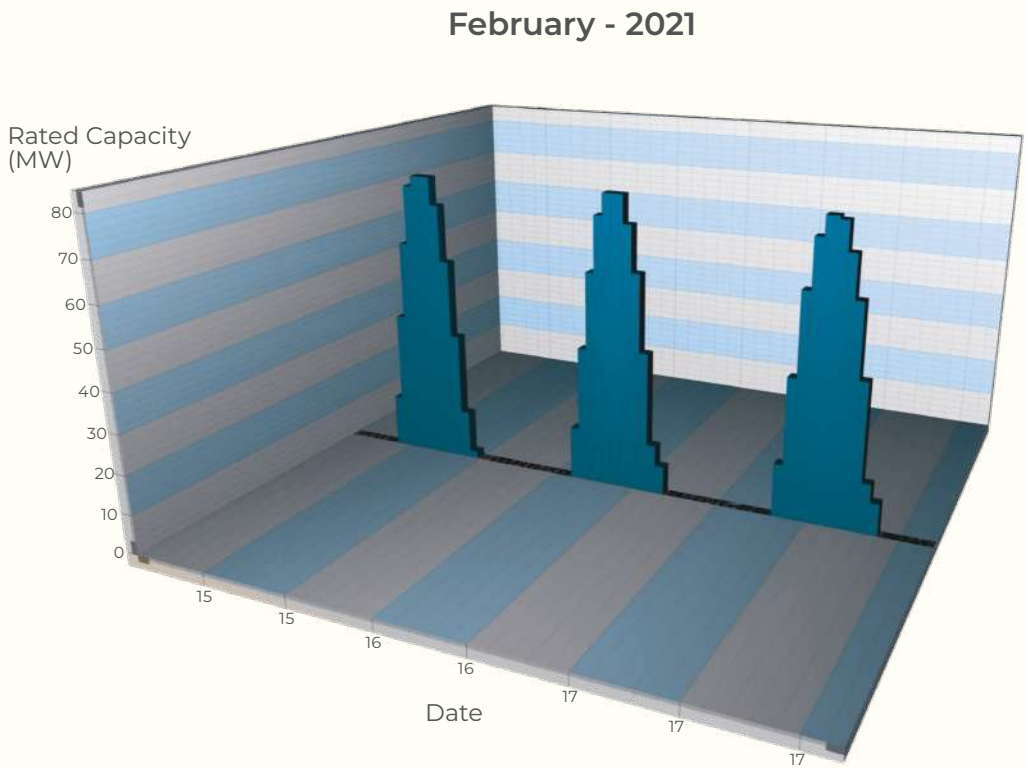


Figure 25 – Daily demand profiles for Peak Support 10, S15, 15, S20, 20, 30 (MW)

### 5.2.1. Peak Support – Solar

The Solar power plant (100MW) coupled with 80 MWh BESS can effectively meet the daily peak requirements as shown in the figure below:

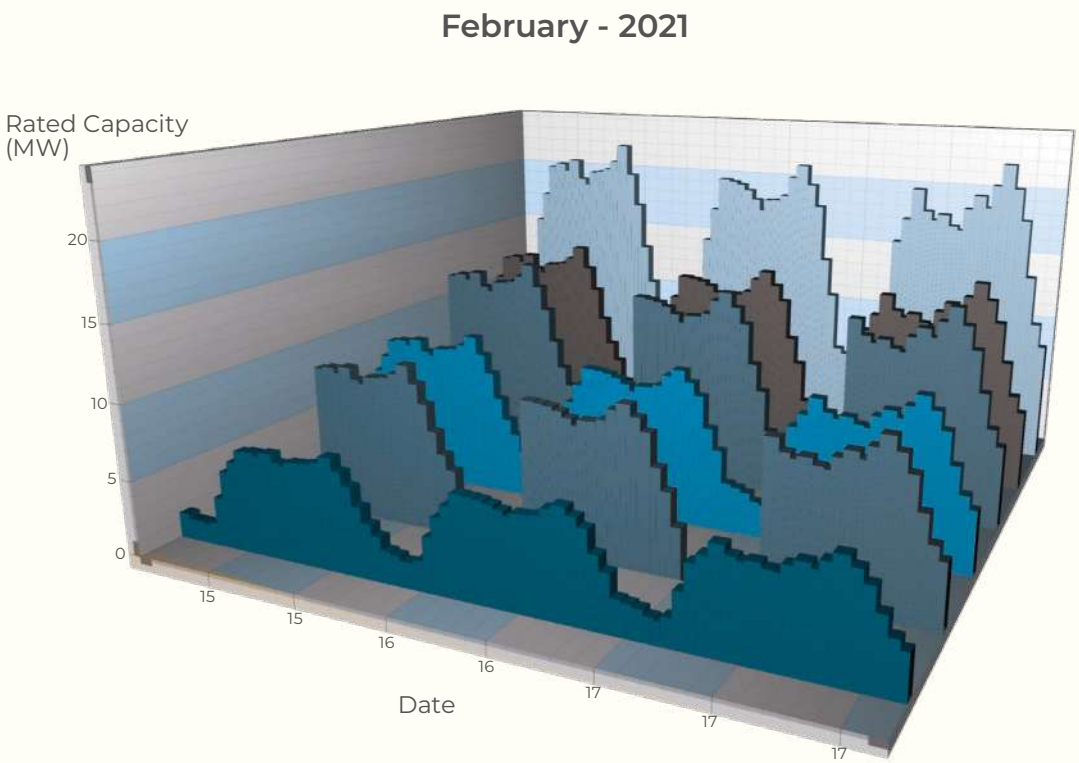


Figure 26 - Generation Profile of Solar plant coupled with 80MWh BESS

### 5.2.2. Peak support – Wind

In contrast to the Solar power plant coupled with the BESS, the wind power plant coupled with the BESS cannot meet the daily peak requirements due to wind seasons and inconsistencies in the generation during summer months. This is clear from the generation profile of 100 MW wind coupled with 80 MWh of BESS, as shown in Figure below:

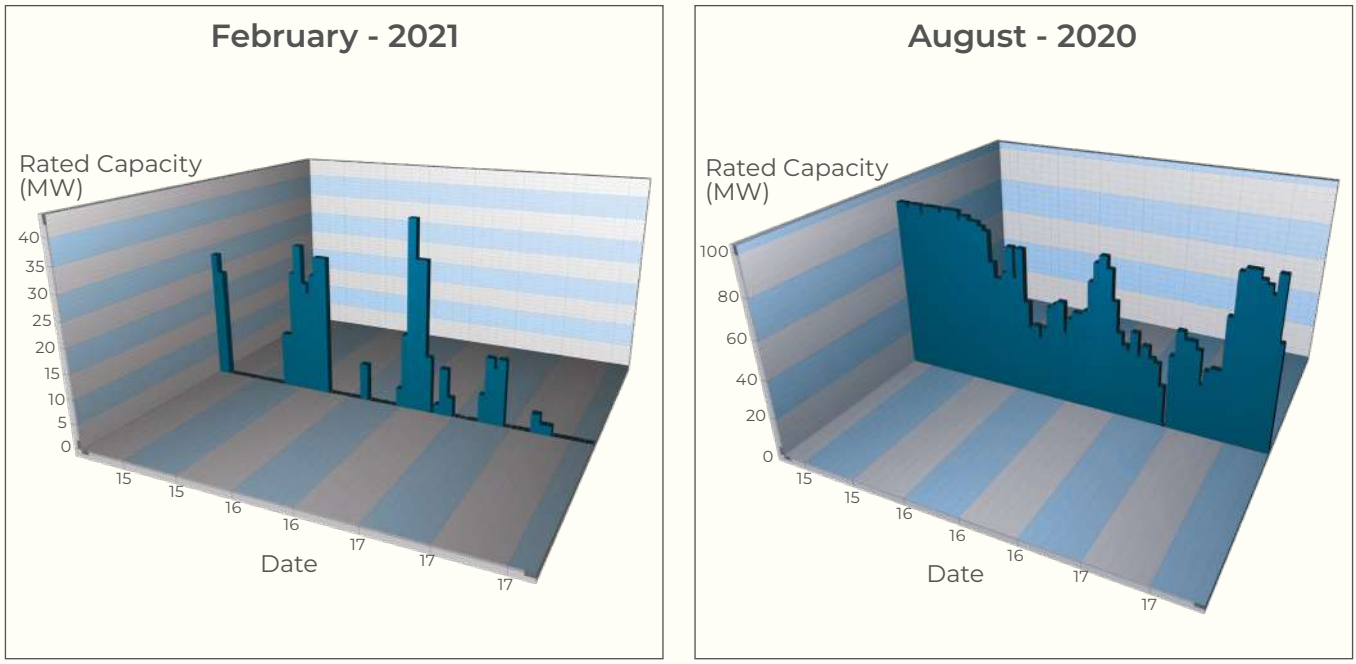


Figure 27 – Generation Profile of 100MW Wind with 80 MWh BESS

### 5.2.3. Peak support – Hybrid

The 100 MW hybrid power plant with a wind-to-solar ratio of 3:1, when coupled with 80 MWh BESS, effectively met the daily demand. However, the LCOE of this hybrid system is higher than standalone other REs (Wind and Solar) with BESS settings. The generation profile of this hybrid setting is shown in Figure below:

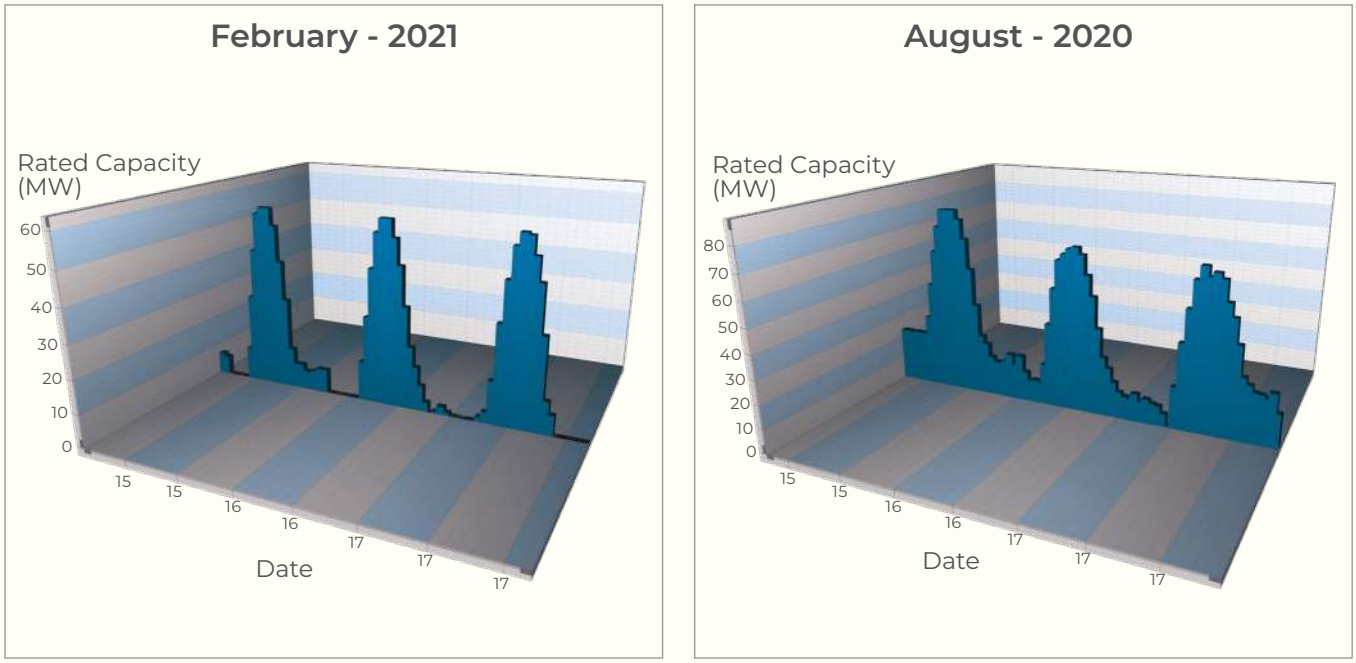


Figure 28 - Generation Profile of 100 MW Hybrid plant with 80MWh BESS

### 5.2.4. Peak support – Summer & Winter Variation

The overall analysis of use case cases reveals that the hybrid proposition of a wind power plant coupled with the BESS is well suited for peak demand during summer; the hybrid power plant with a 3:1 wind-to-solar ratio coupled with the BESS can provide peak support during winter months. Whereas, owing to a consistent generation profile throughout the year (ideally, the sun is shining all days), a solar power plant coupled with the BESS is well suited for annual firm capacity and wind with the BESS for daily firm capacity support. Similarly, a wind power plant coupled with the BESS is suitable for annual peak support, and a solar power plant with the BESS is well suited for daily peak support.

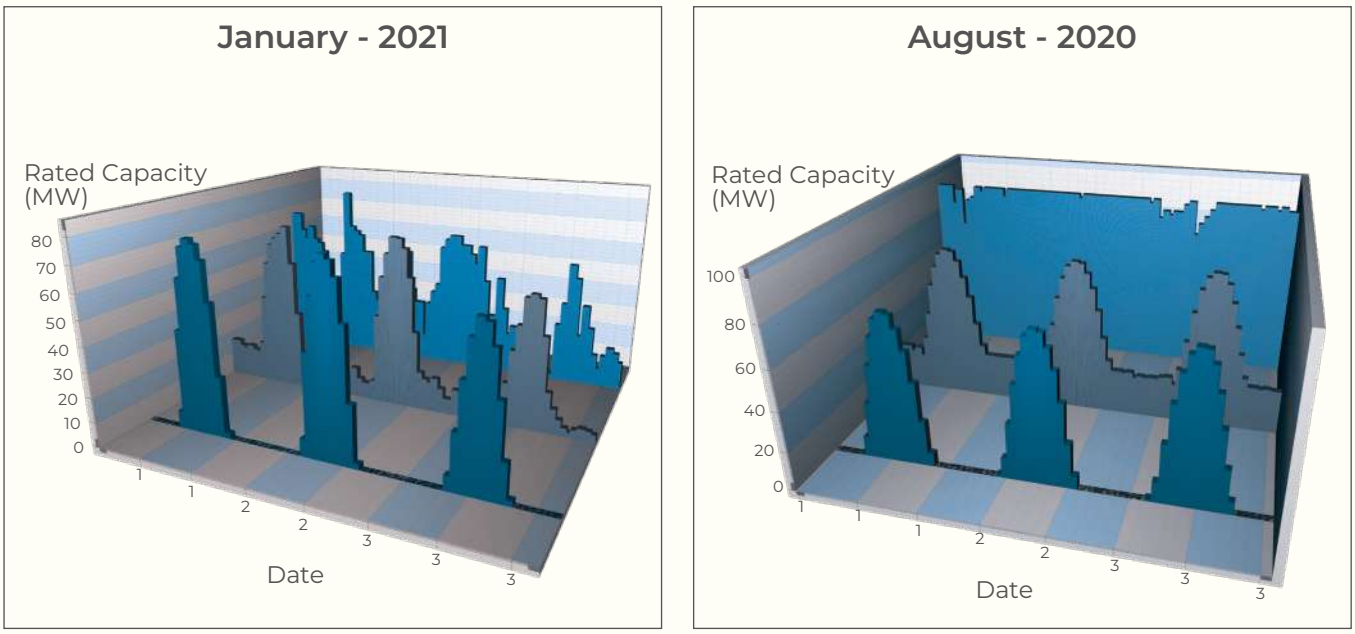


Figure 29 – Systems suitable for Seasonal peak support

| | Yearly | | Daily | |
|---|---|---|---|---|
| **Use Case** | **Solar** | **Wind** | **Solar** | **Wind** |
| Firm Capacity | Ideal | | | Ideal |
| Peak Support | | Ideal | Ideal | |

Table 4: Ideal use case based on diurnal and yearly variation

## 6. The Techno-economic analysis (TEA)

Techno-economic analysis (TEA) is a technique for evaluating the economic feasibility of specific technologies. TEA combines the technical and financial parameters to assess the overall performance, costs, and benefits of implementing a technology. In electrical power generation projects, TEA helps decision-makers better understand the economic implications and benefits associated with adopting such technologies. In this study, the scope of TEA encompasses understanding a vital metric, i.e., the Levelized cost of energy (LCOE) of integrating the BESS with REs, which is described below:

### 6.1 Levelized Cost of Energy (LCOE)

The LCOE is a critical metric used to assess the per unit cost of generating energy from a power plant throughout its life. It is a standard way of evaluating and comparing the cost competitiveness of different energy generation technologies (such as solar, wind, coal, natural gas, etc.)

LCOE is the lowest cost at which the electricity must be sold to cover all costs incurred over a plant's lifespan and achieve a break-even point. The electricity tariff allows the power plant to recover its initial investment and ongoing operational expenses.

For a clearer insight into how LCOE reflects the costs associated with generating electricity and meeting investor expectations, it's essential to familiarize ourselves with critical terms such as:

- **Equity:** Ownership stake in a company representing shareholders' residual interest in assets after deducting liabilities.
- **Debt:** Borrowed funds that require repayment with interest over a specific period, usually involving a

creditor-lender relationship.

- **Return on Equity:** Measure of profitability showing the percentage return earned on shareholders' investment in a company.
- **Return on Debt:** Indicator of the efficiency of debt utilization, representing the interest payments made on borrowed funds relative to the debt amount.
- **Discount rate:** The rate used to find the present value of future cash flows, considering the time value of money and risk factors.

### 6.1.1 Methodology: LCOE calculation for Solar/Wind and Batteries

LCOE considers all the expenses related to setting up a power plant, encompassing capital expenditure, operational and maintenance costs, and fuel expenses, and then computes a levelized cost for each unit of electricity the power plant generates throughout its lifespan. The LCOE comprises two main components: fixed costs and variable costs. The variable costs include fuel cost and the variable component of operations and maintenance cost (VO&M), usually expressed in $/MWh (US Dollars per Megawatts). Whereas fixed costs include capital cost, fixed operations and maintenance cost (FO&M), return on equity (ROE), return on equity during construction (ROEDC), insurance, and debt servicing, usually expressed in $/kW-year (US Dollars per kilowatt-year). The levelized cost calculation for the BESS is analogous to that for wind/solar. The only difference is that the levelized cost for BESS uses charging costs as fuel costs and takes the discharged electricity instead of generated electricity. For the analysis of LCOE, all the cost assumptions/per unit rate of electricity generated by REs has been taken from NEPRA latest tariff determinations. The USD to PKR exchange rate considered is 1USD=280 PKR. These assumptions are summarized in the table below:

| # | Technology | LCOE (Cents/kWh) | LCOE Rs/kWh) |
|---|---|---|---|
| 1 | Solar | 4 | 11.2 |
| 2 | Wind | 4.5 | 12.6 |

Table 5 – Reference Tariff determinations by NEPRA

For the optimization of BESS, key input parameters are taken from the energy consultants and publicly available report by World Bank "Guidelines to implement battery energy storage systems under public-private partnership structures" as listed in the table below:

| Parameter | Value |
|---|---|
| Build Cost ($/kWh) * | 300 |
| VO&M Charge ($/kWh) | 7 |
| Economic life (yrs.) | 10 |
| Technical life (yrs.) | 10 |
| Max Cycles | 8000 |
| Capacity Degradation (%/1000 cycles) | 10.9 |
| Max SoC (%) | 95 |
| Min SoC (%) | 5 |
| Charge Efficiency (%) | 95 |
| Discharge Efficiency (%) | 95 |
| *Inclusive of DC/AC and EPC costs | |

Table 6 – Input parameters for BESS Modelling

The formula to calculate LCOE involves dividing the total present value of costs (both initial and ongoing) by the total electricity output over the plant's lifespan. The result is the cost of generating each unit of electricity averaged over the plant's lifetime.

$$LCOE = \frac{\sum[Capital\ cost + O\&M\ cost + Fuel\ cost) \times (1+r)^{-t}]}{\sum Energy\ (MWh)\ \times (1+r)^{-t}}$$

The LCOE is calculated as formulated in the following equation:

Were,

**Capital cost**: Total capital expenditures in year t;

**O&M cost:** Total operations and maintenance costs in year t;

**Fuel cost:** Fuel costs in year t;

**Energy:** The amount of electrical energy produced in MWh in year t

**r:** Discount rate (the rate at which money is discounted over a period, i.e., the rate that time adds or drops value to the money per time period); and

**$(1 + r)^{-t}$:** The discount factor for year t.

### 6.1.2. Scenario Wise LCOE

The LCOE for firm support scenario of solar power plant paired with BESS are mentioned in the table below, the exchange rate is considered as 1USD = 280PKR.

| **Solar 100 MW** | | |
|---|---|---|
| BESS (MWh) | LCOE (US cents/kWh) | Firm Support |
| 137 | 6.84 | 10 MW all day |
| 205 | 8.25 | 15 MW all day |
| 277 | 9.70 | 20 MW all day |
| 127 | 6.62 | 30 MW 14 hours |
| **Solar 70 MW** | | |
| BESS (MWh) | LCOE (US cents/kWh) | Firm Support |
| 137 | 8.1 | 10 MW all day |
| 205 | 10 | 15 MW all day |
| 85 | 6.5 | 20 MW 14 hours |
| **Solar 50 MW** | | |
| BESS | LCOE (US cents/kWh) | Firm Support |
| 137 | 9.6 | 10 MW all day |
| 64 | 6.6 | 15 MW 14 hours |

Table 7 – Scenario-wise costs for Solar-Battery Systems

The LCOE for each peak support scenario (REs coupled with BESS) is calculated with various capacity factors of REs and sizes of BESS as shown in Table below:

| **LCOE (US Cents/kWh)** | | | |
|---|---|---|---|
| Scenarios | With 20 MW BESS | With 40 MW BESS | With 60 MW BESS |
| Solar Net Meter | 7.35 | 9.5 | 11.65 |
| Utility-scale solar | 6.65 | 7.3 | 8.9 |
| Utility-scale wind | 5.42 | 6.34 | 7.25 |
| Hybrid 1 | 5.55 | 6.93 | 8.3 |
| Hybrid 2 | 5.48 | 6.6 | 7.8 |
| Hybrid 3 | 5.45 | 6.48 | 7.5 |

Hybrids 1, 2, and 3 represent hybrid power plants with solar-to-wind ratios of 1:3, 1:1, and 3:1, respectively. Plant size is 100MW

Table 8 – Scenario-wise costs for Solar-Wind-Battery Systems

# 7. Conclusion and Future scope

The integration of Renewable Energy sources, complemented by Battery Energy Storage Systems (BESS), stands as a transformative solution to address the evolving dynamics of electricity demand in Pakistan. This study highlights the importance of tailoring electricity generation strategies to suit the unique requirements of various sectors and the broader national context, thereby ensuring both operational efficiency and long-term sustainability. The use case on Firm Capacity highlights the opportunity for RE developers to develop customized products that can be sold bilaterally in the power market. The firm capacity can be ensured by optimal combination of REs especially solar power plant with BESS that can be customized as per the demand profiles of industries. This hybridization offers cleaner energy and an opportunity to reduce reliance on thermal power plants. The surplus energy can be utilized either by direct selling into power market or to charge the battery that can be used to take advantage of market arbitrage. However, this requires enactment of open market regulations entailed in the Competitive Trading Bilateral Contracts Markets (CTBCM) that are still pending in Pakistan. The use case on Peak support suggests that optimum hybridization of REs and BESS is well suited for peak shaving instead of using costly peaking plants that are majorly based on take or pay contracts.

Central to the success of this hybrid approach is the precise determination of BESS size and the establishment of optimal charging and discharging schedules. These pivotal steps not only guarantee cost competitiveness but also empower the system to seamlessly meet predefined demand profiles, offering a reliable and consistent power supply.

Moreover, the hybridization of REs with BESS presents an unparalleled opportunity to craft tailored solutions for bulk power consumers, notably those within the export industry striving to meet stringent Scope 2 and Scope 3 emissions targets. Through the adoption of REs coupled with BESS, these consumers can enhance their energy autonomy, reduce dependency on fossil fuels, and champion environmental stewardship.

Looking forward, intensified research and development endeavors should prioritize the refinement of BESS sizing and scheduling methodologies within hybrid RE configurations. Rigorous feasibility assessments and pilot initiatives are indispensable to confirm the scalability and efficacy of these integrated solutions in real-world scenarios.

Furthermore, fostering collaboration among governmental entities, industry stakeholders, and research institutions is imperative to propel innovation and foster widespread adoption of REs coupled with BESS. This collaborative ethos will foster the creation of bespoke energy solutions tailored to the diverse needs of various sectors and consumers, ultimately ushering in a resilient, sustainable, and low-carbon energy landscape for Pakistan.

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Newspaper articles for the advocacy of cost potential cost savings in Pakistan's power sector can be accessed via following links:

- https://www.thenews.pk/print/1204918-shortcomings-in-igcep-2024-34-call-for-reassessment
- https://tribune.com.pk/article/97856/the-energy-conundrum-a-call-for-sustainable-solutions-in-pakistan
- https://www.brecorder.com/authors/102236/ubaid-ullah-khan
- https://www.dawn.com/news/1877255/power-efficiency